本节内容

外部排序

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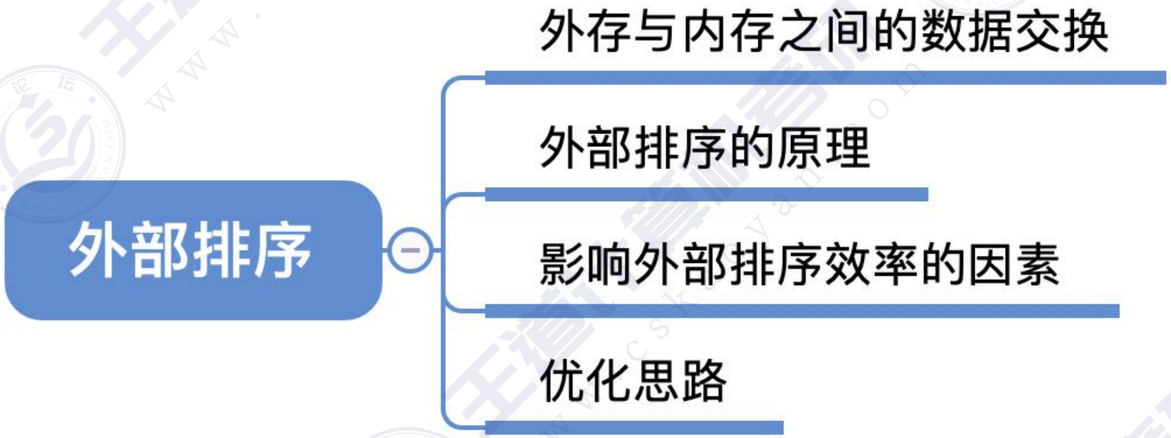
知识总览



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外存、内存之间的数据交换

操作系统以“块”为单位对磁盘存储空间进行管理，如：每块大小 1KB
各个磁盘块内存放着各种各样的数据

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磁盘的读/写以“块”为单位
数据读入内存后才能被修改
修改完了还要写回磁盘

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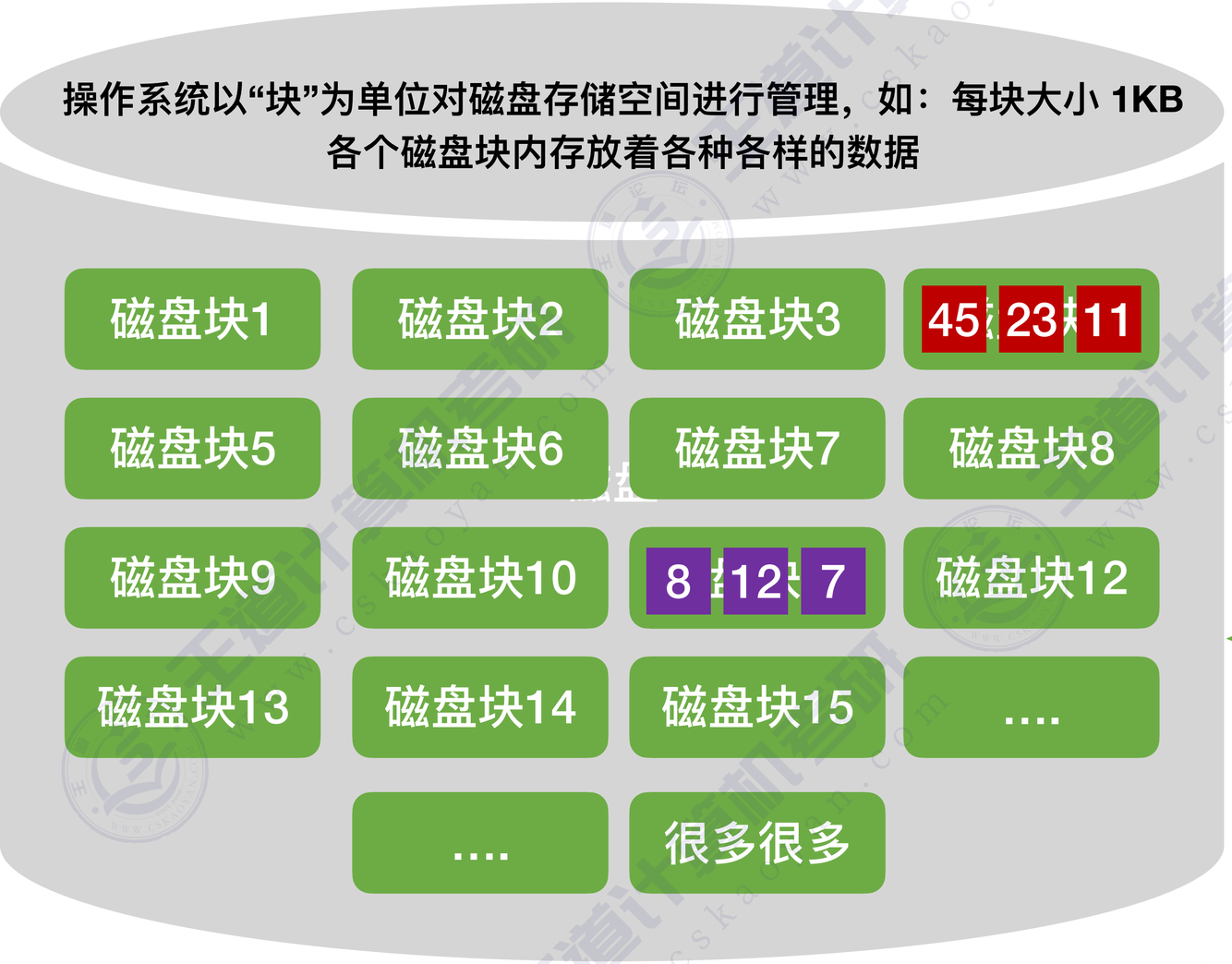

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外部排序原理

外部排序：数据元素太多，无法一次全部读入内存进行排序

使用“归并排序”的方法，最少只需在内存中分配3块大小的缓冲区即可对任意一个大文件进行排序

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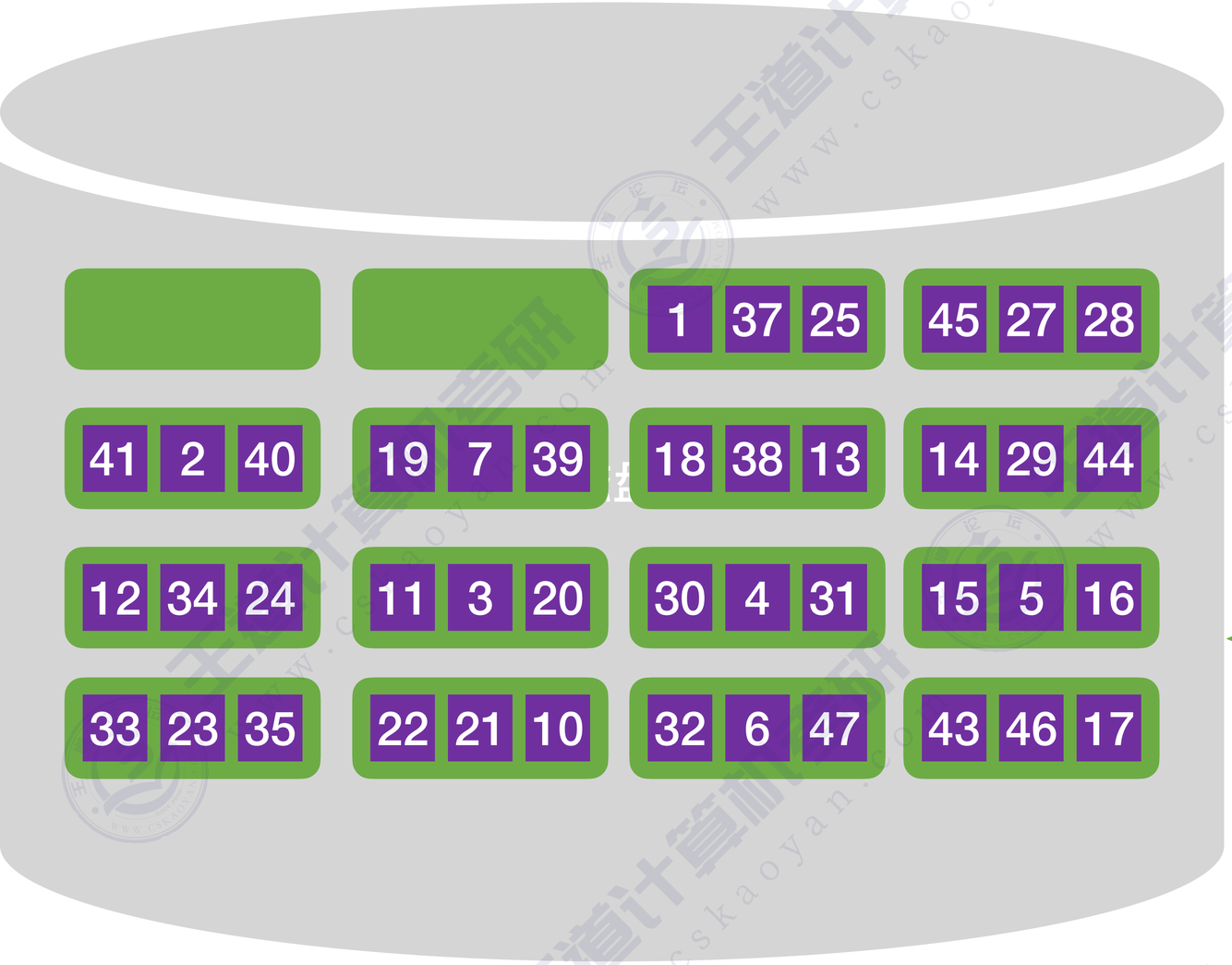
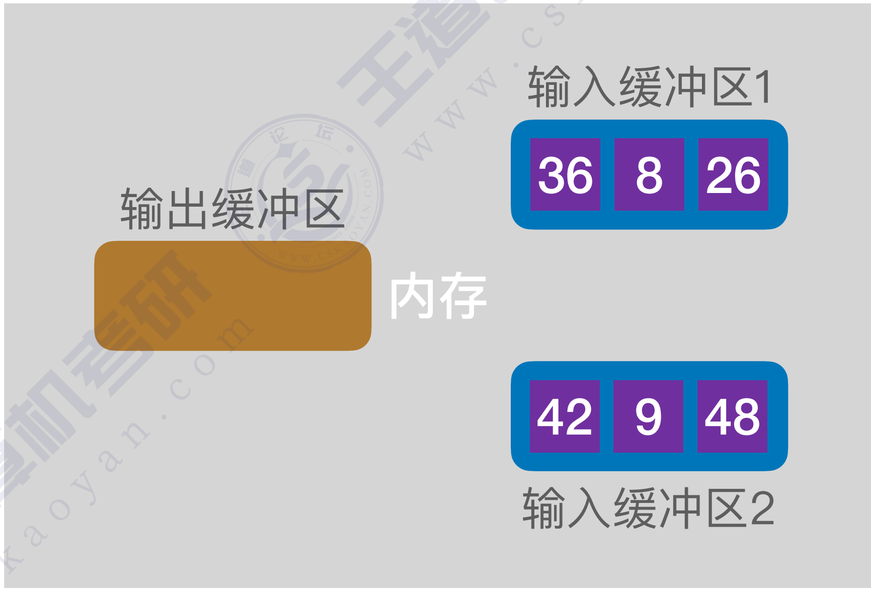
构造初始“归并段”



读磁盘

写磁盘

“归并排序”要求各个子序列有序，每次读入两个块的内容，进行内部排序后写回磁盘



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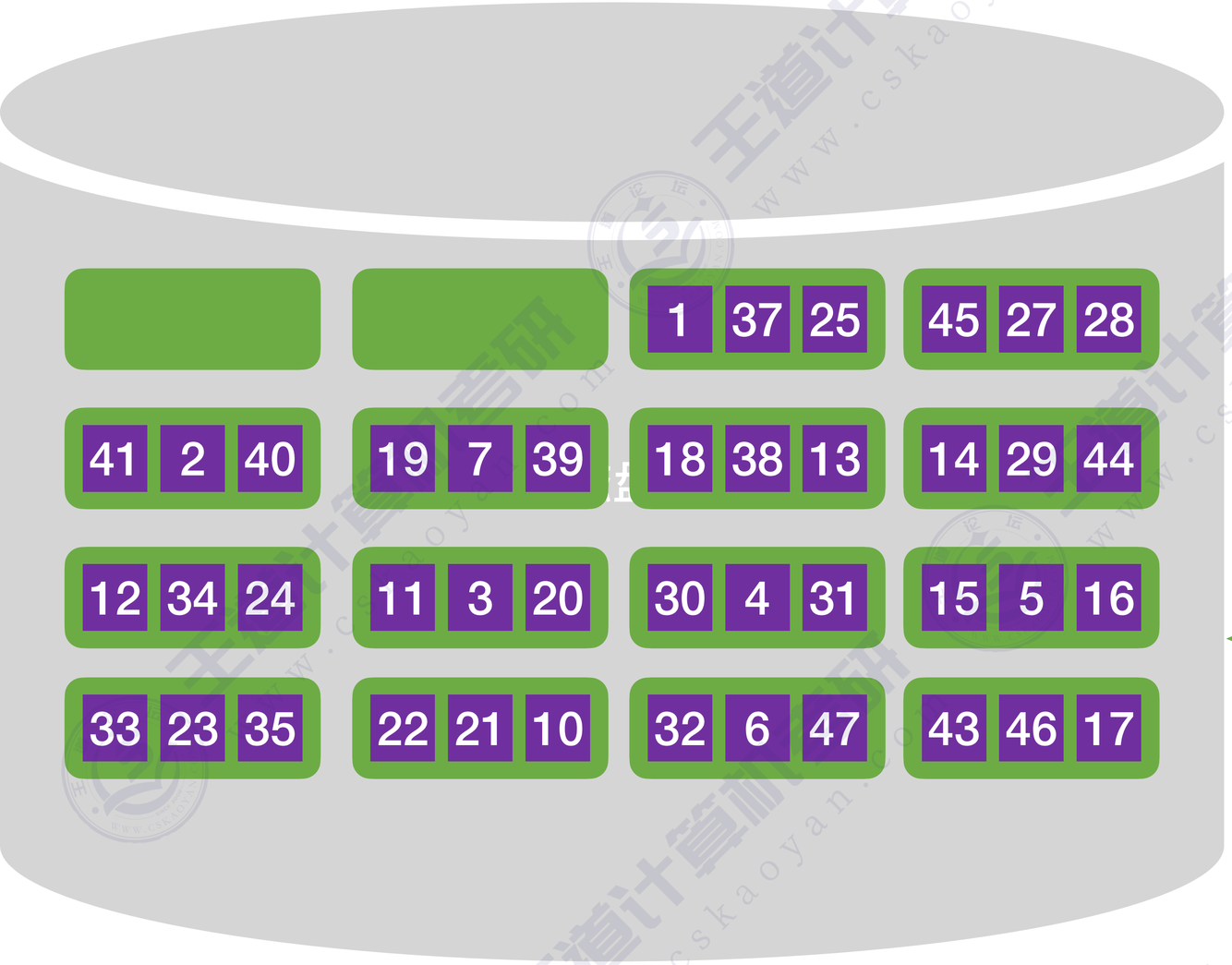

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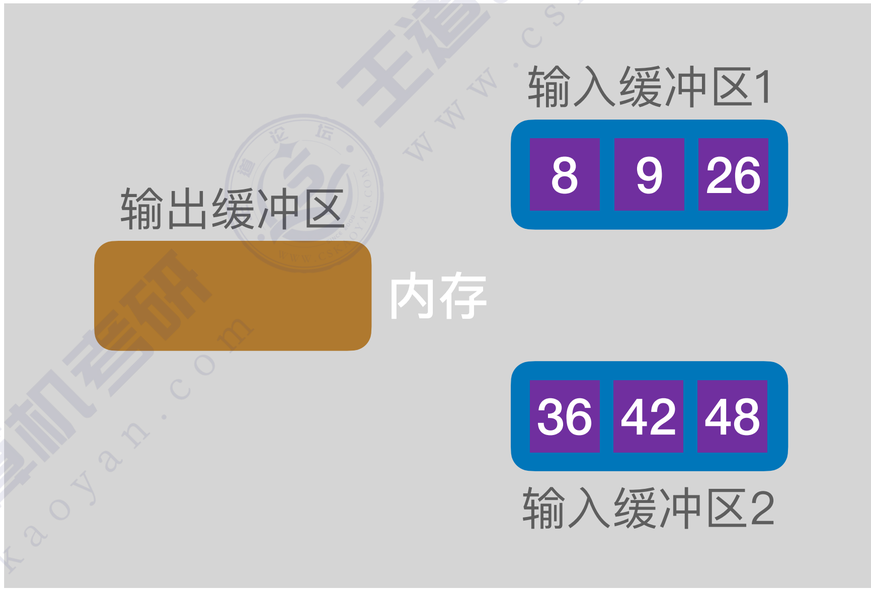

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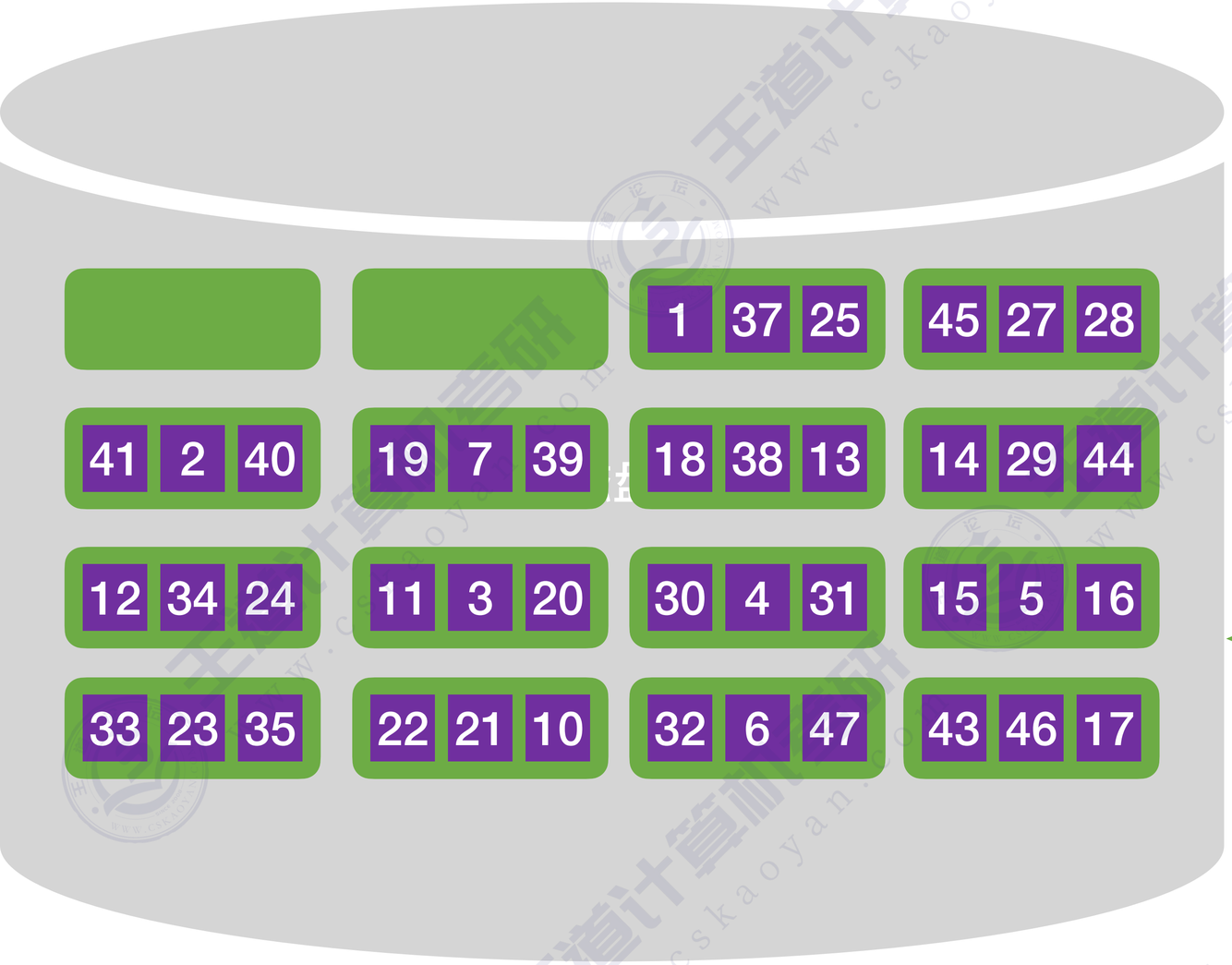

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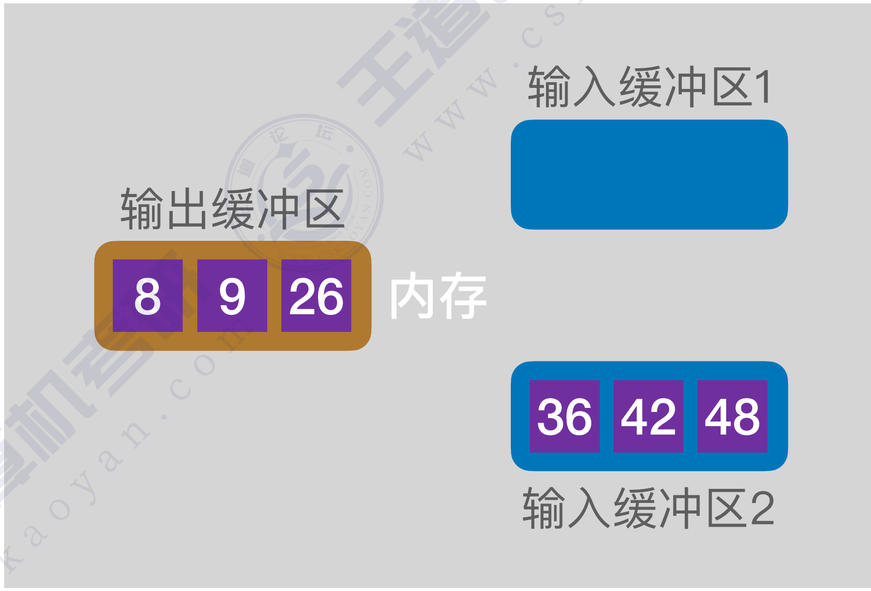

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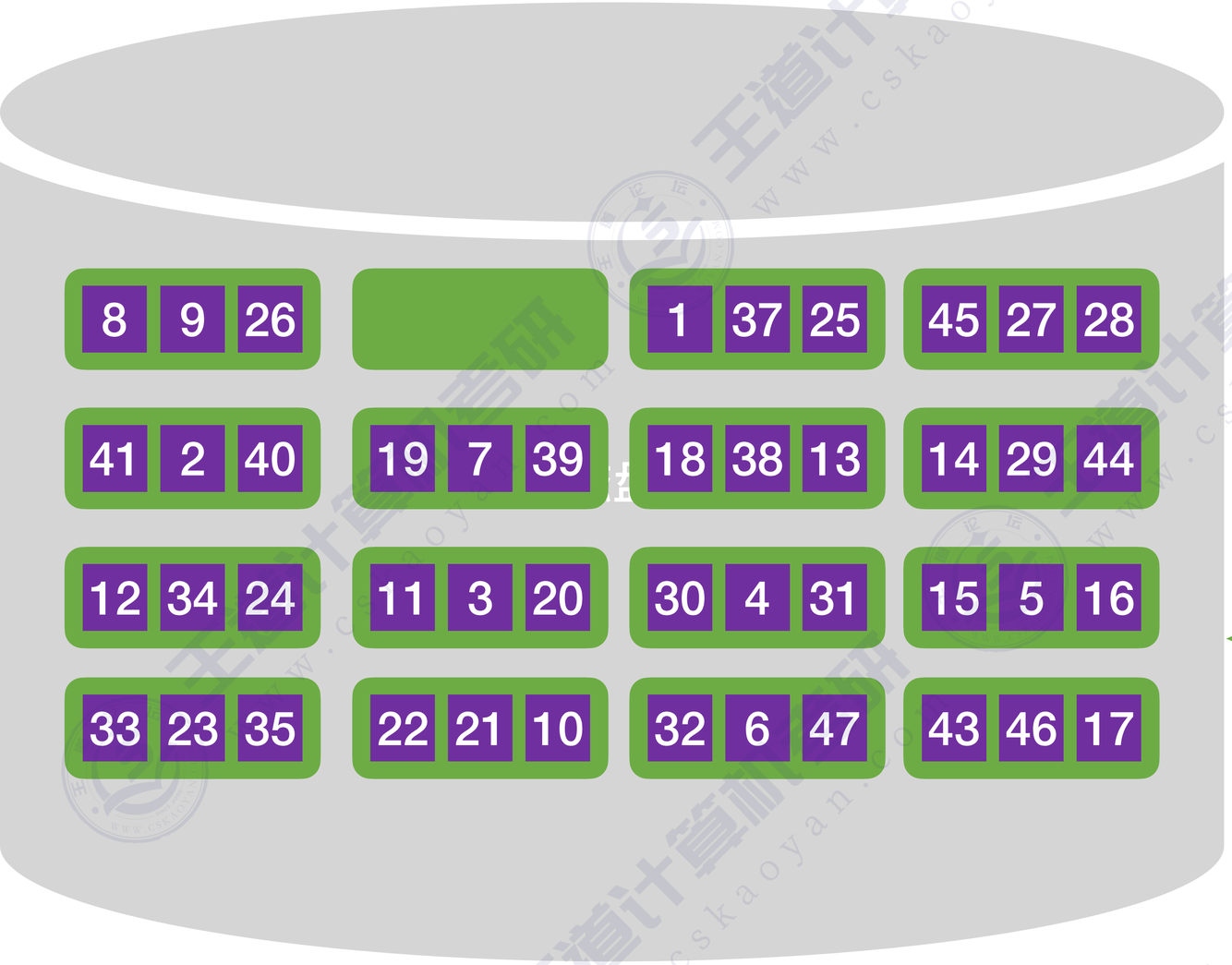
构造初始“归并段”



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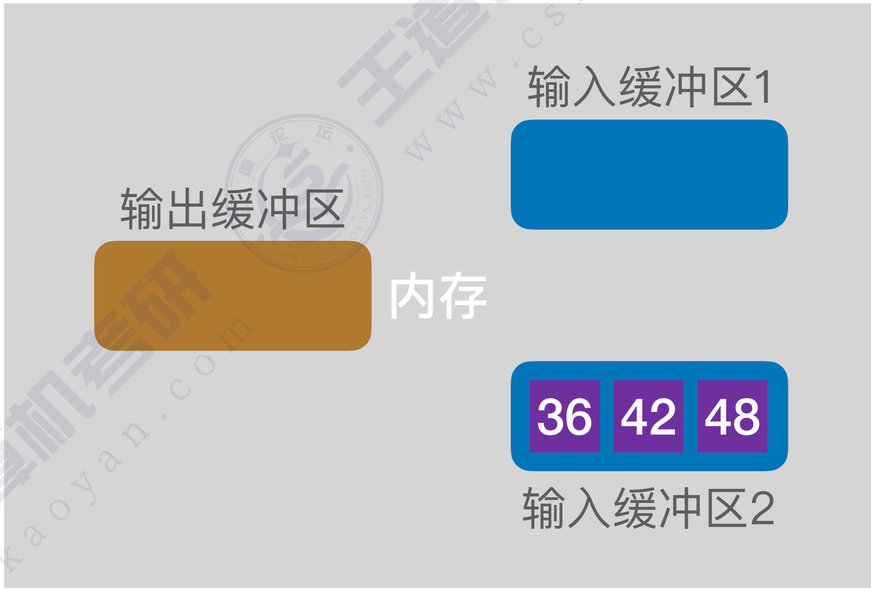

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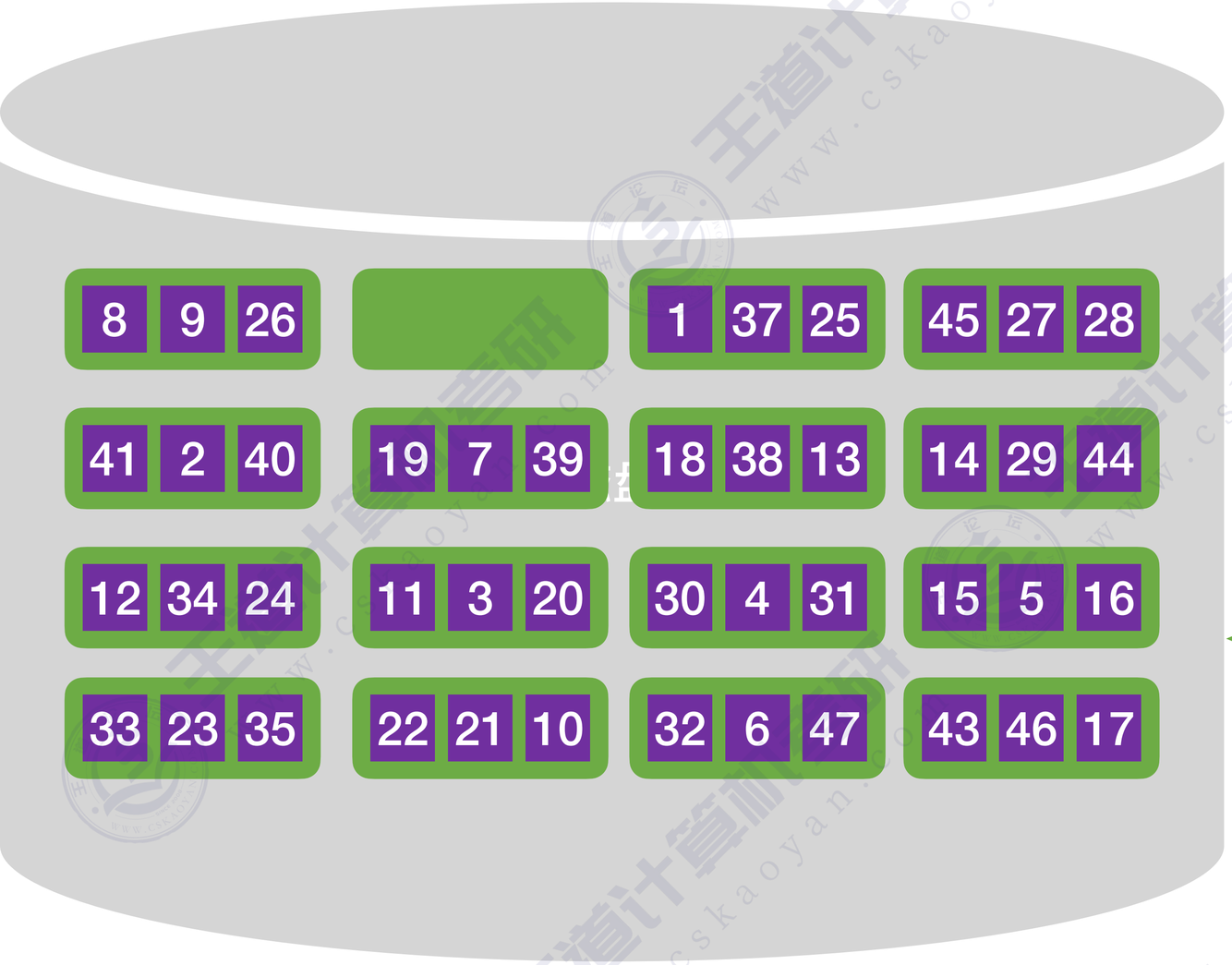
构造初始“归并段”



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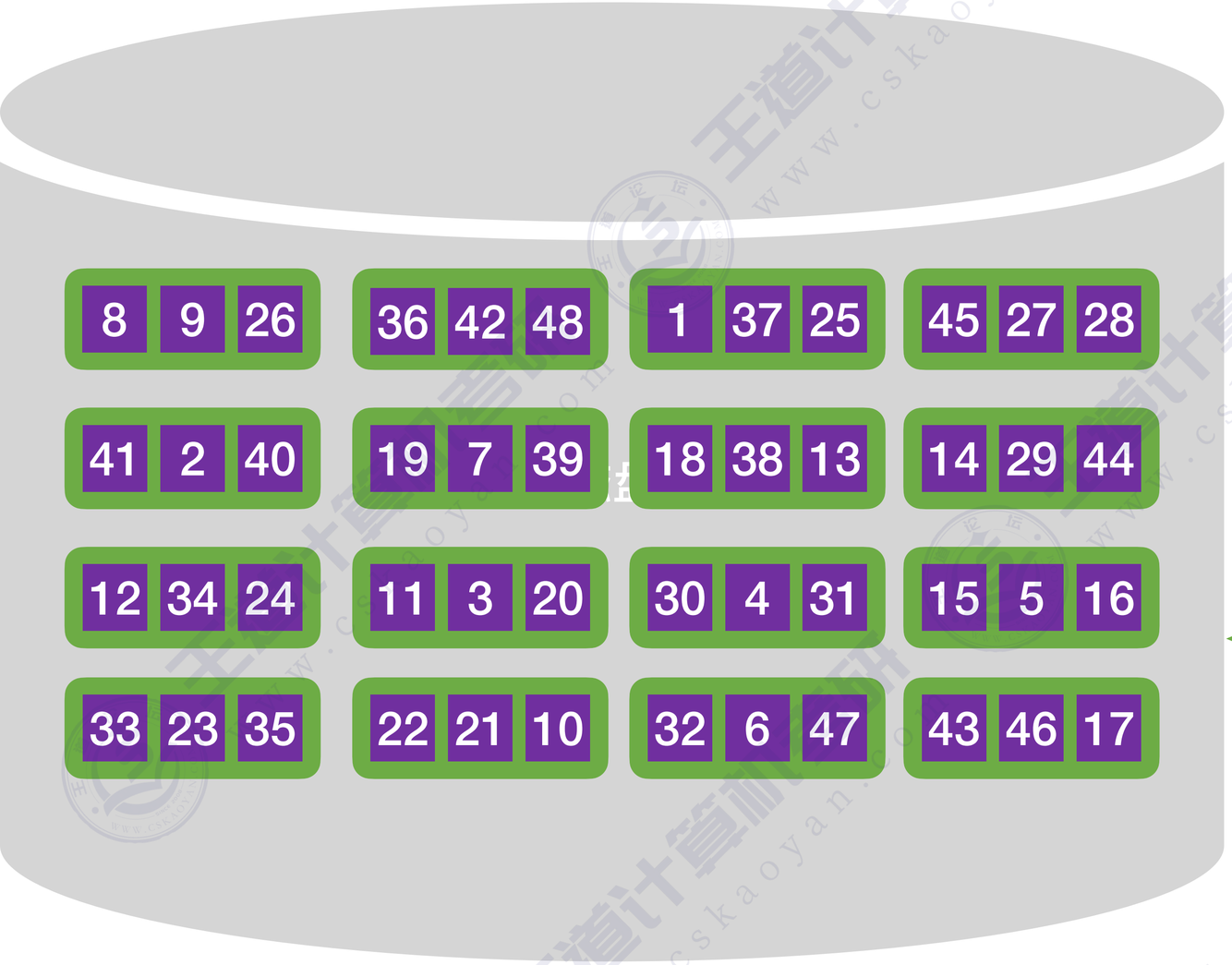

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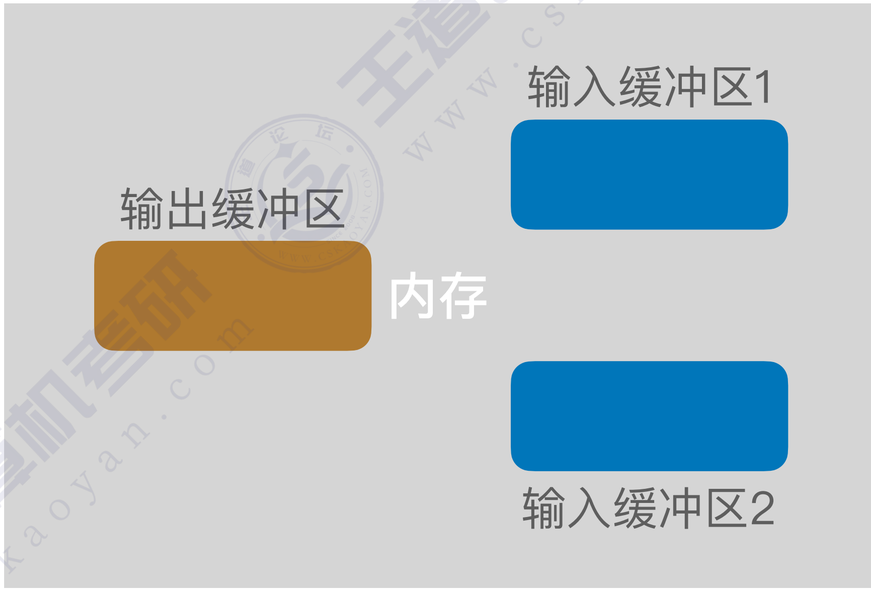

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构造初始“归并段”

一个有序的“归并段”

8 9 26	36 42 48	1 37 25	45 27 28
41 2 40	19 7 39	18 38 13	14 29 44
12 34 24	11 3 20	30 4 31	15 5 16
33 23 35	22 21 10	32 6 47	43 46 17

读磁盘

写磁盘

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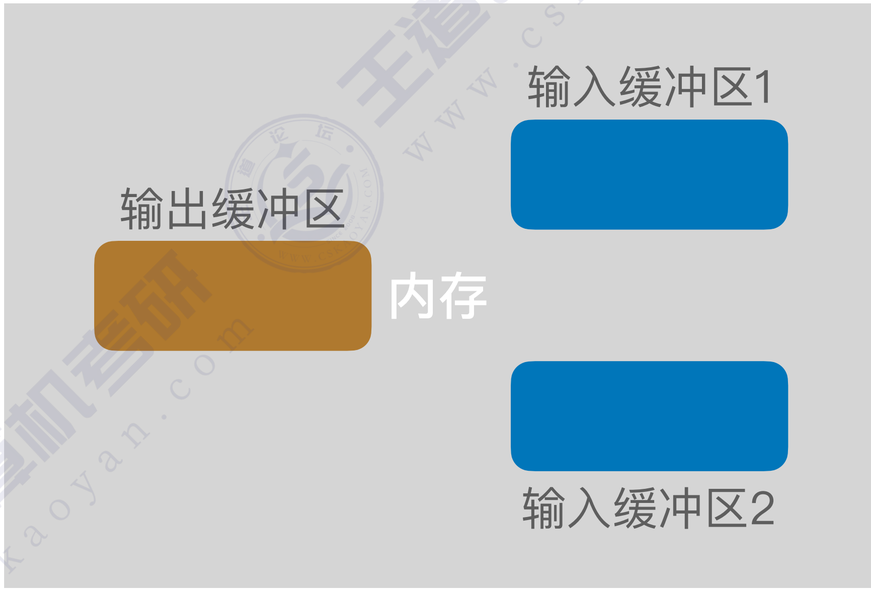

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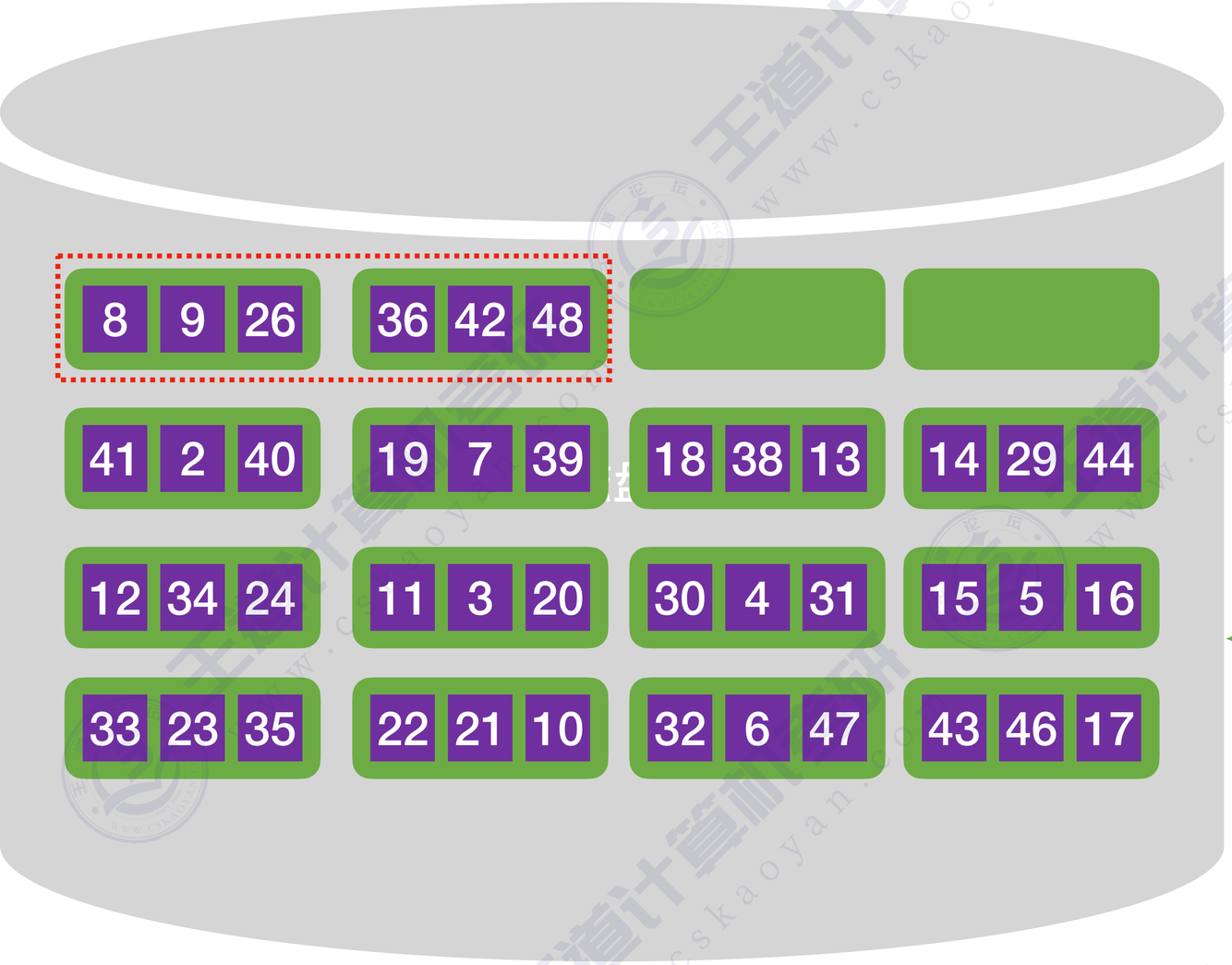

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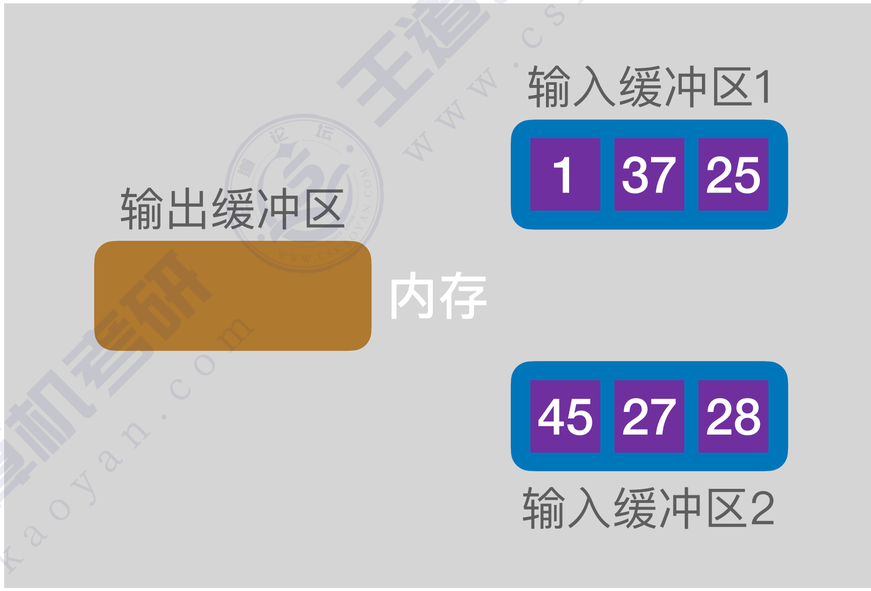

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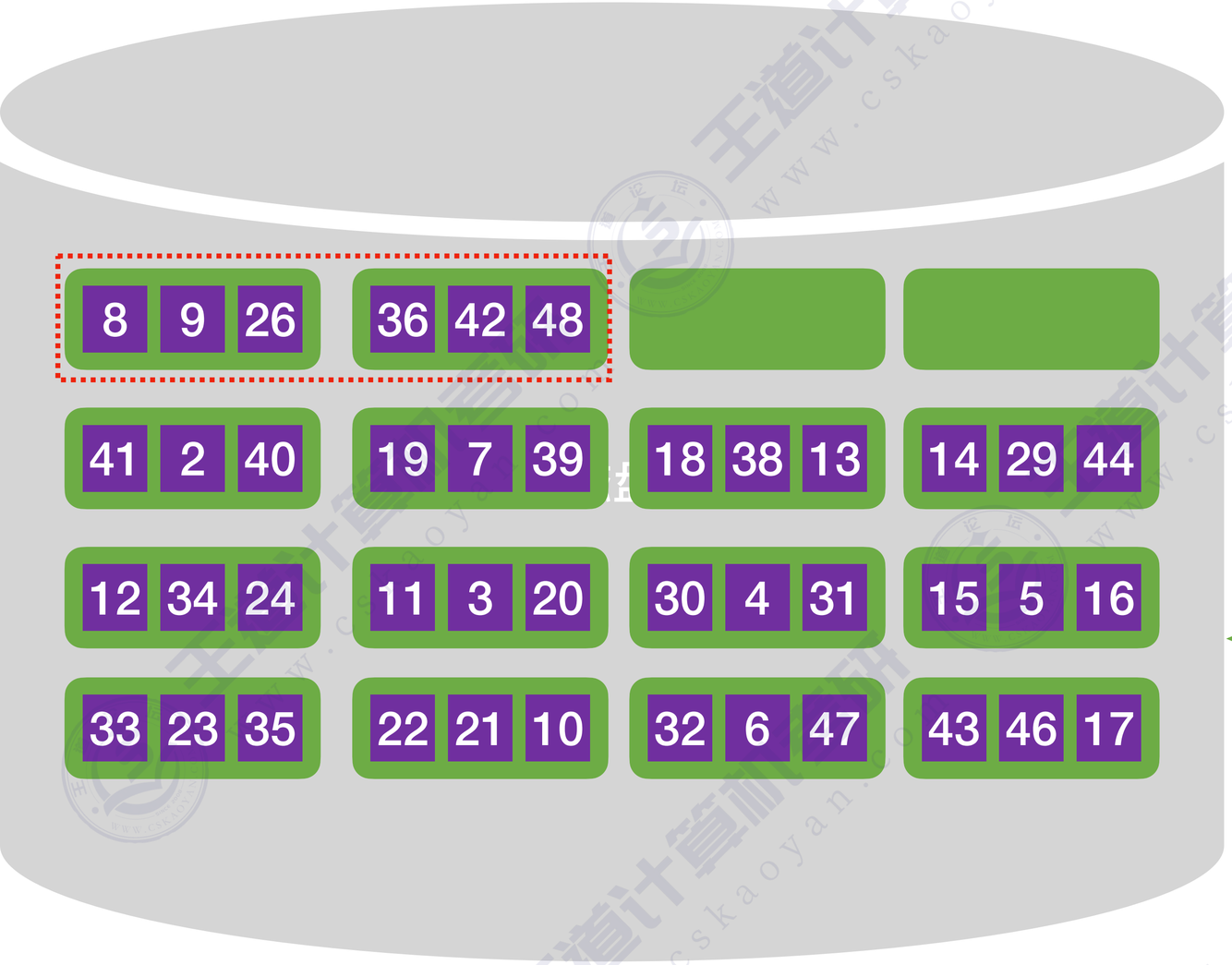

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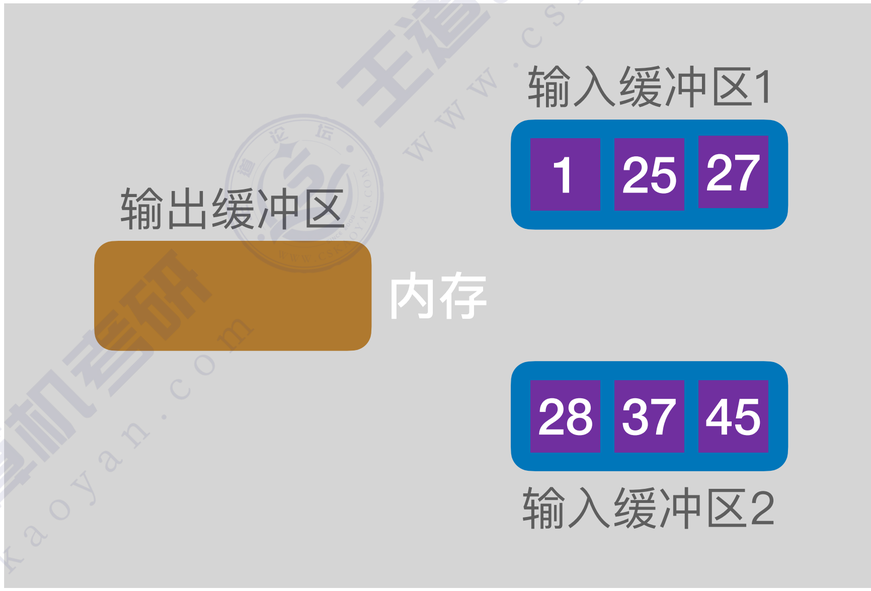

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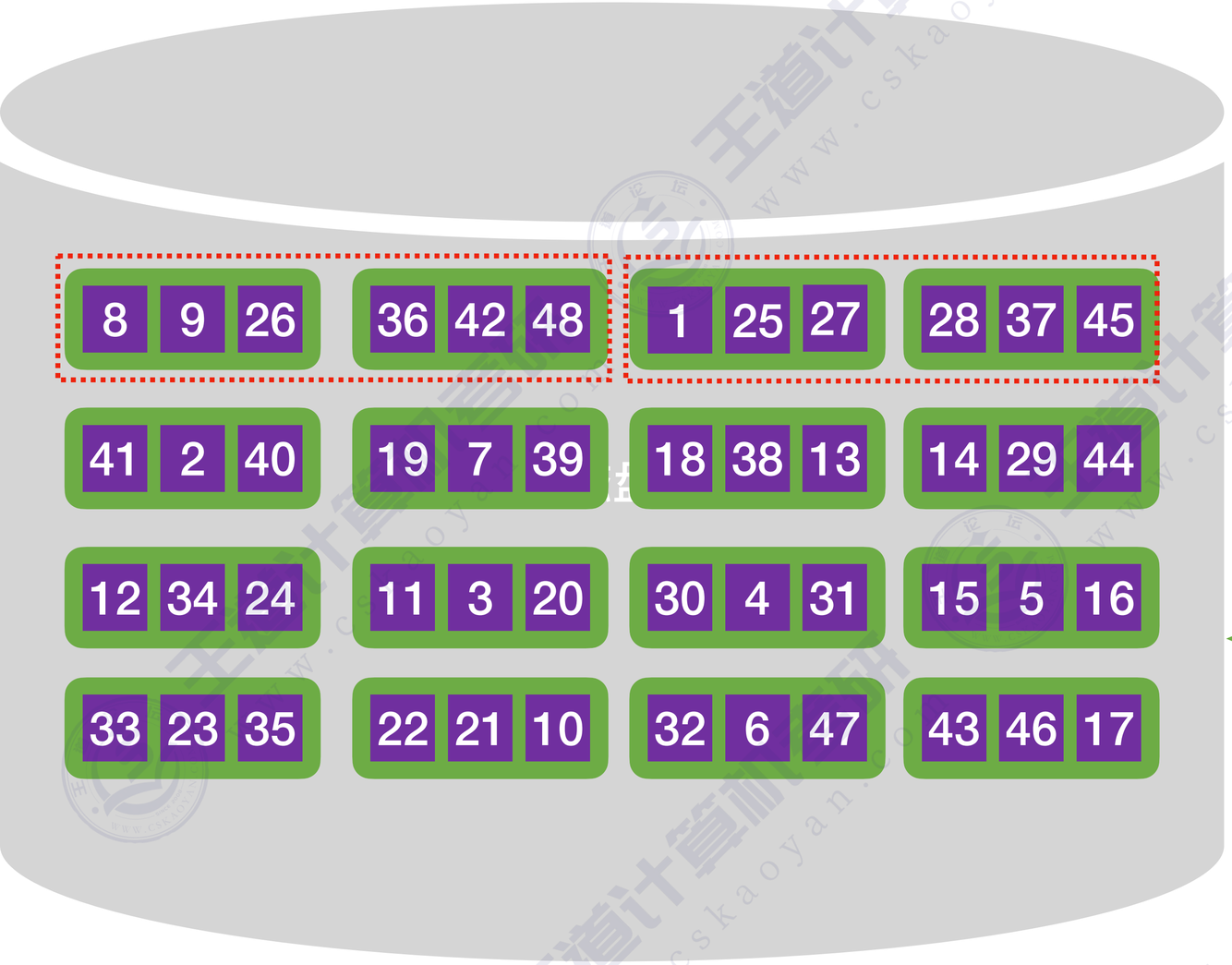

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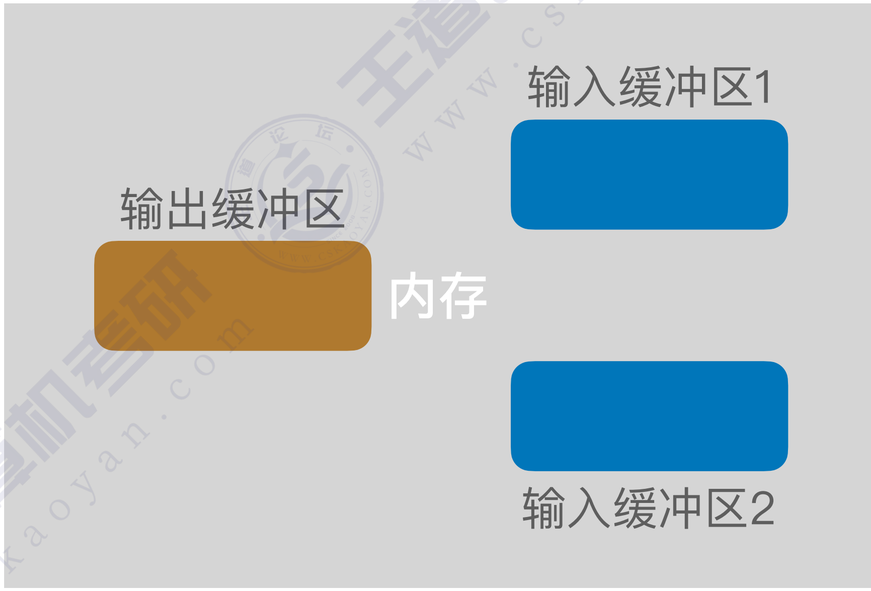

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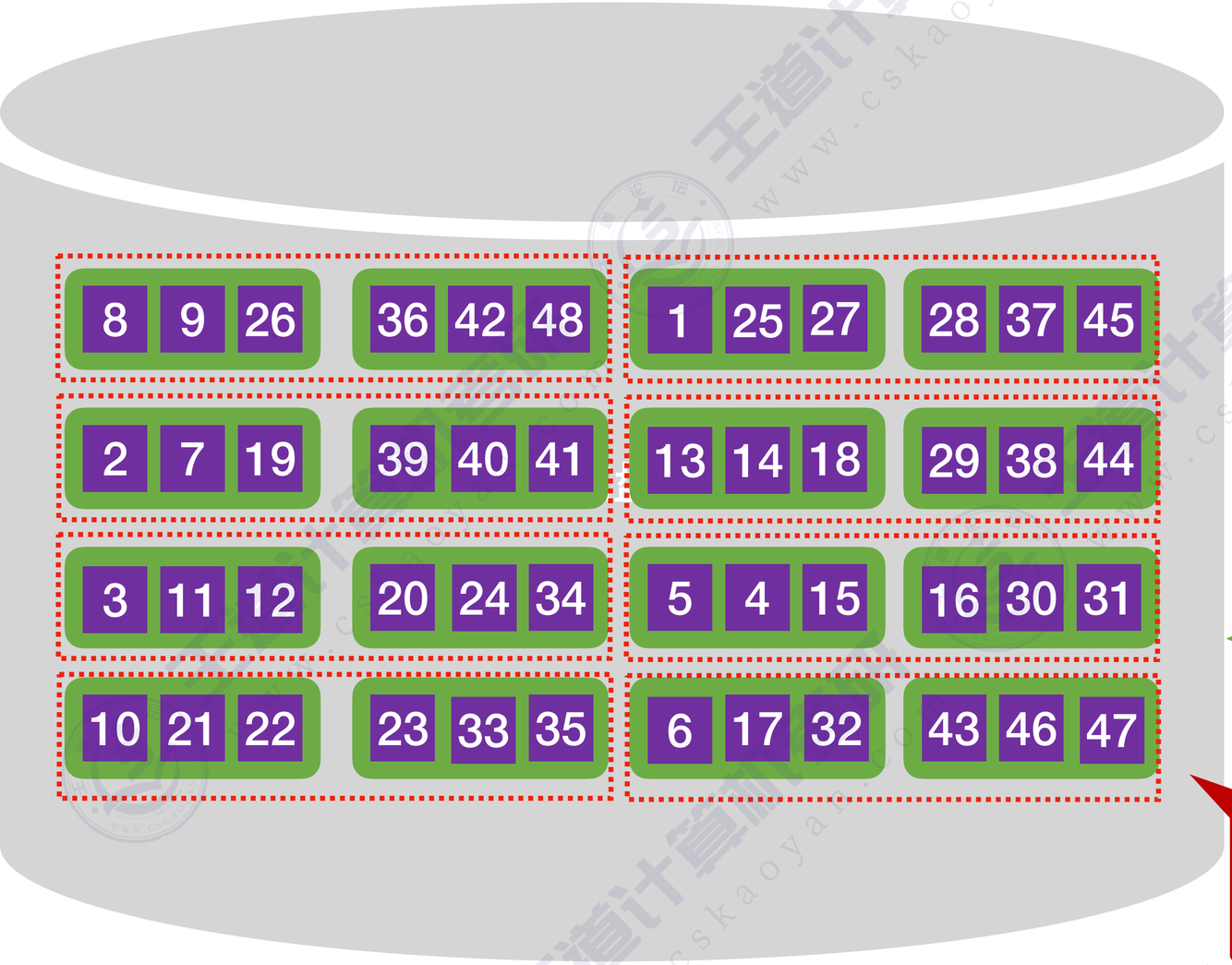

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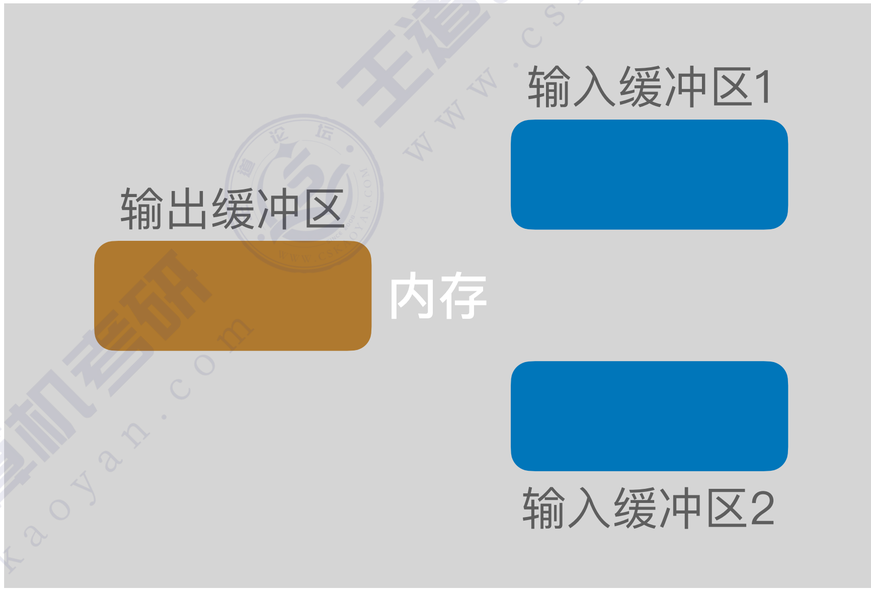

构造初始归并段：
需要16次“读”和16次“写”

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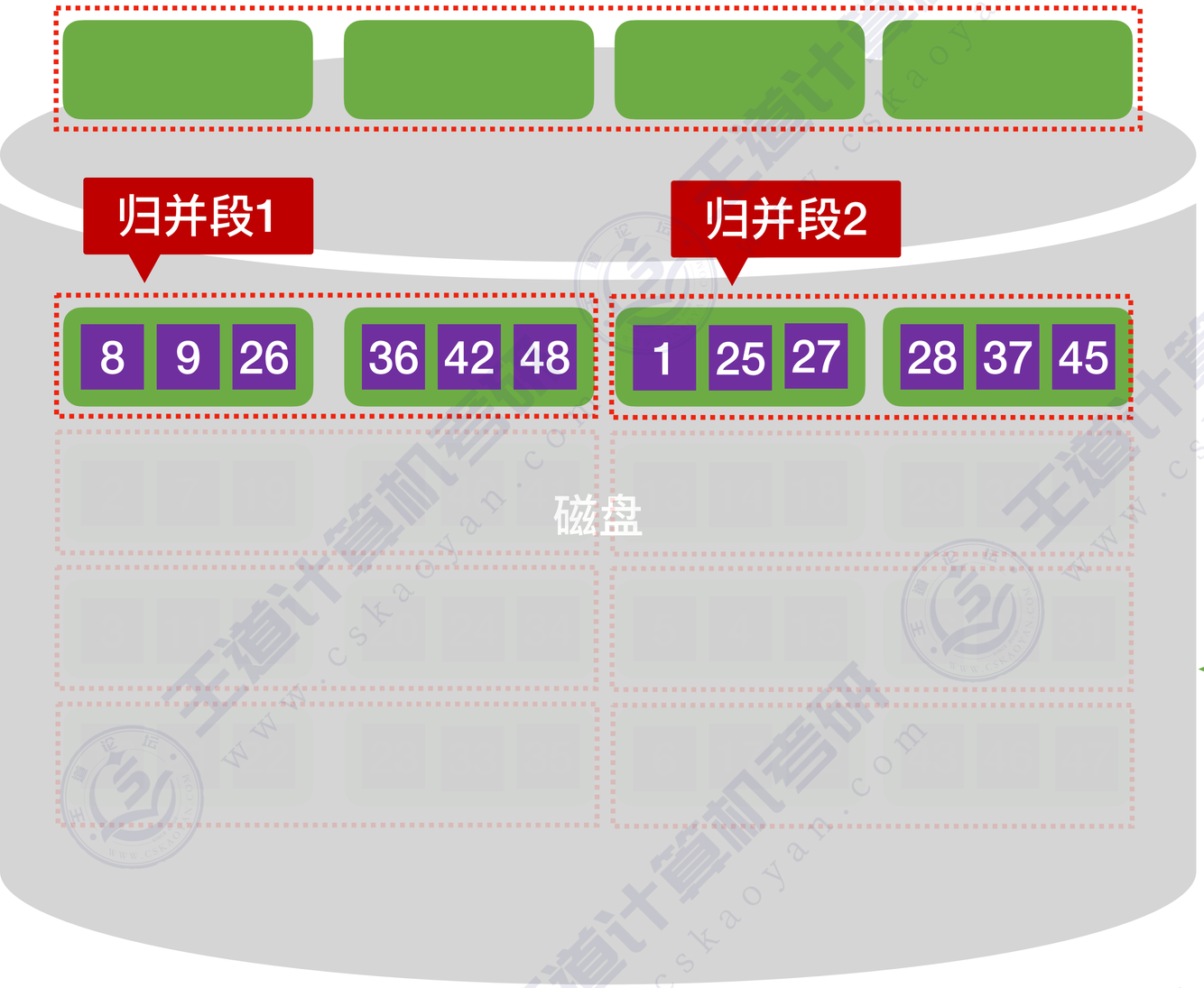
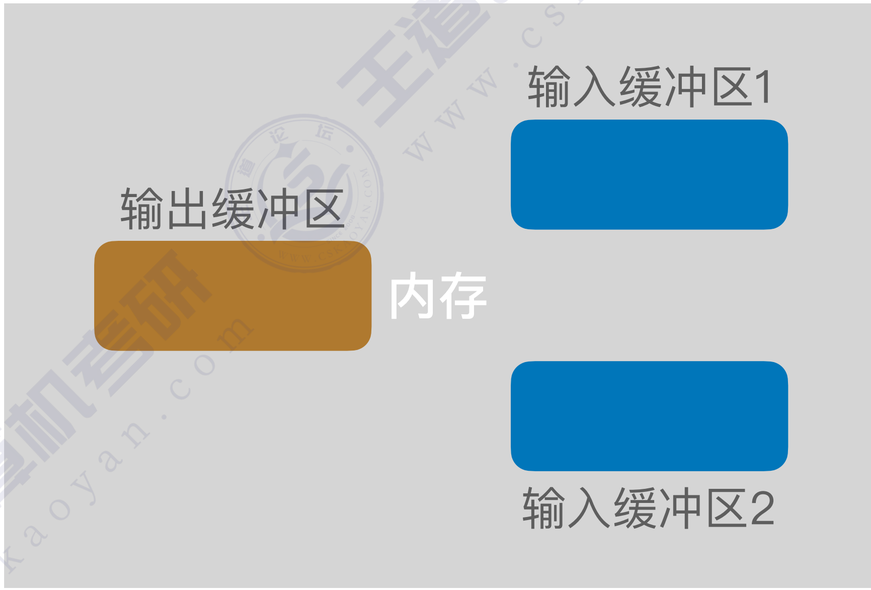
第一趟归并

把8个有序子序列（初始归并段）两两归并





将两个有序归并段归并为一个

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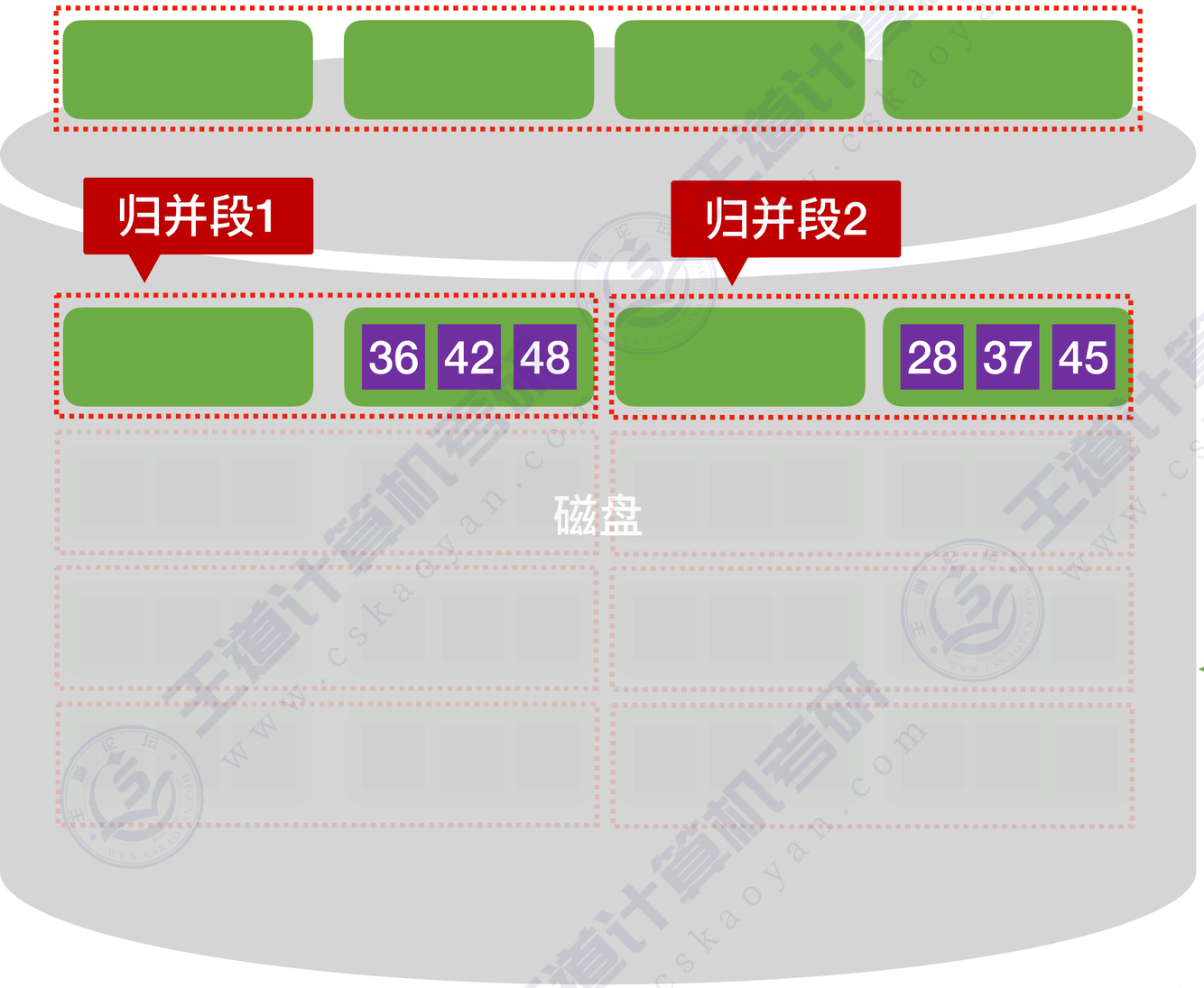
第一趟归并



读磁盘

写磁盘

将两个有序归并段归并为一个

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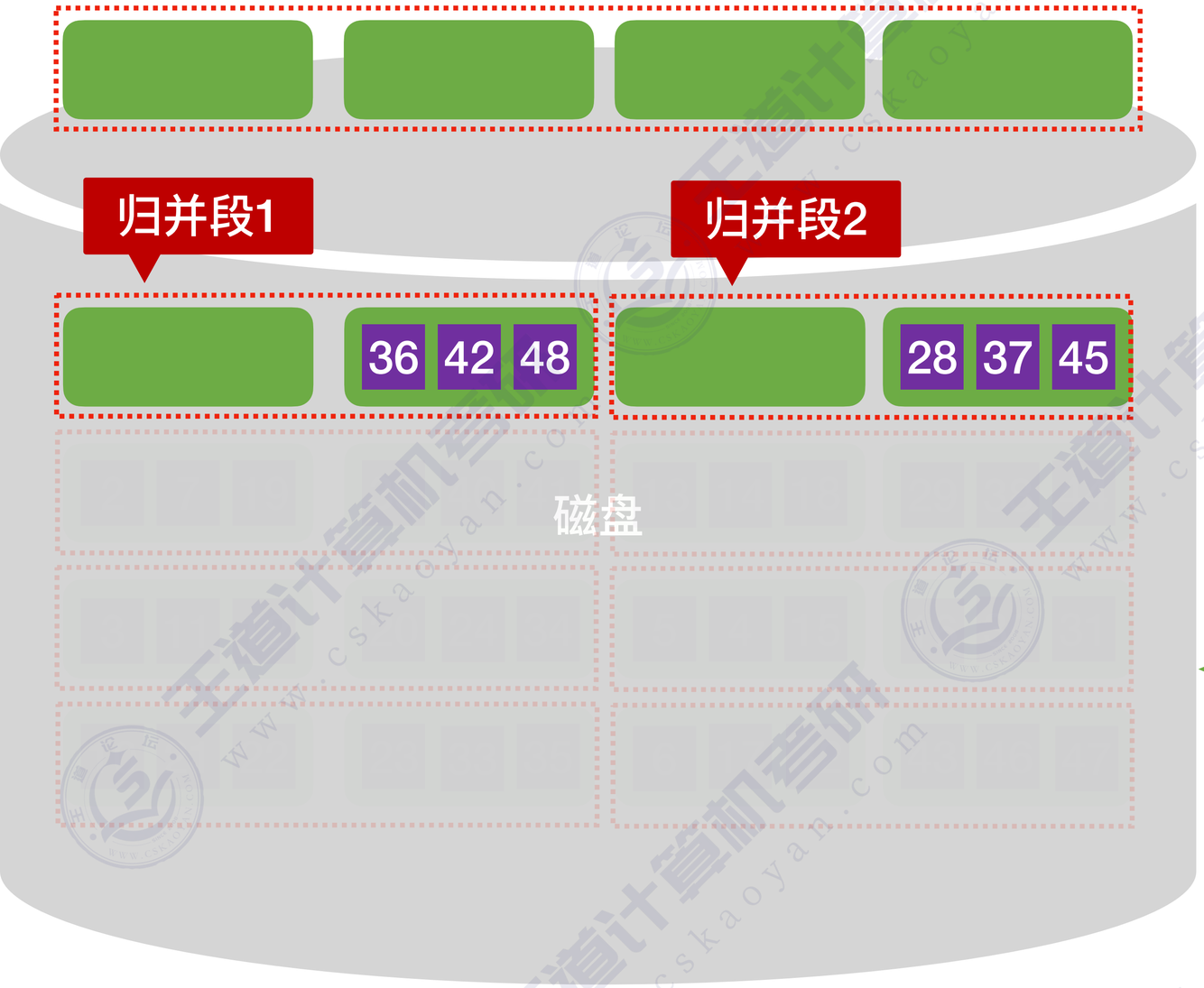
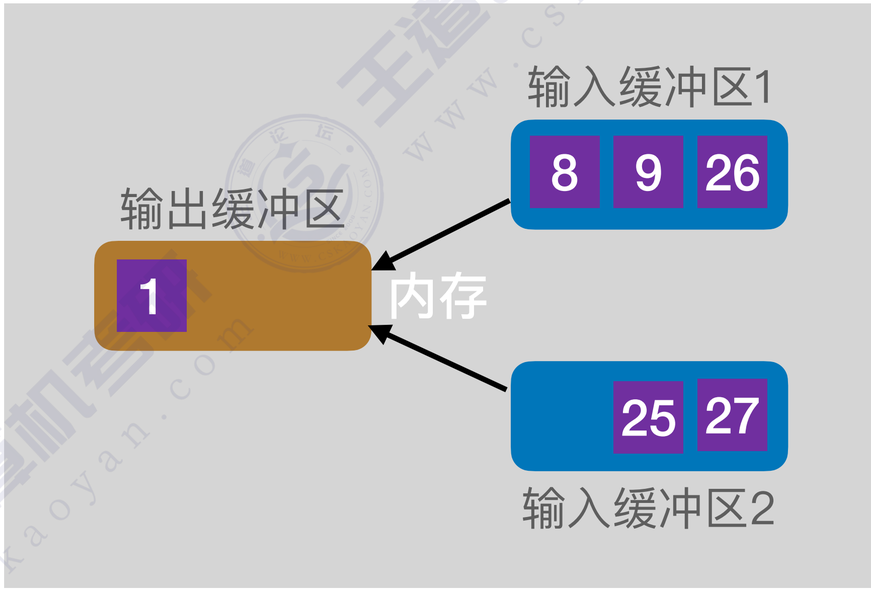
第一趟归并



读磁盘

写磁盘



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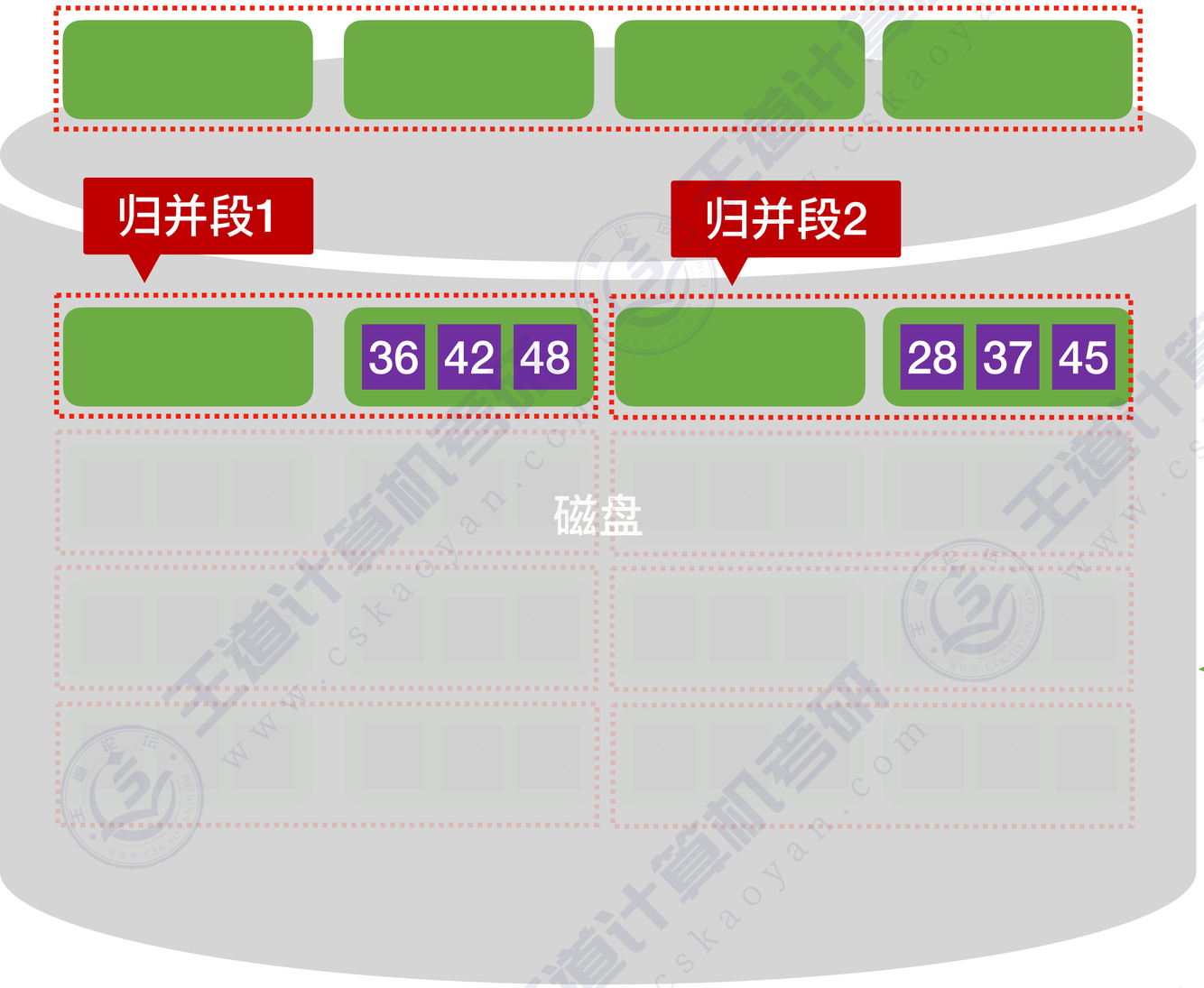
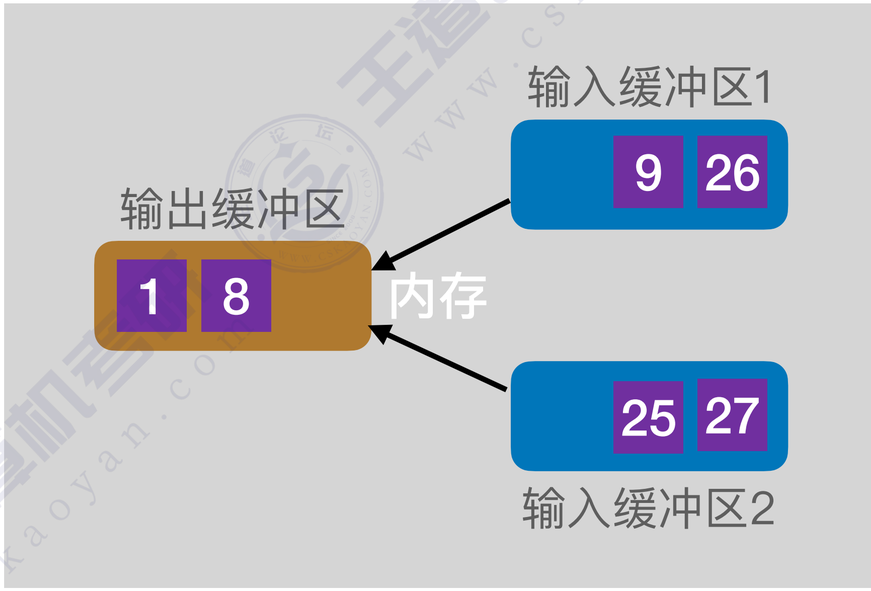
第一趟归并



读磁盘

写磁盘



将两个有序归并段归并为一个

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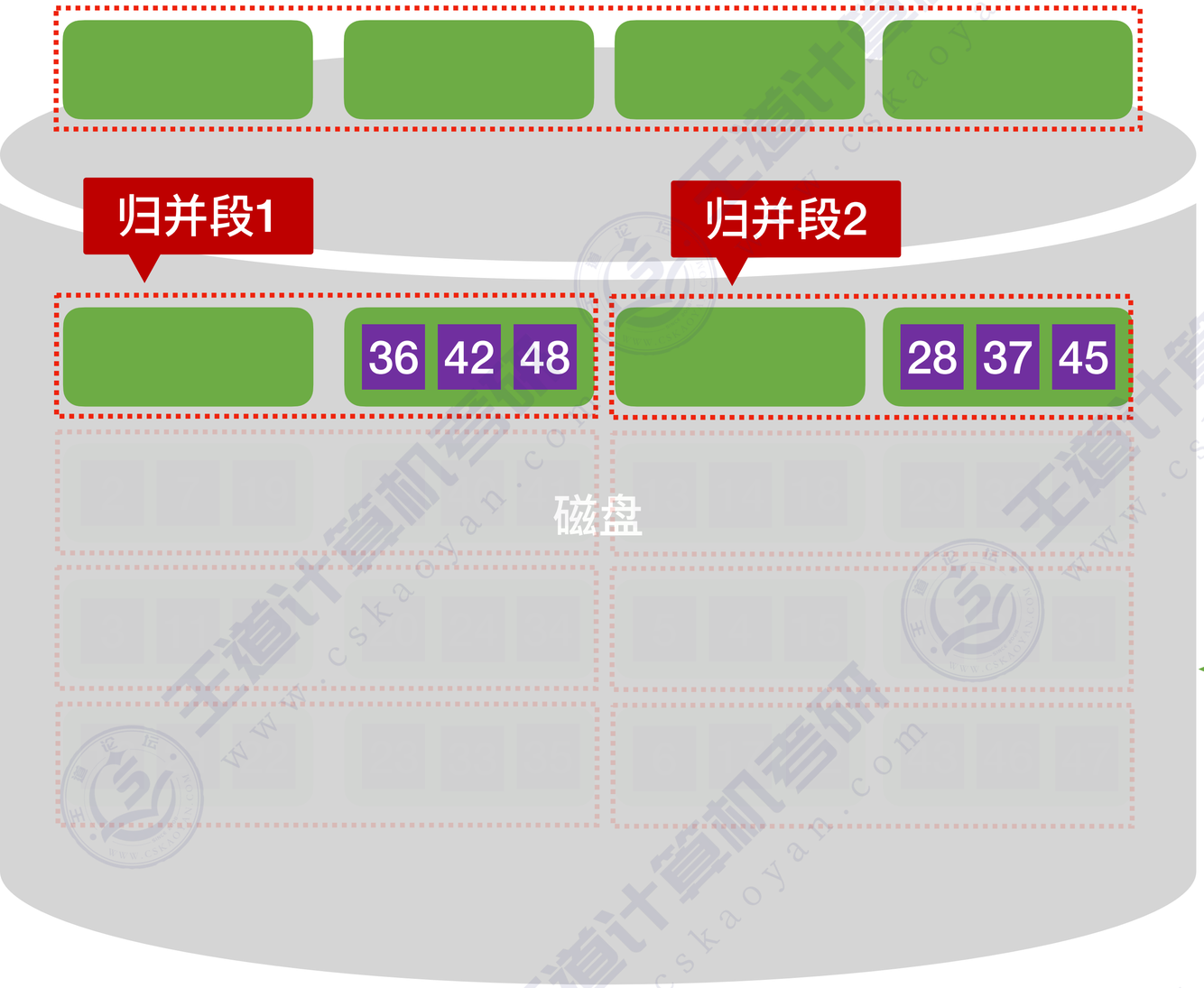
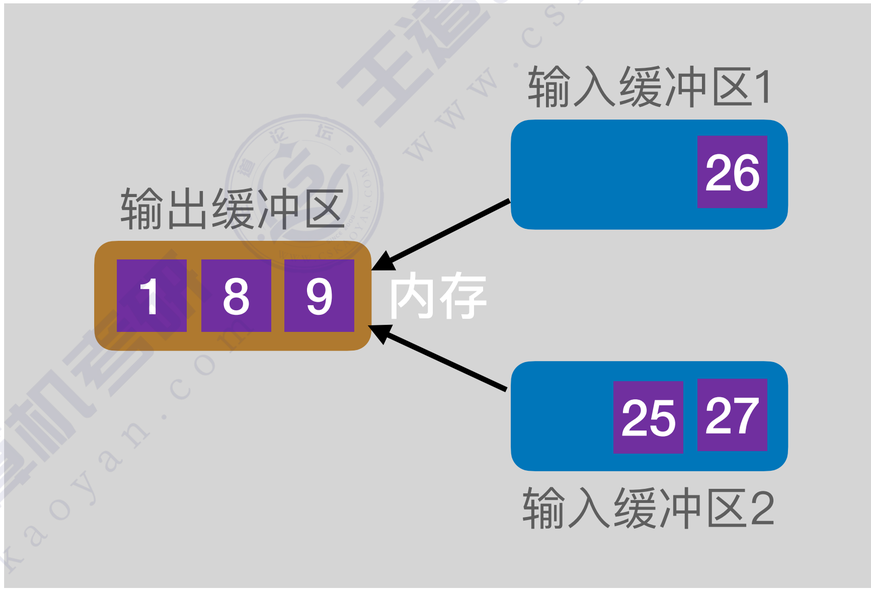
第一趟归并



读磁盘

写磁盘



将两个有序归并段归并为一个

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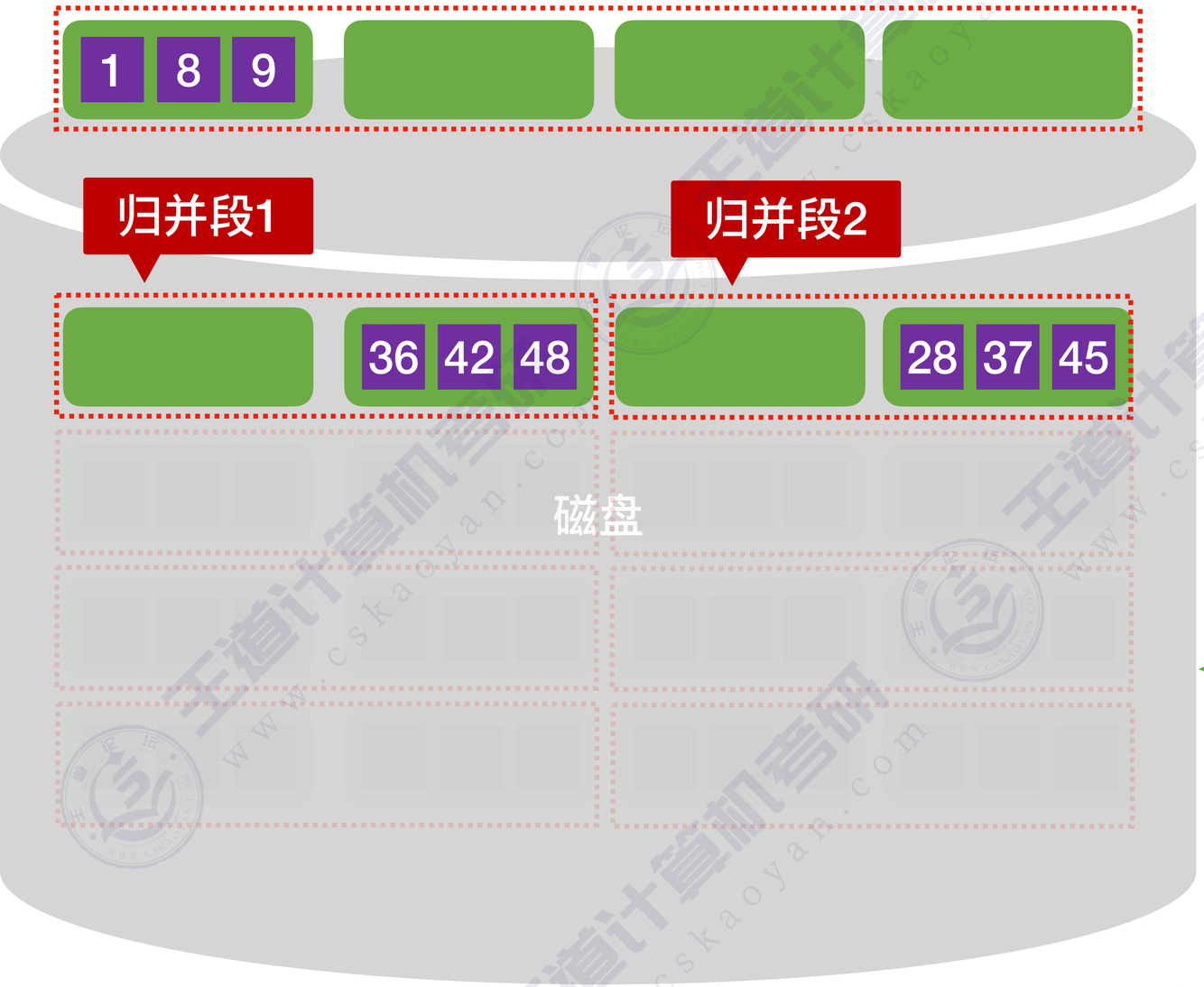
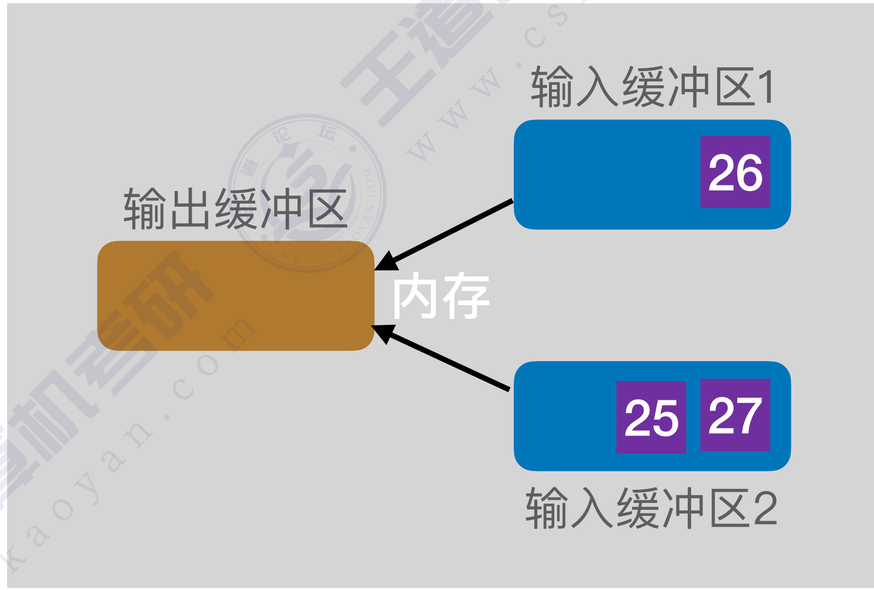
第一趟归并





将两个有序归并段归并为一个

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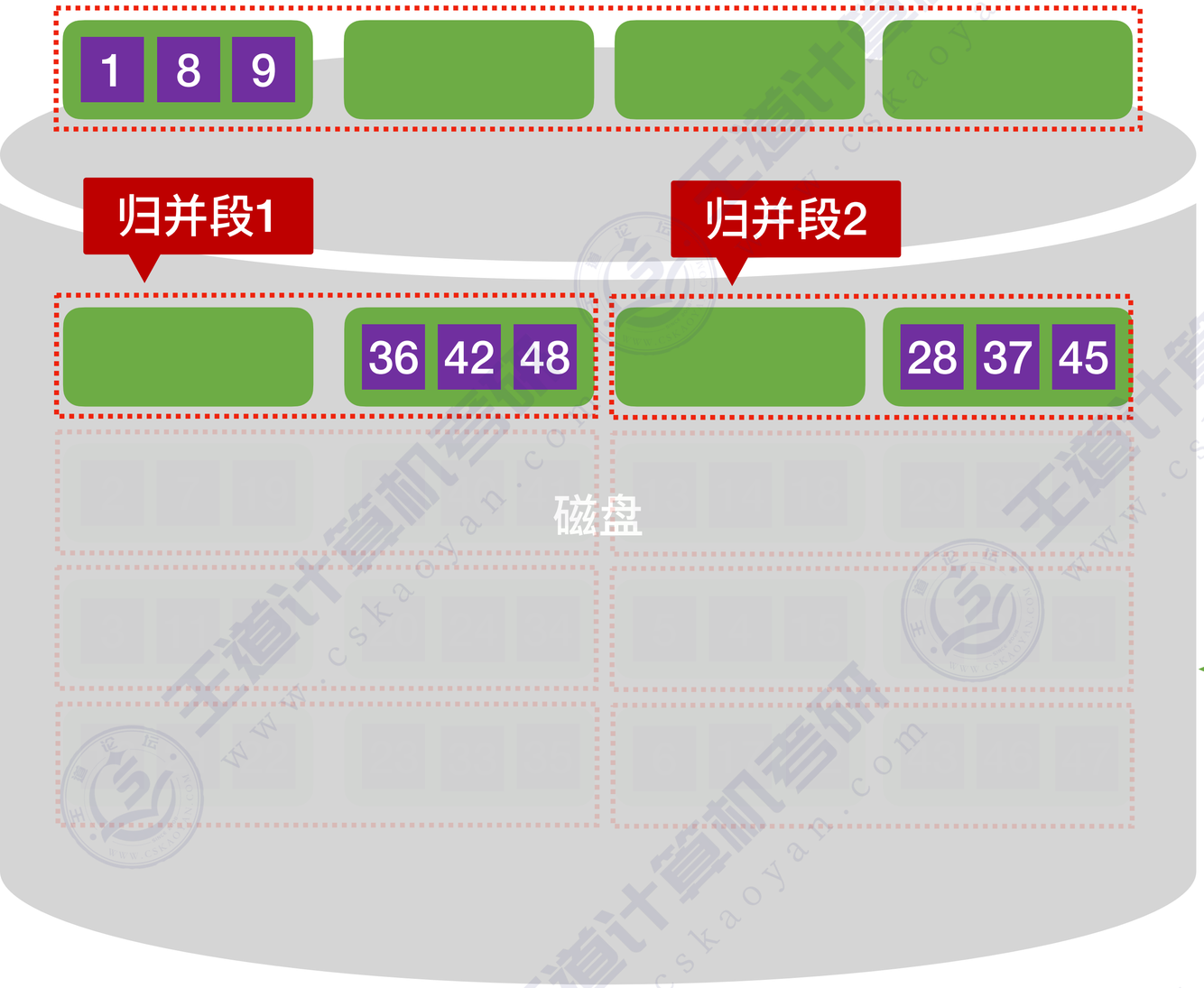
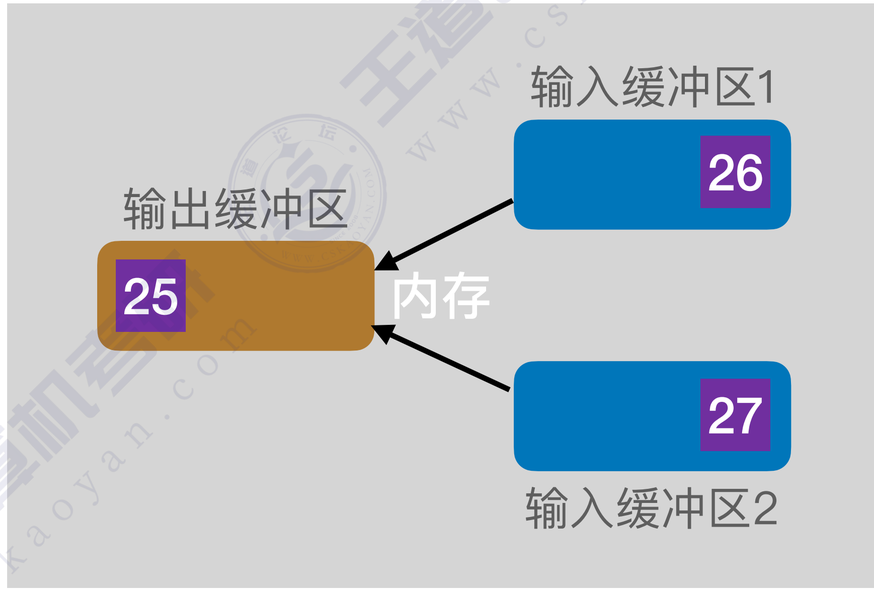
第一趟归并

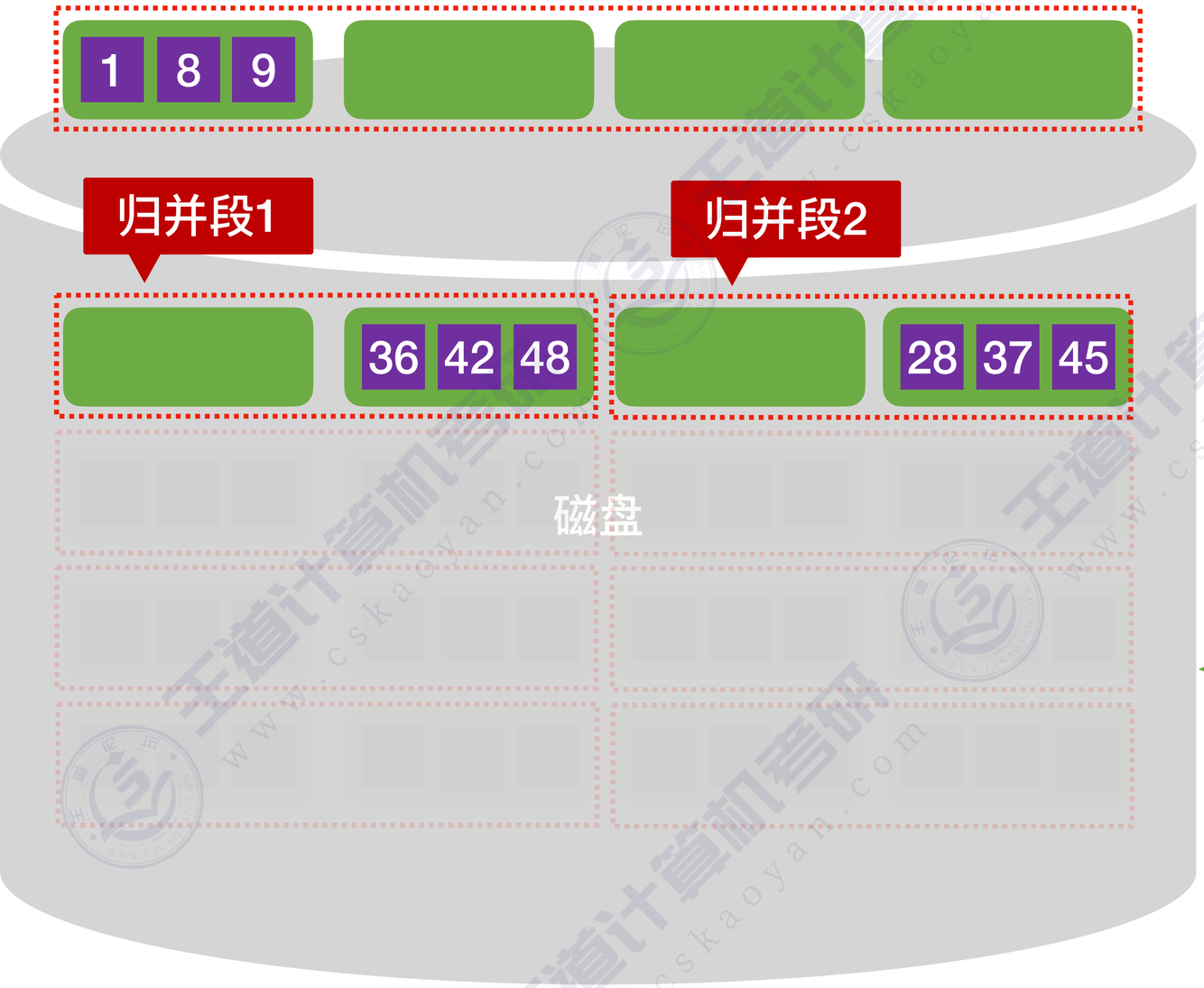




将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

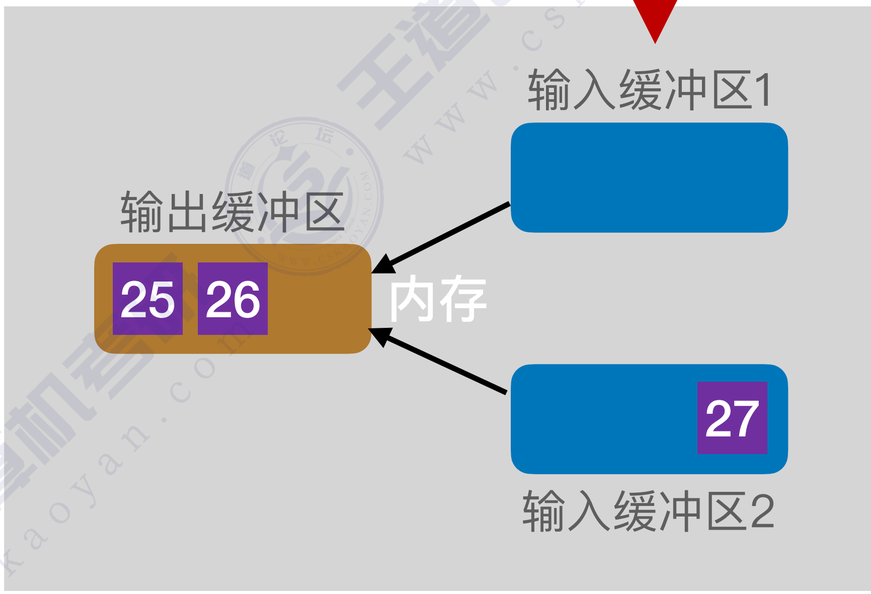
第一趟归并



读磁盘

写磁盘

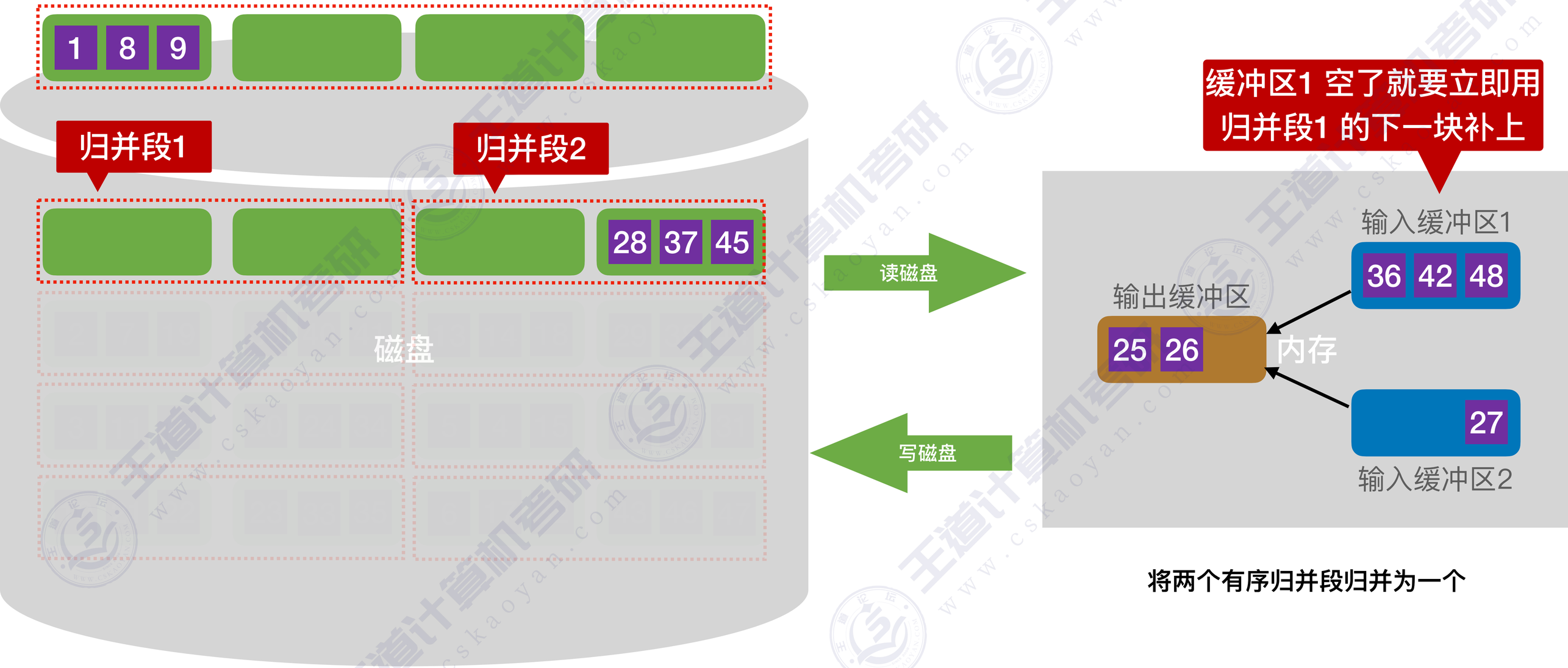
缓冲区1 空了就要立即用
归并段1 的下一块补上



将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

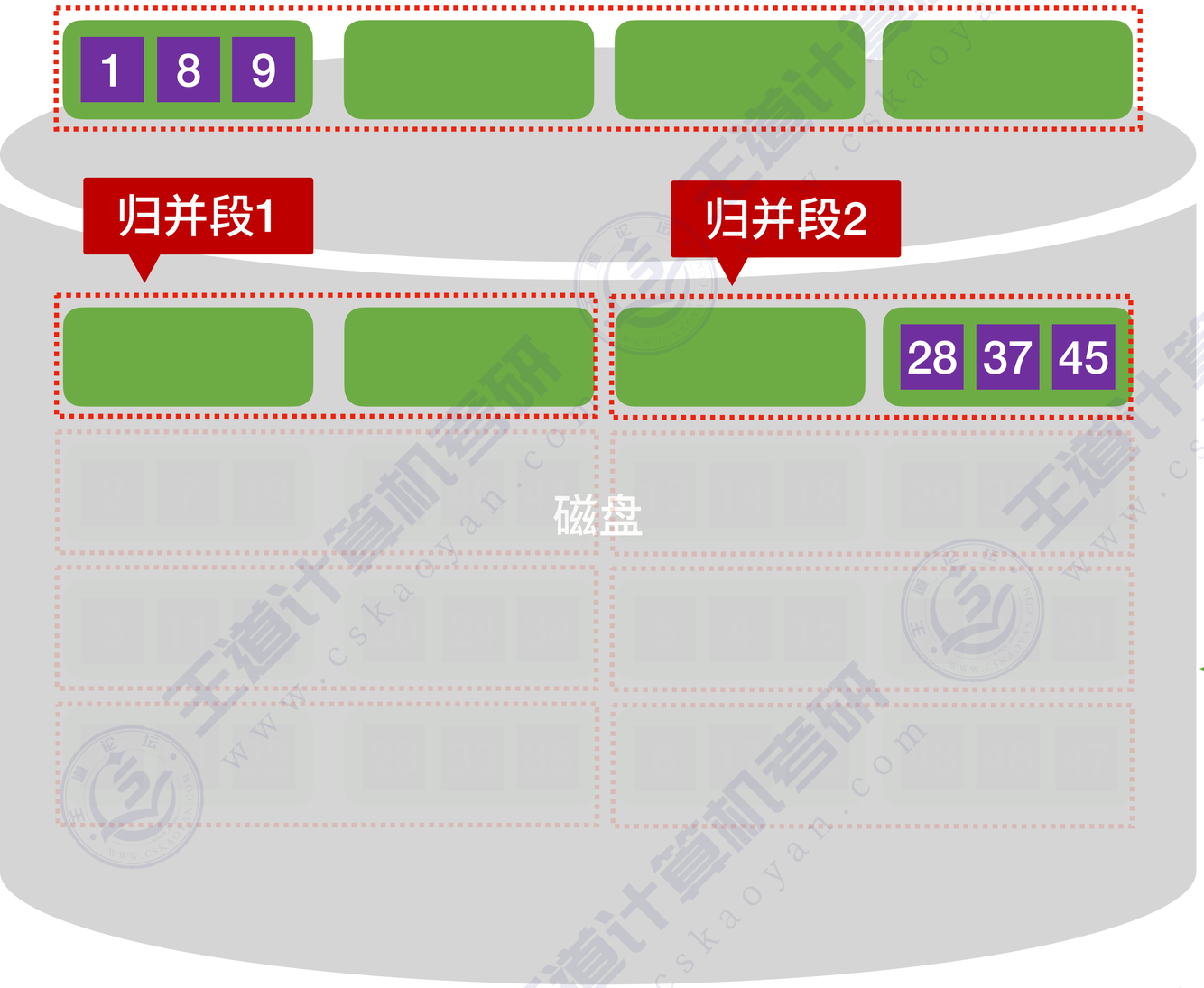
第一趟归并

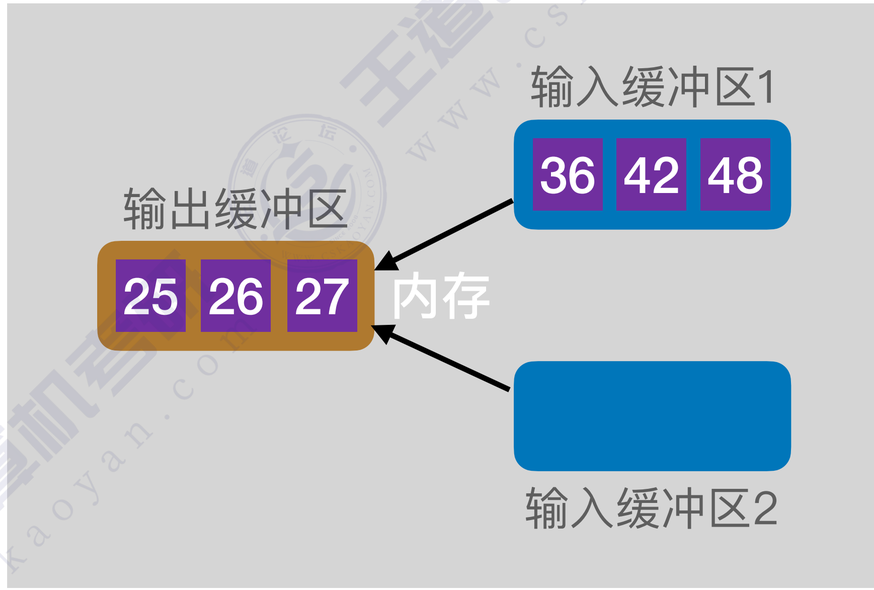


将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第一趟归并





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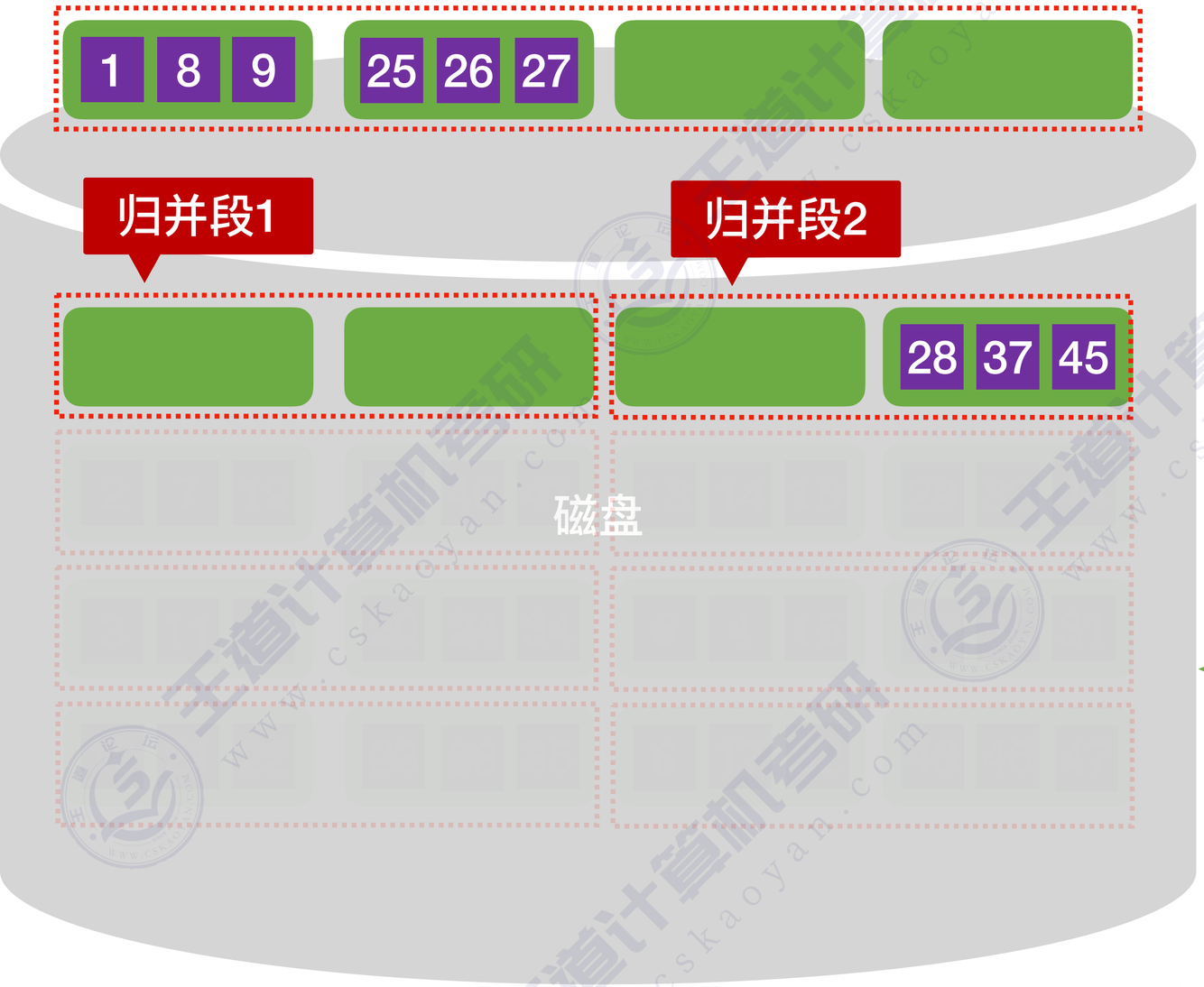
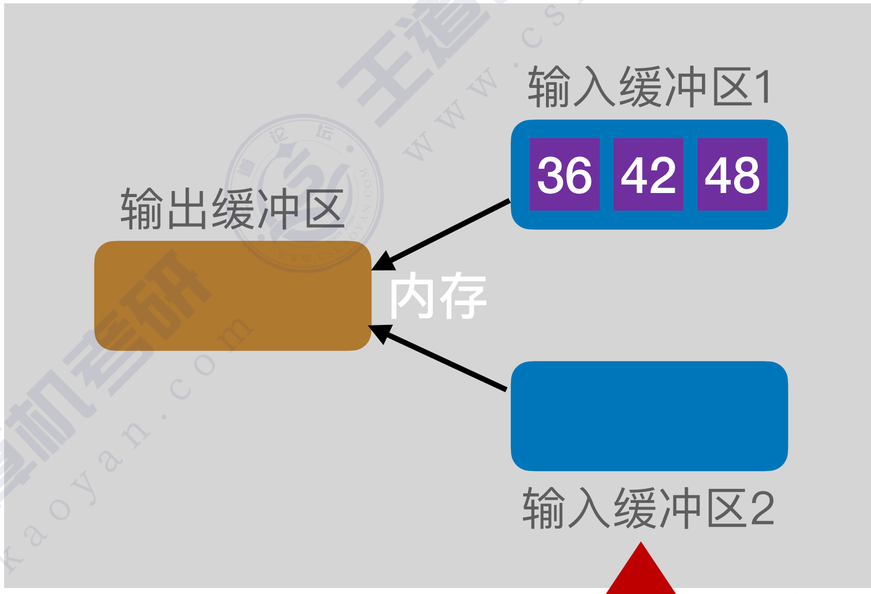
王道考研/CSKAOYAN.COM

第一趟归并



读磁盘

写磁盘

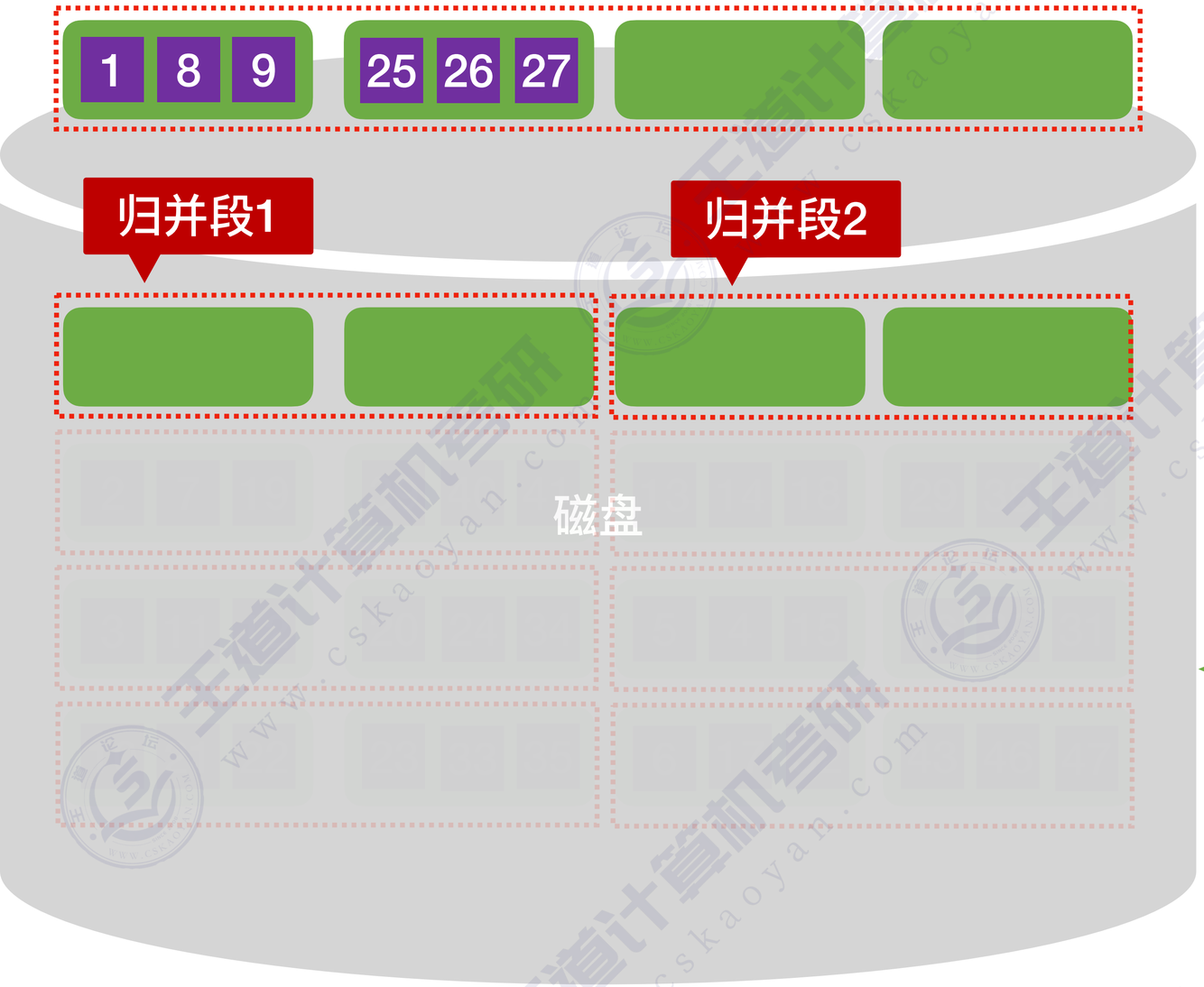


将两

缓冲区2 空了就要立即用归并段2 的下一块补上

王道考研/CSKAOYAN.COM

第一趟归并



读磁盘

写磁盘

输入缓冲区1

36 42 48

输出缓冲区

内存

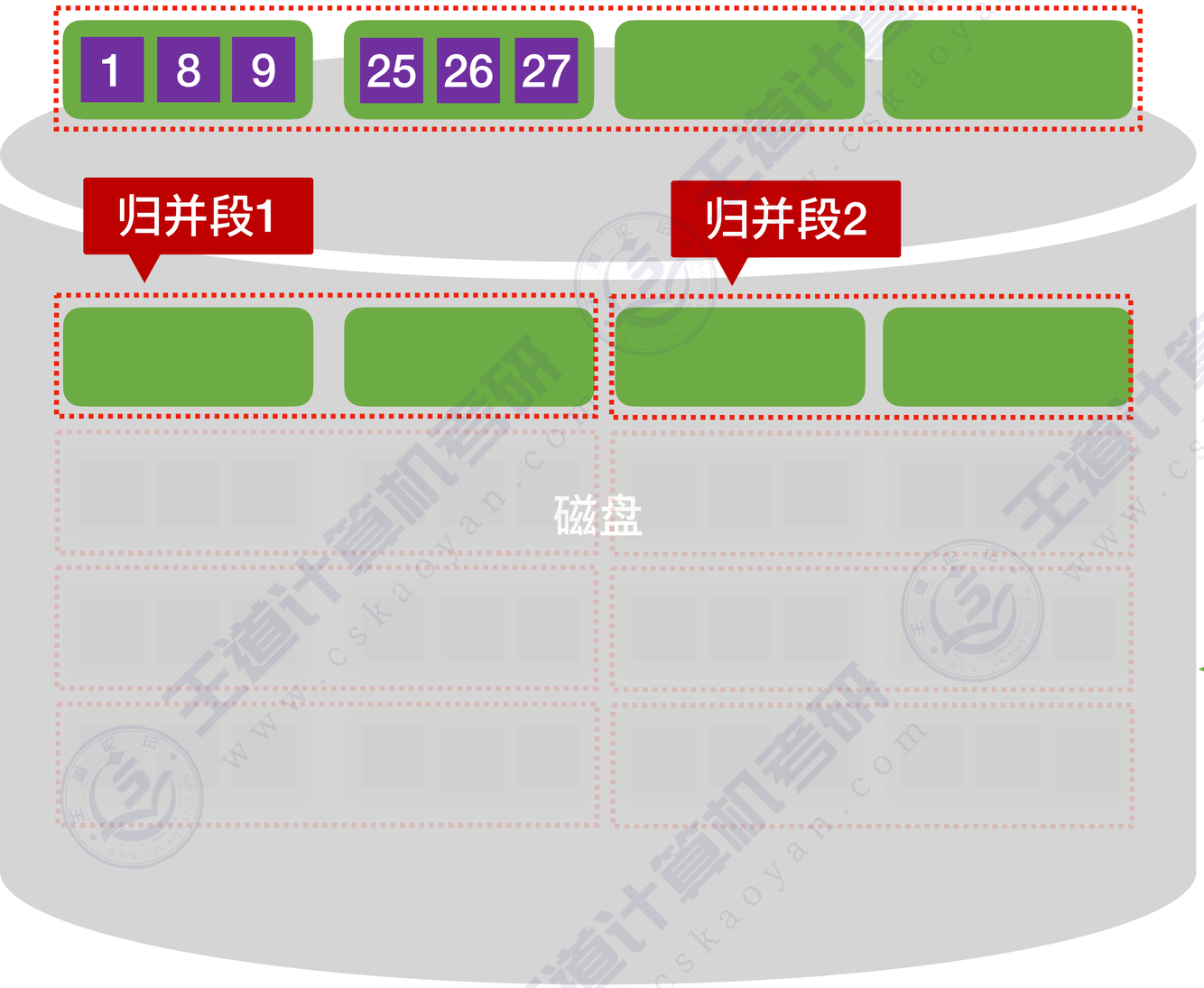
28 37 45

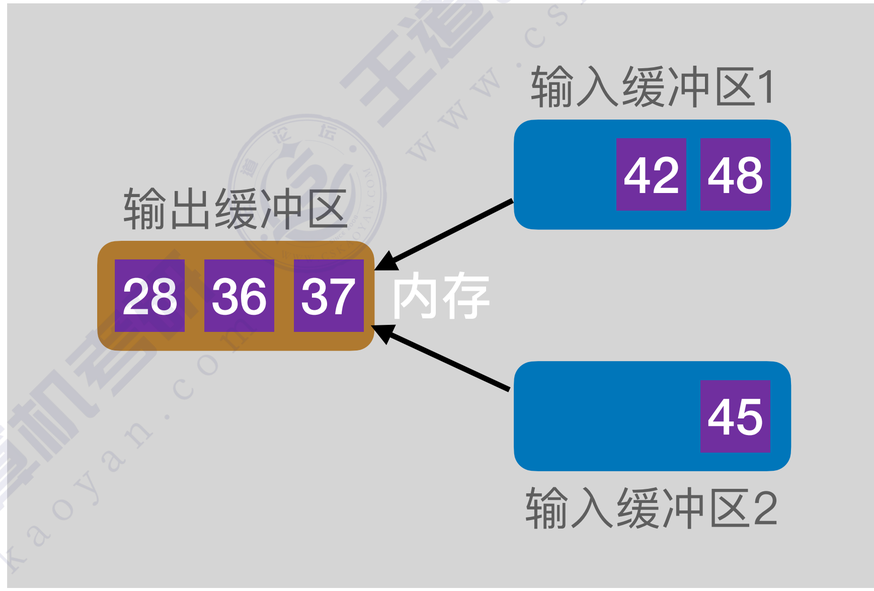
输入缓冲区2

缓冲区2 空了就要立即用
归并段2 的下一块补上

将两

第一趟归并

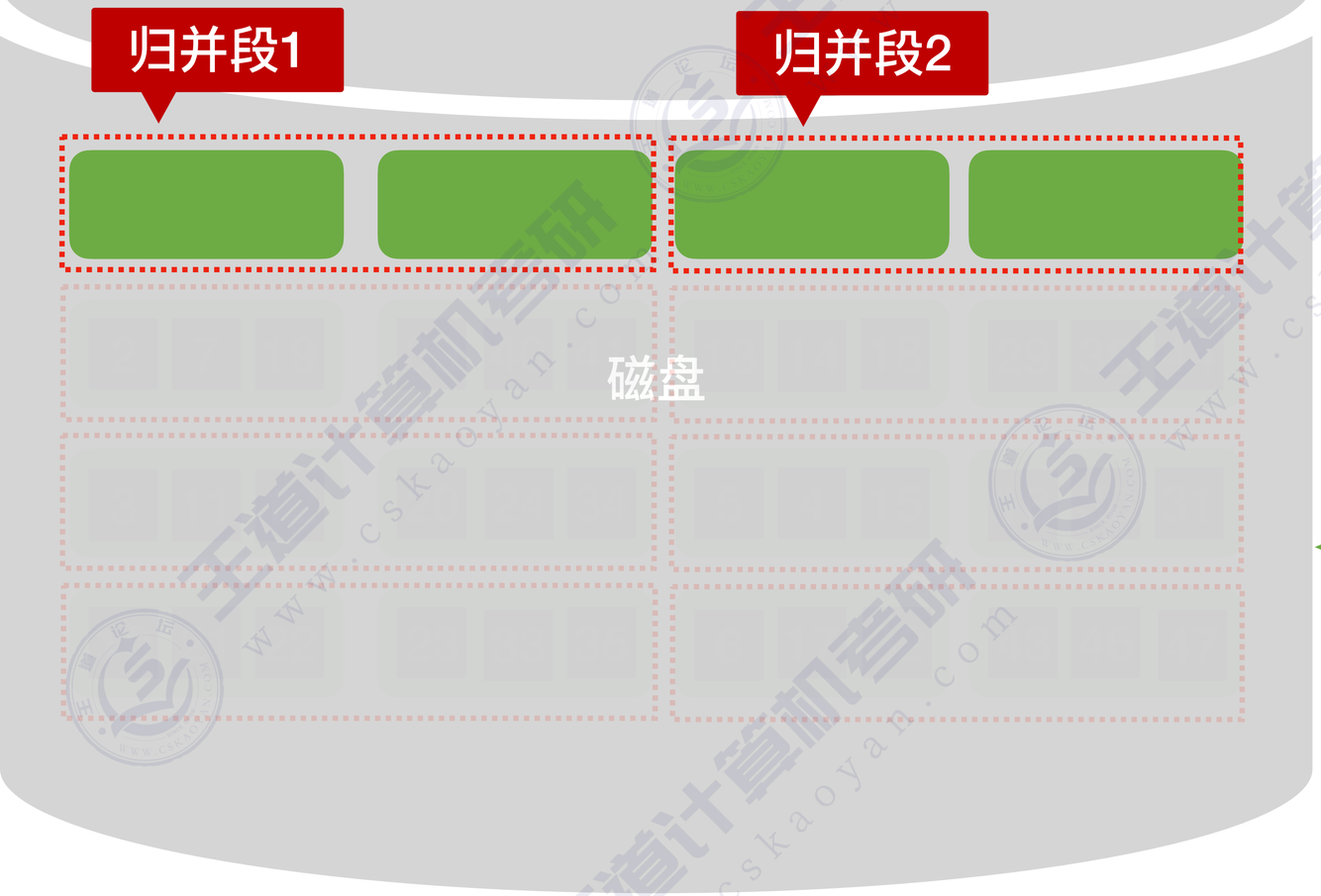


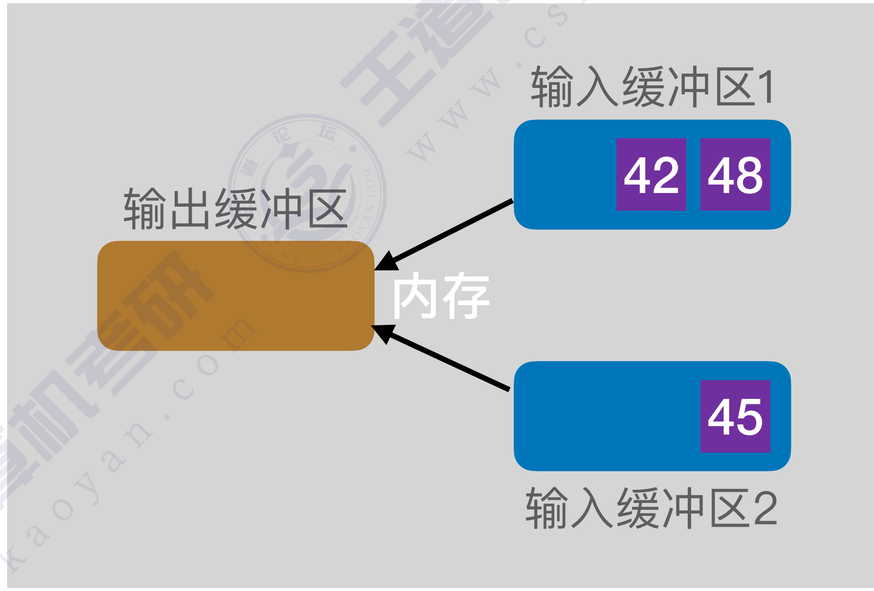


将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第一趟归并

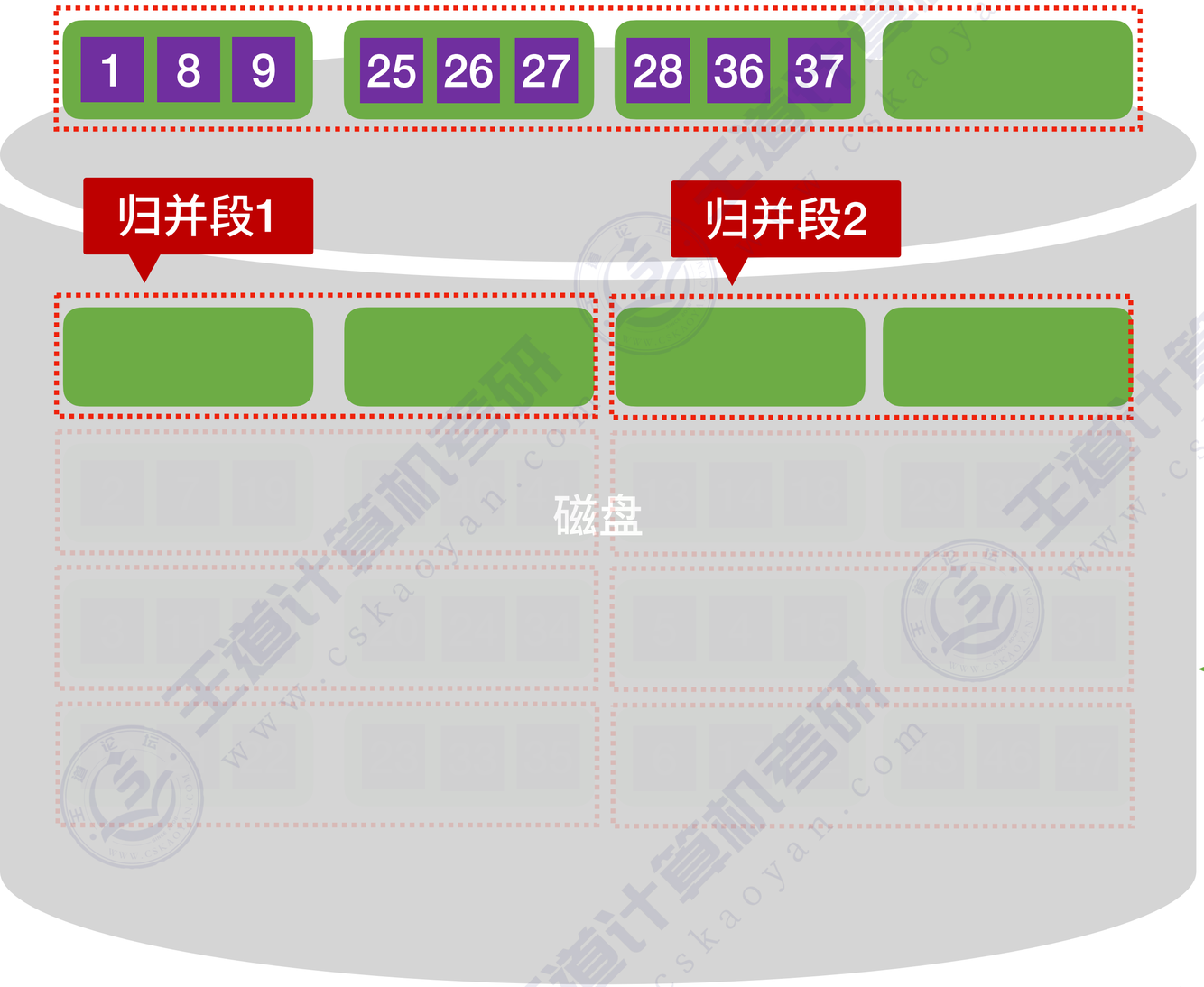


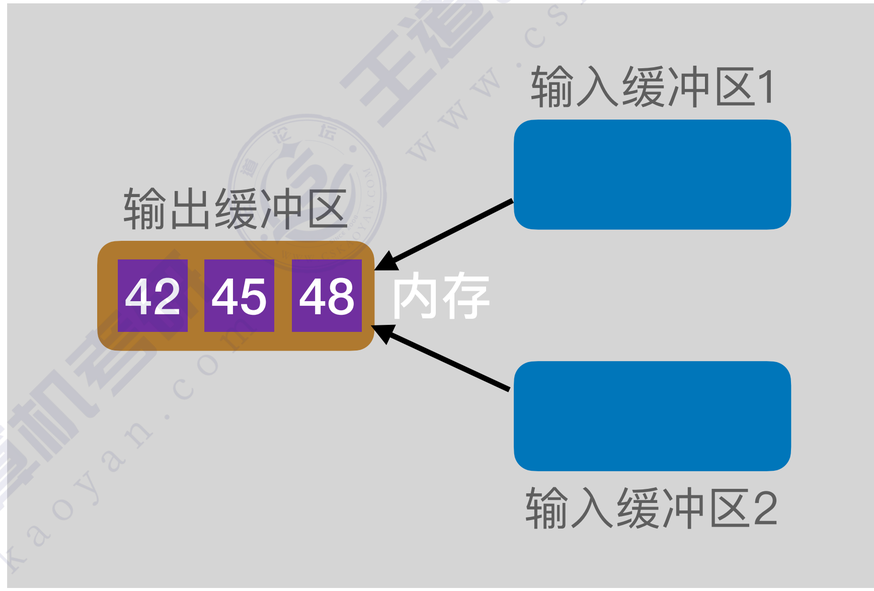


将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第一趟归并

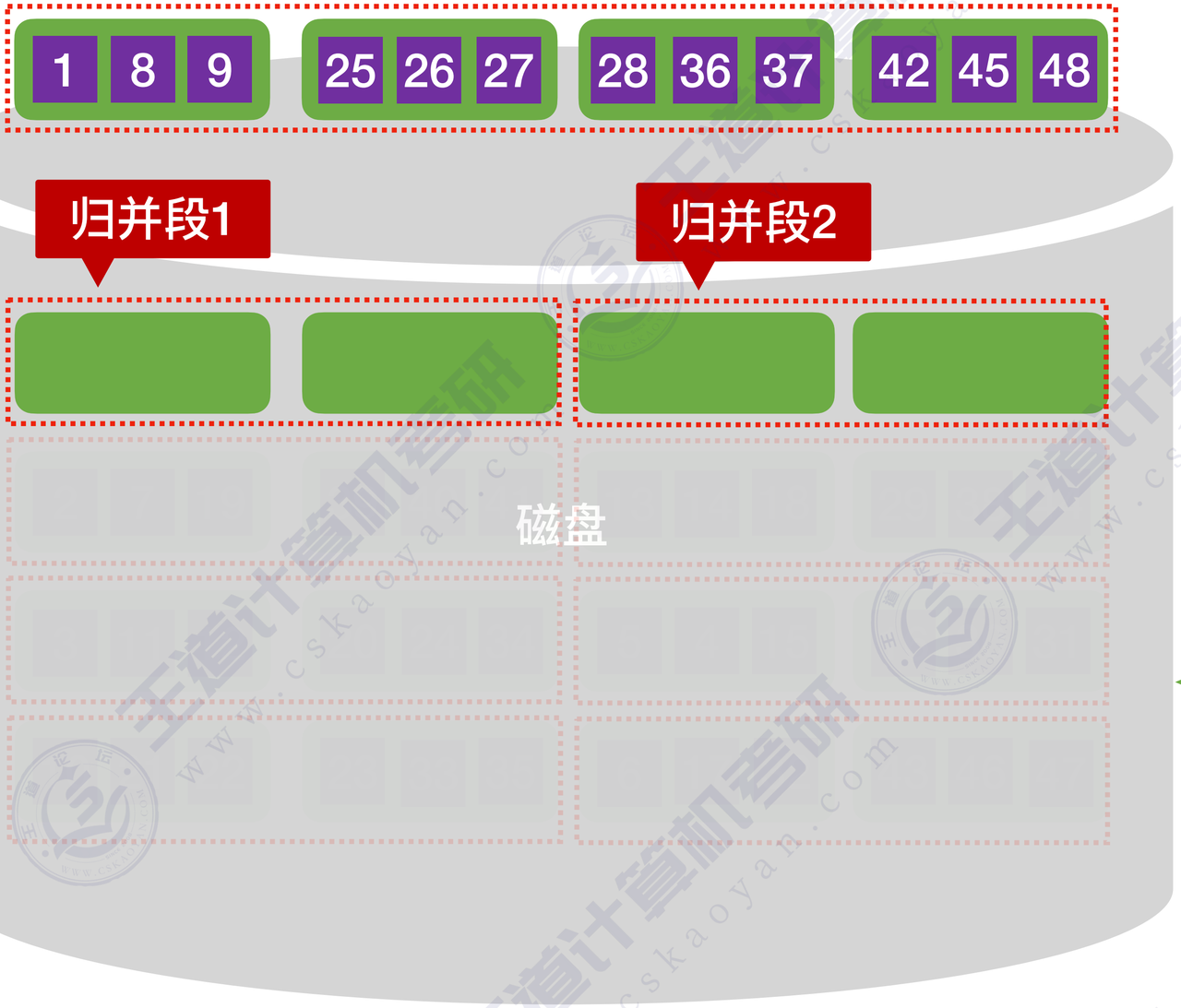


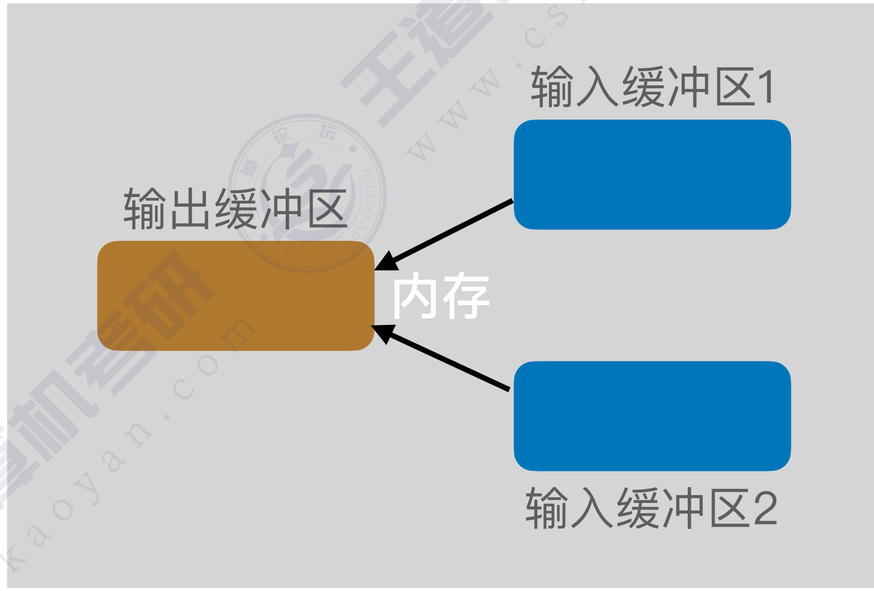


将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第一趟归并

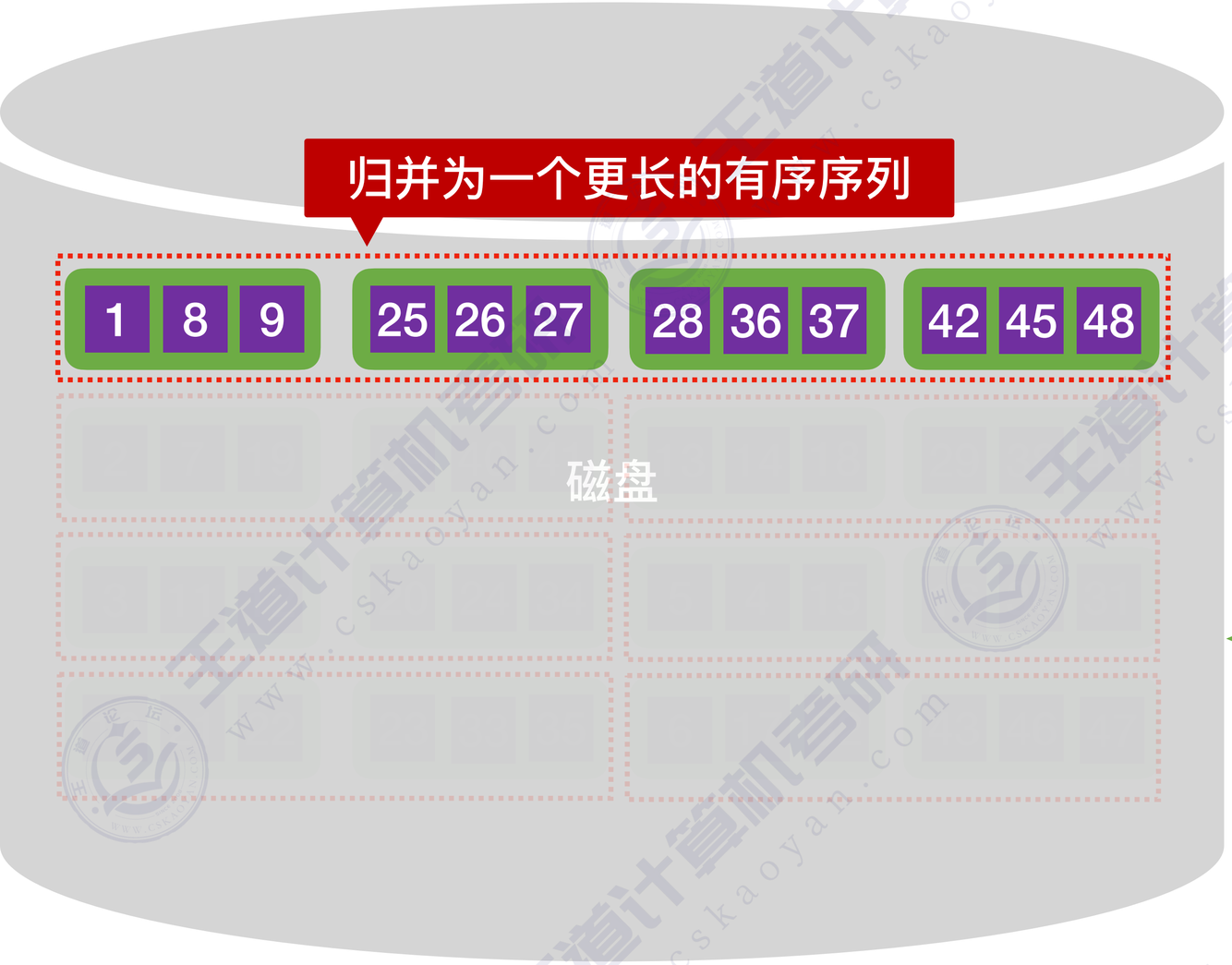


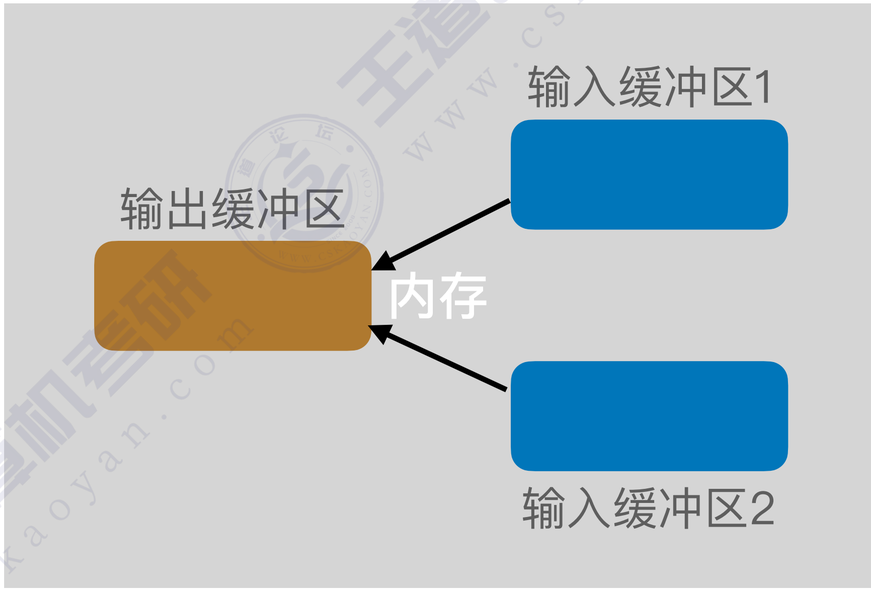


将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第一趟归并

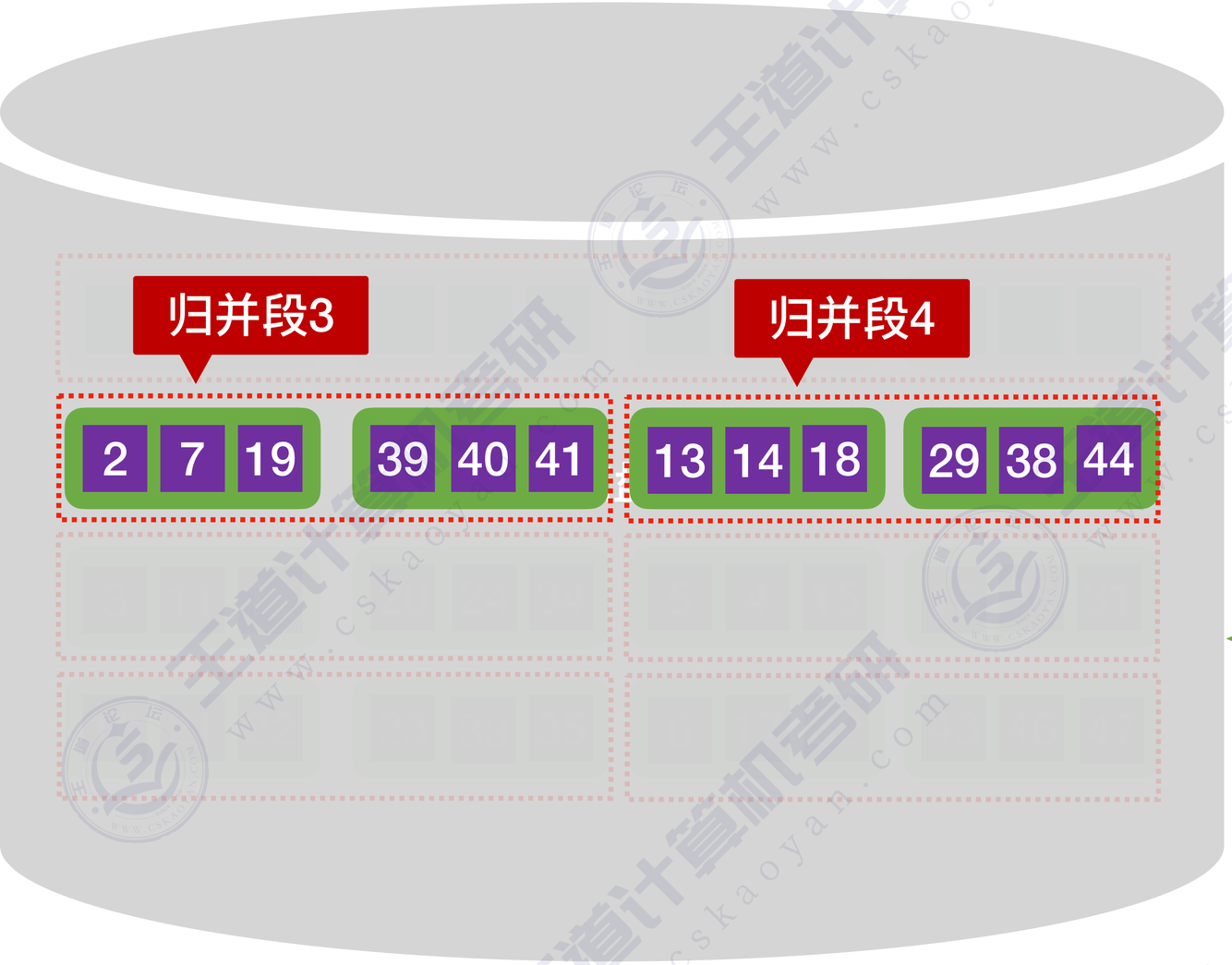


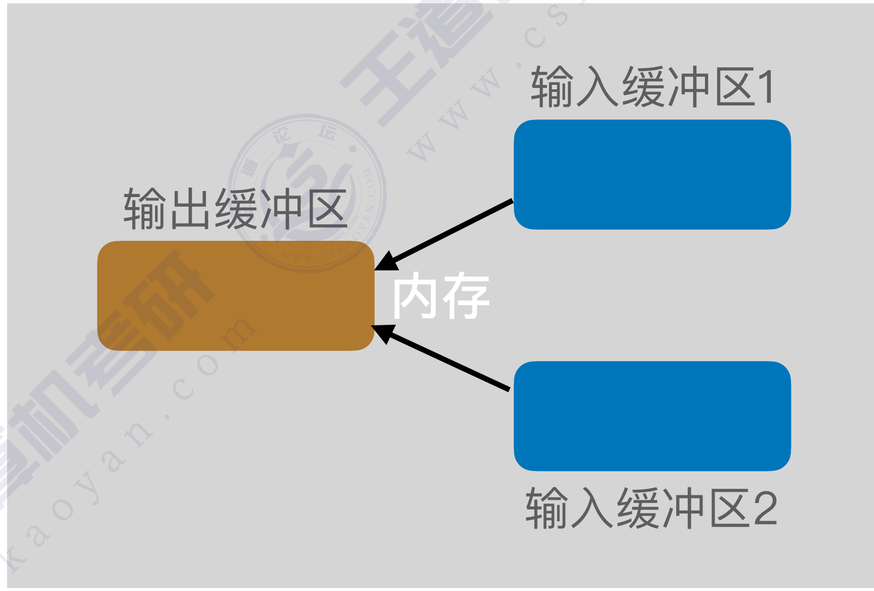


将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第一趟归并

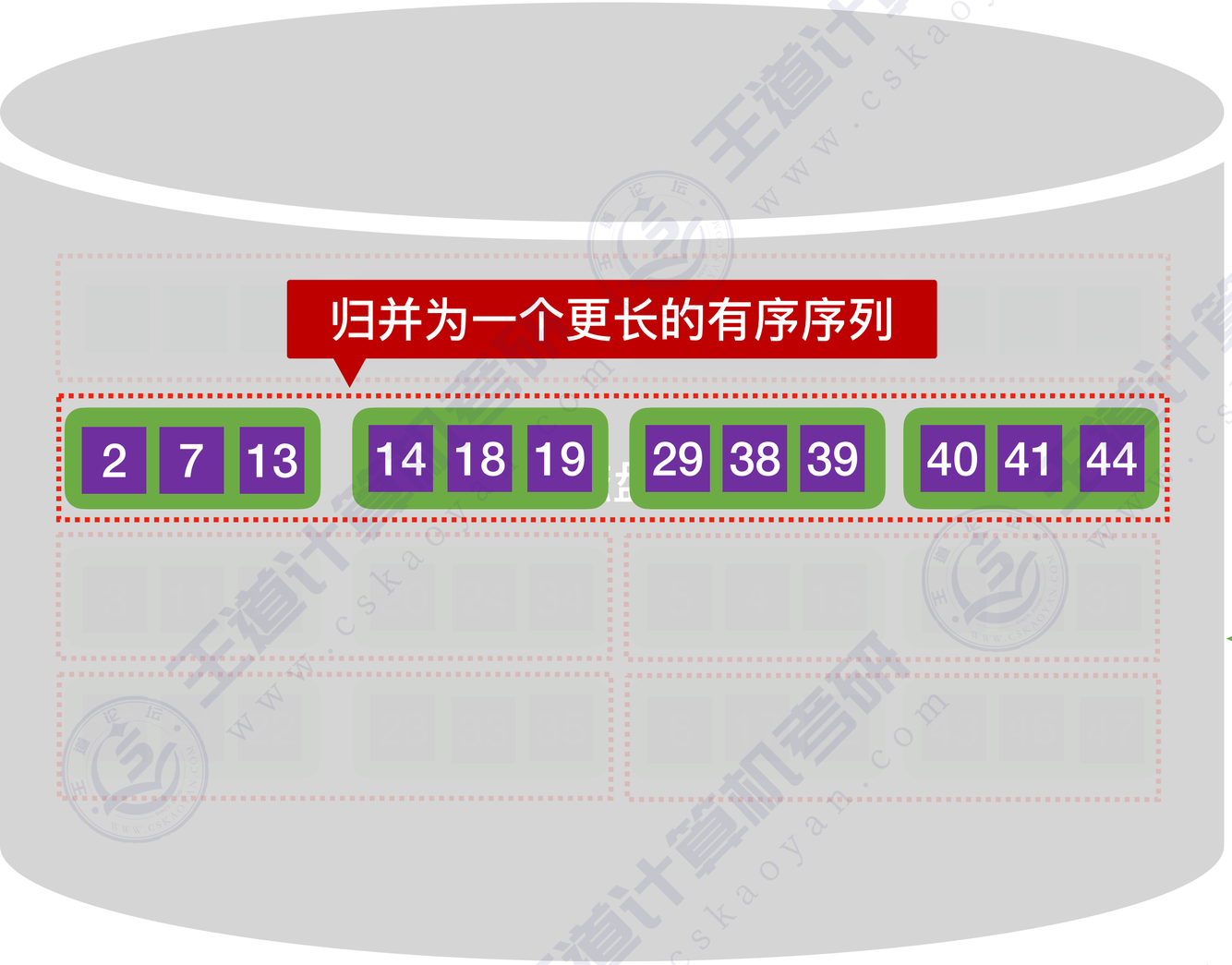


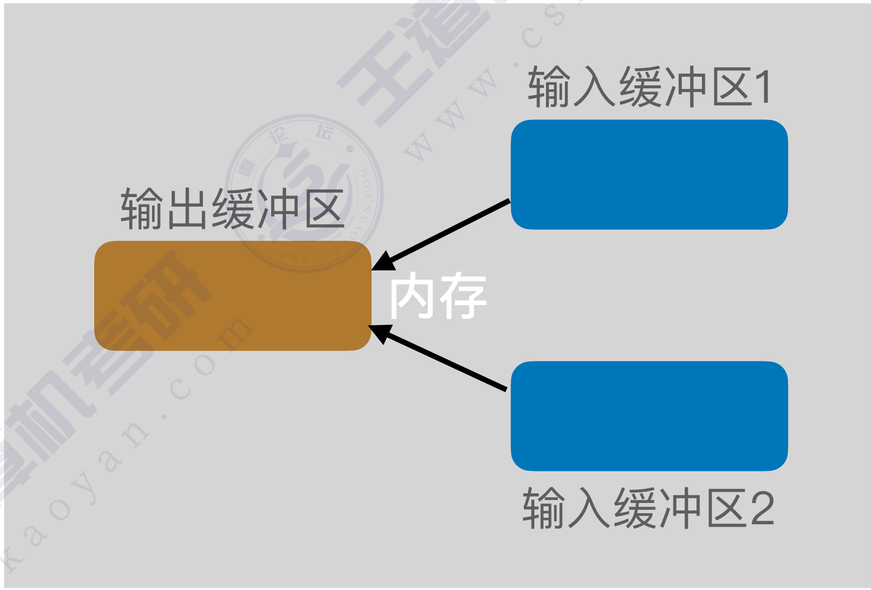


将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第一趟归并

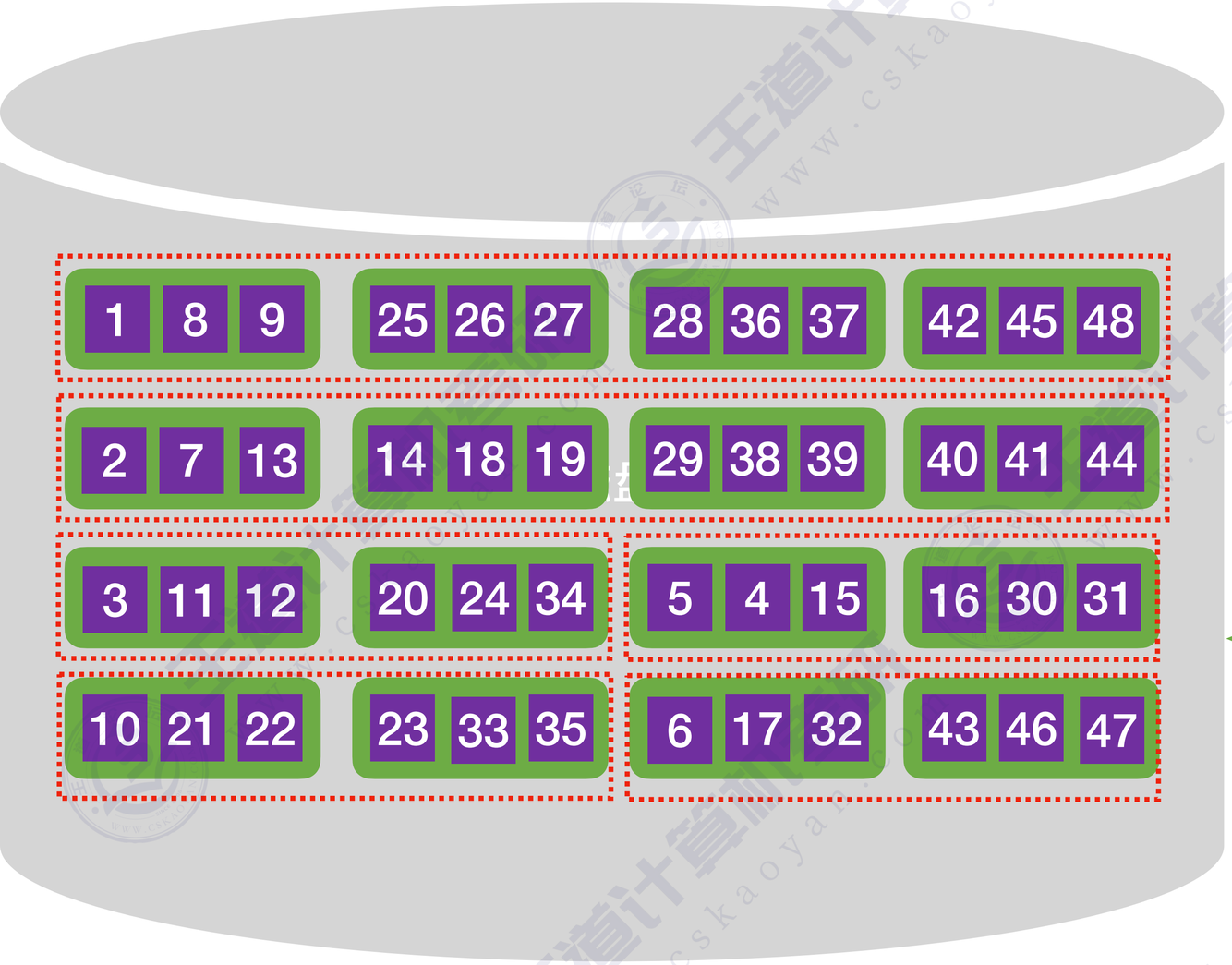


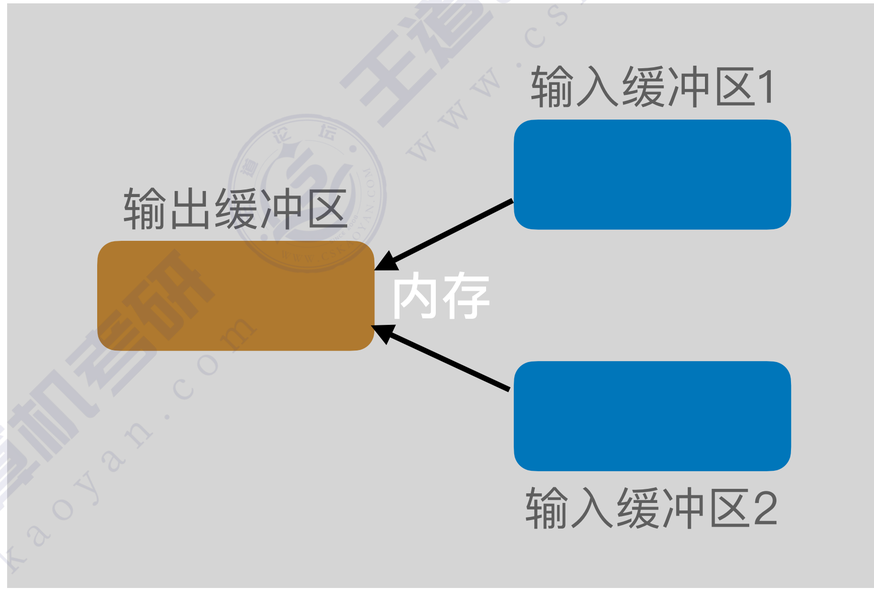


将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第一趟归并

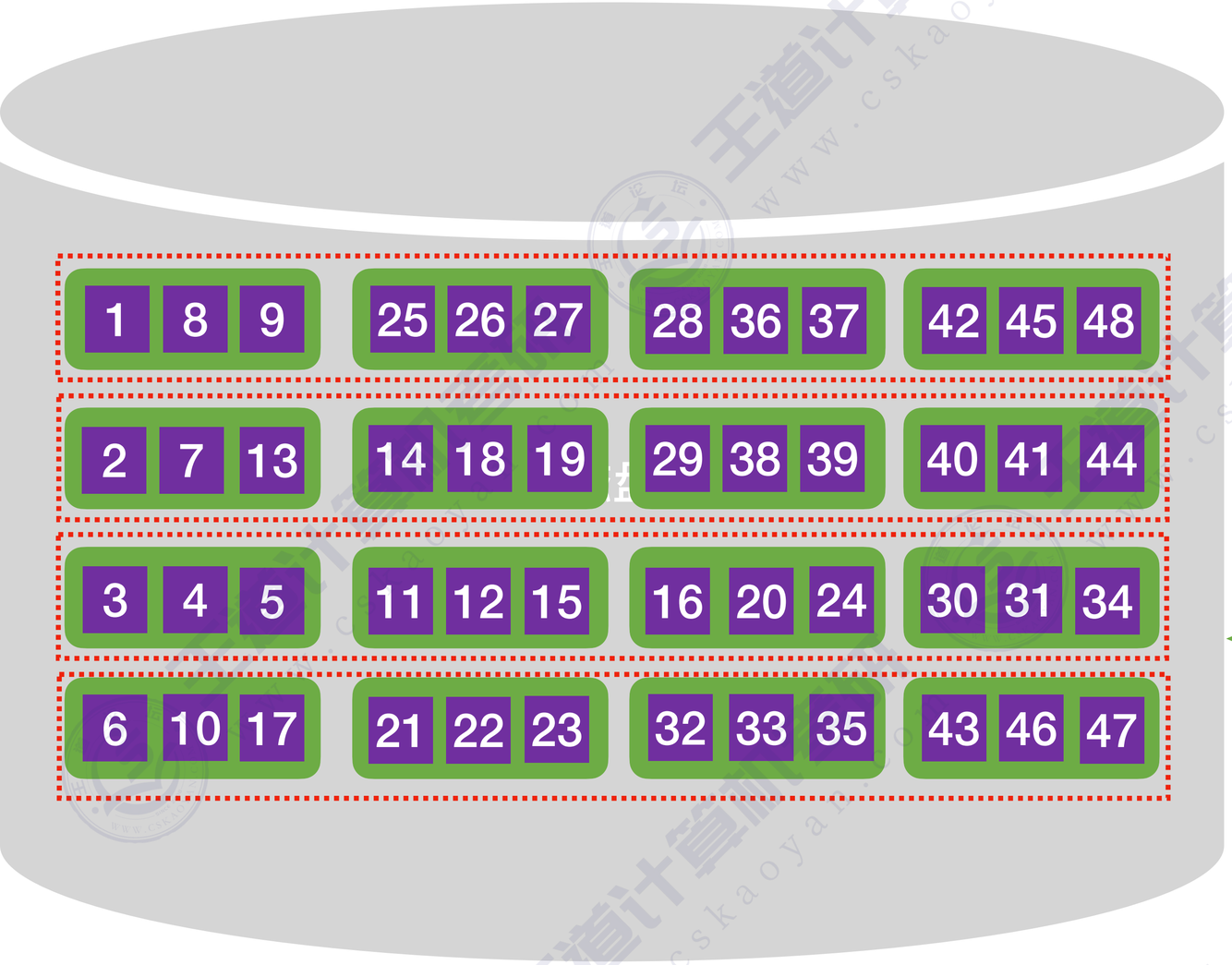


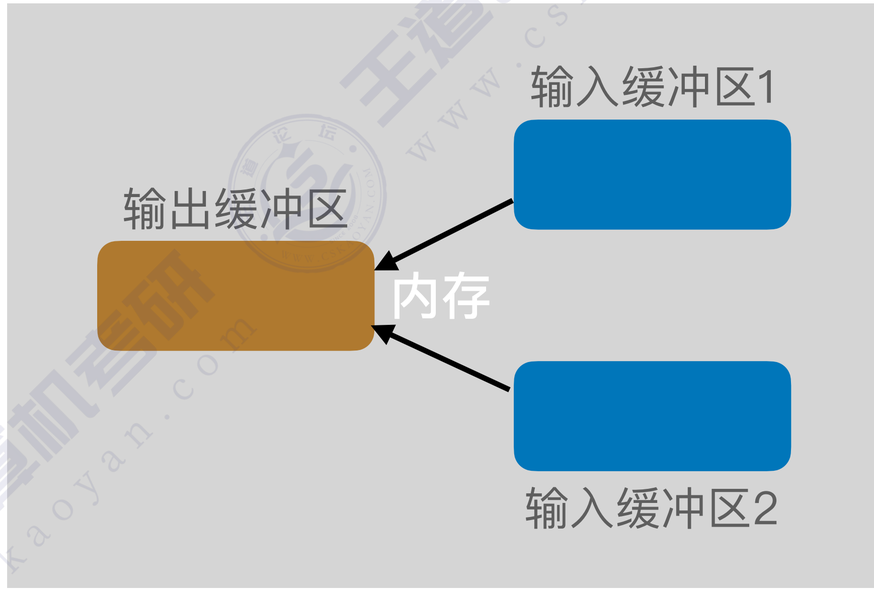


将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第一趟归并



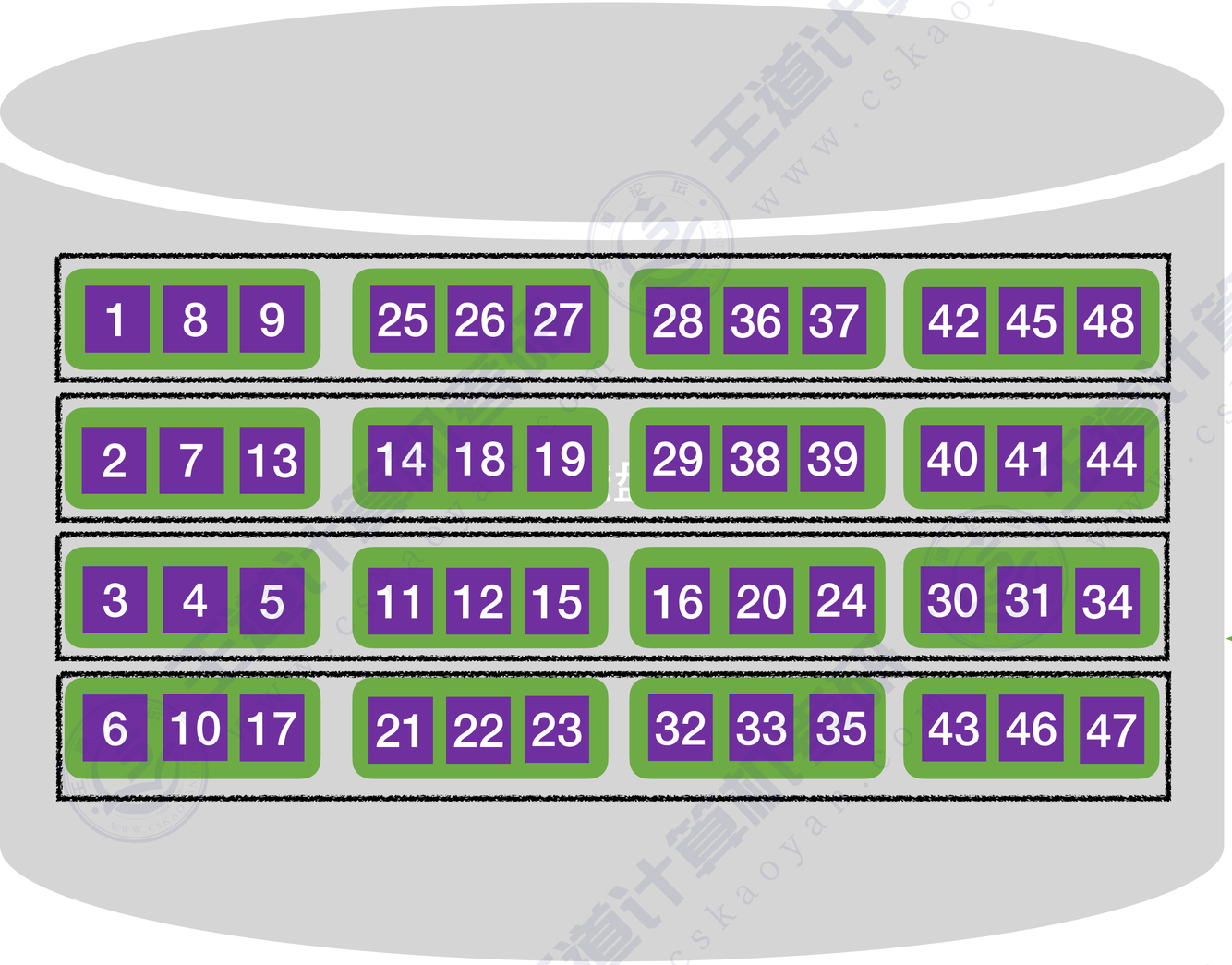


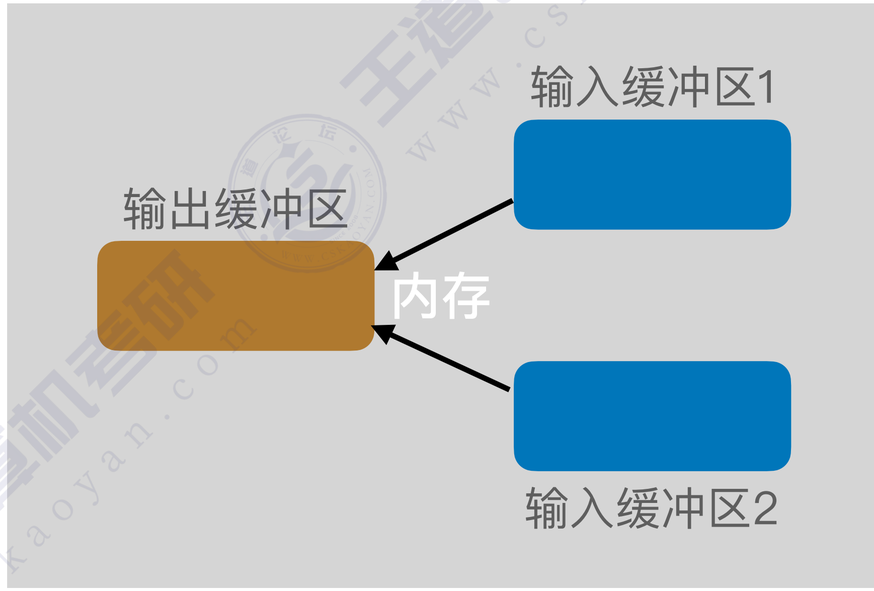
将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并

把4个有序子序列（归并段）两两归并

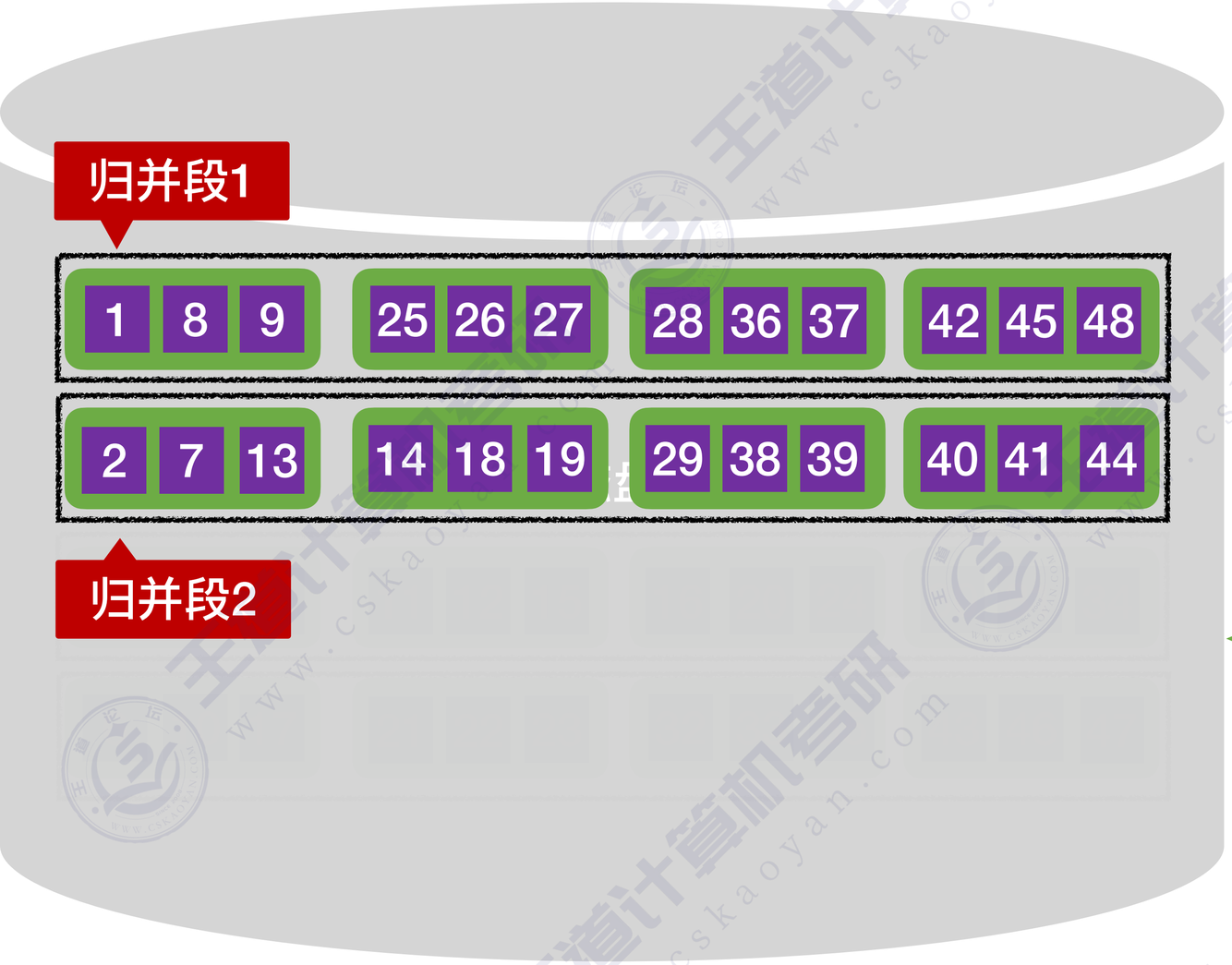


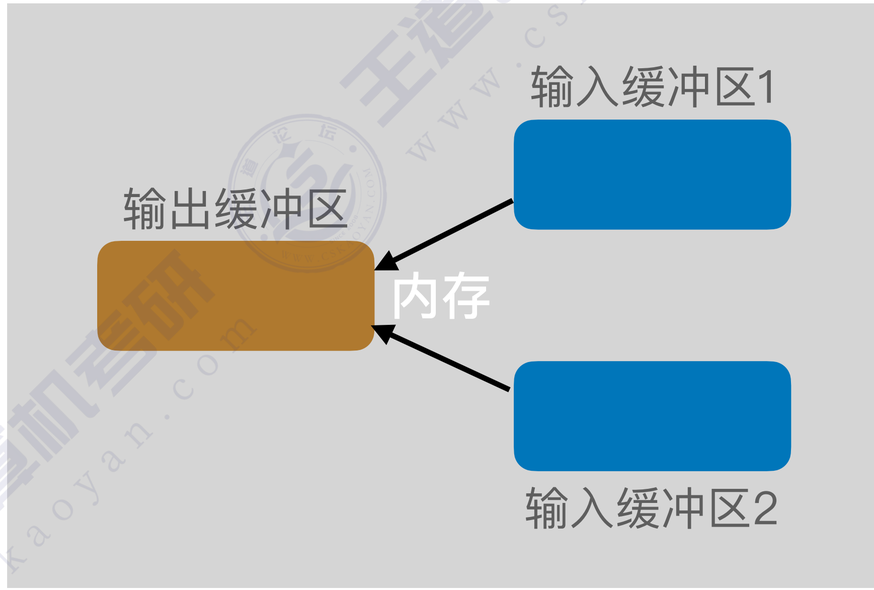


将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并

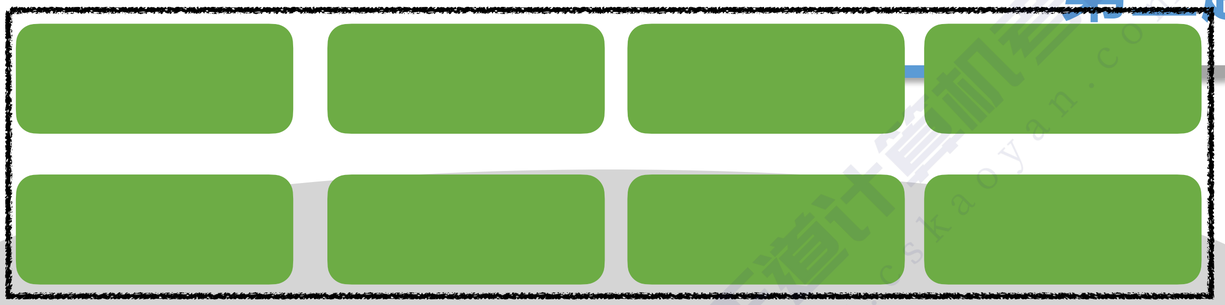




将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并



归并段1

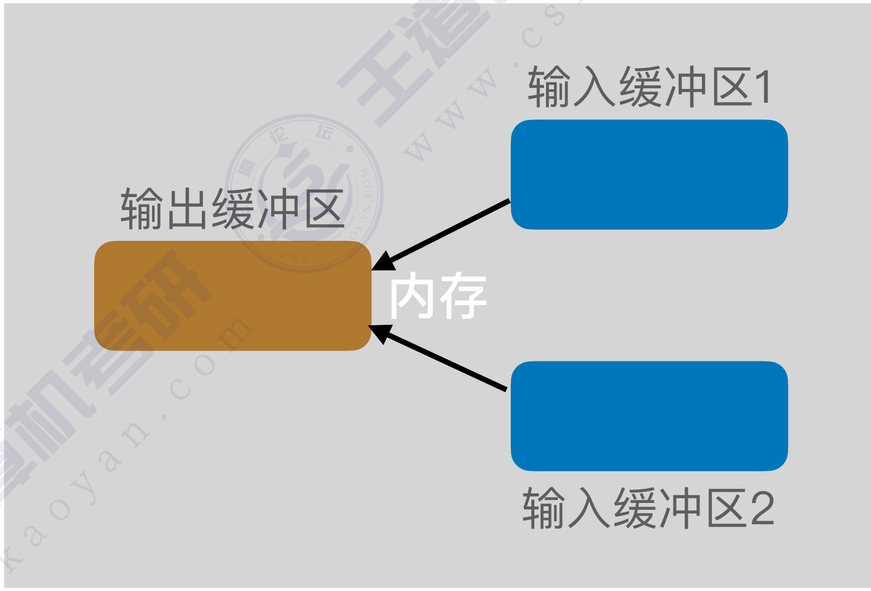
1	8	9	25	26	27	28	36	37	42	45	48

2	7	13	14	18	19	29	38	39	40	41	44

归并段2

读磁盘

写磁盘



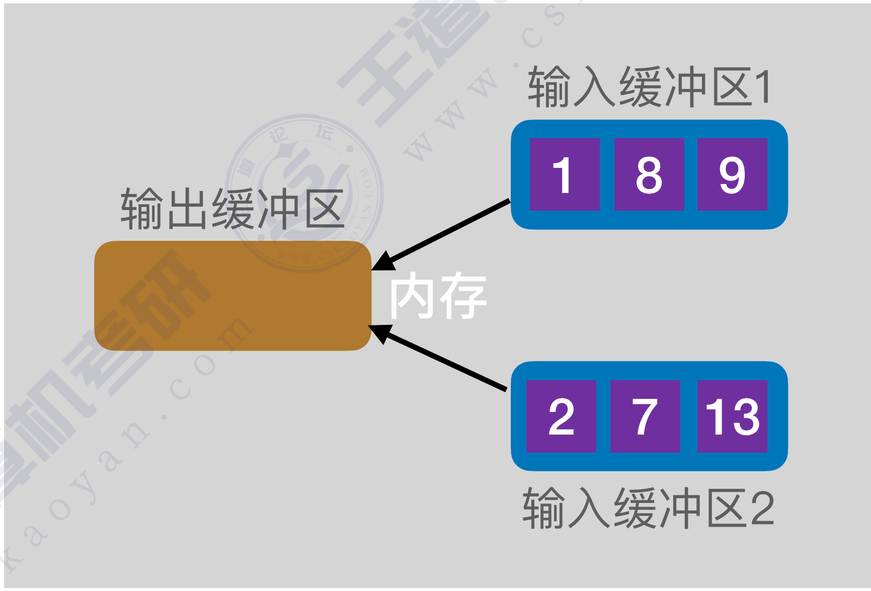
将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并

读磁盘

写磁盘



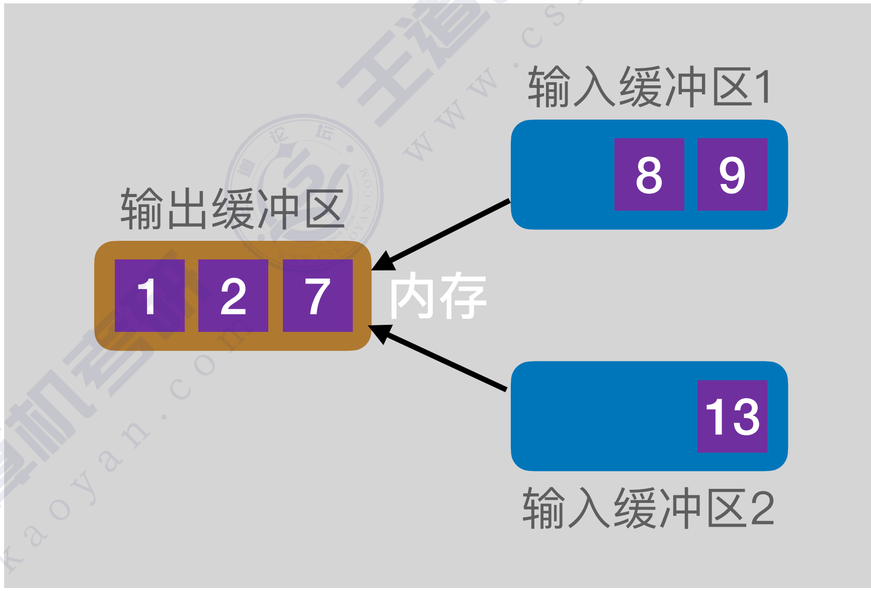
将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并

读磁盘

写磁盘



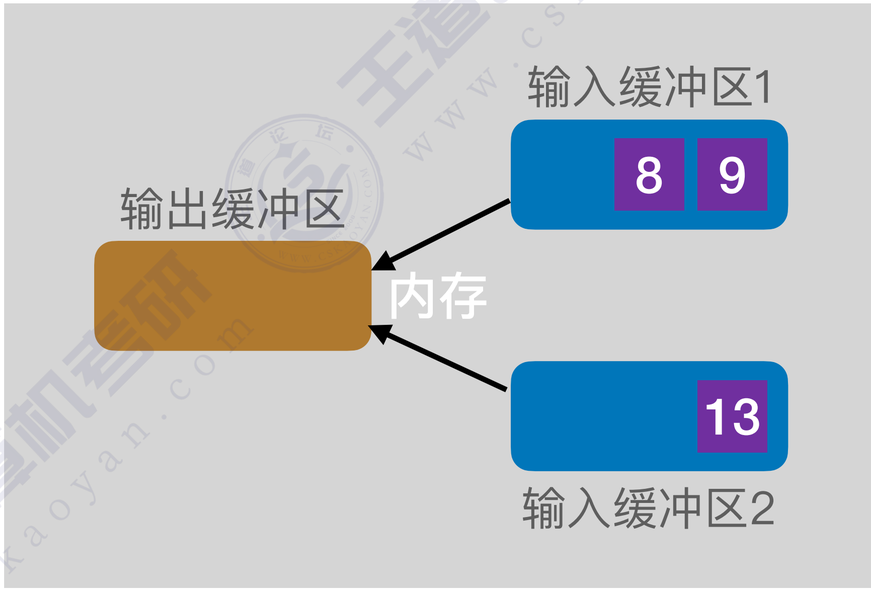
将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并

读磁盘

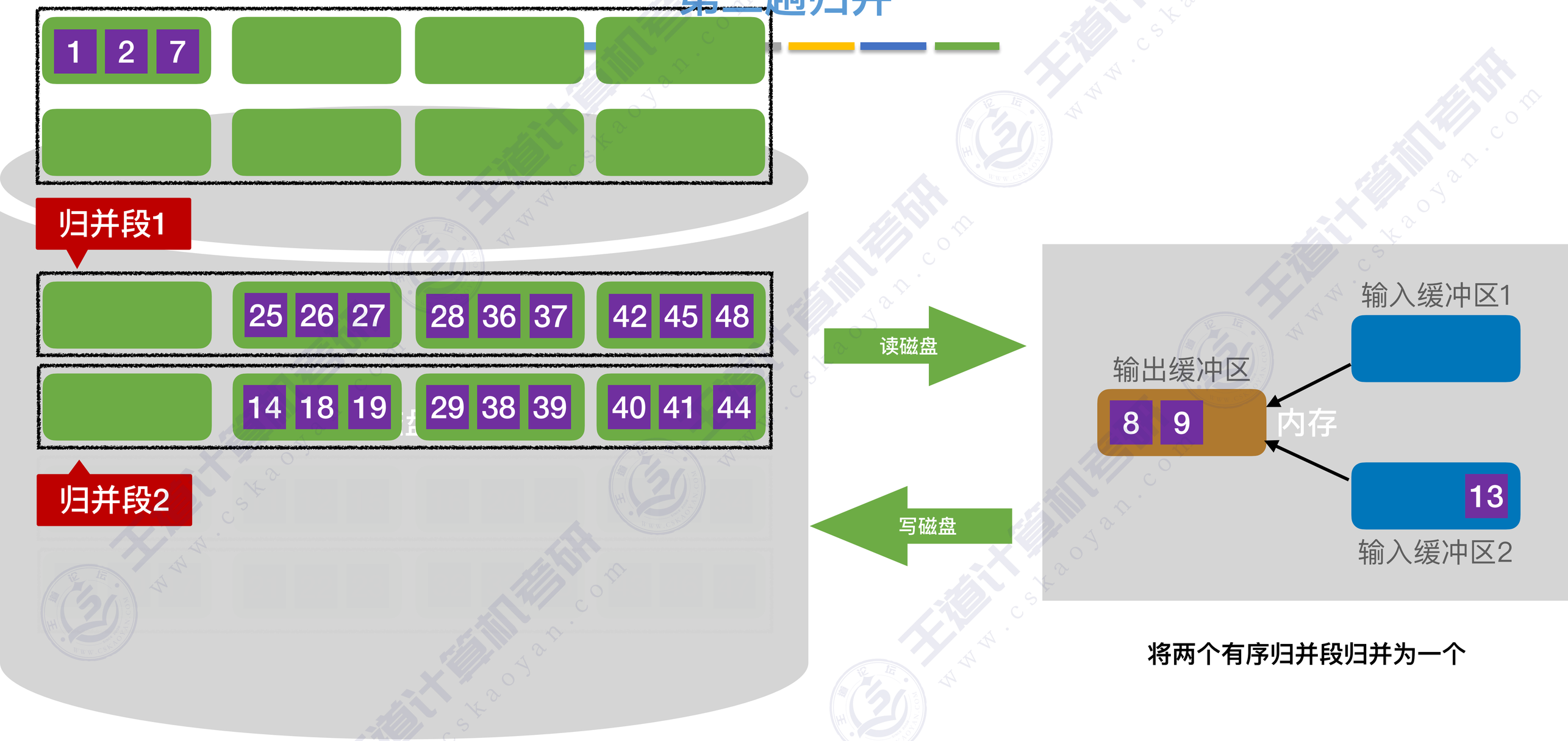
写磁盘



将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

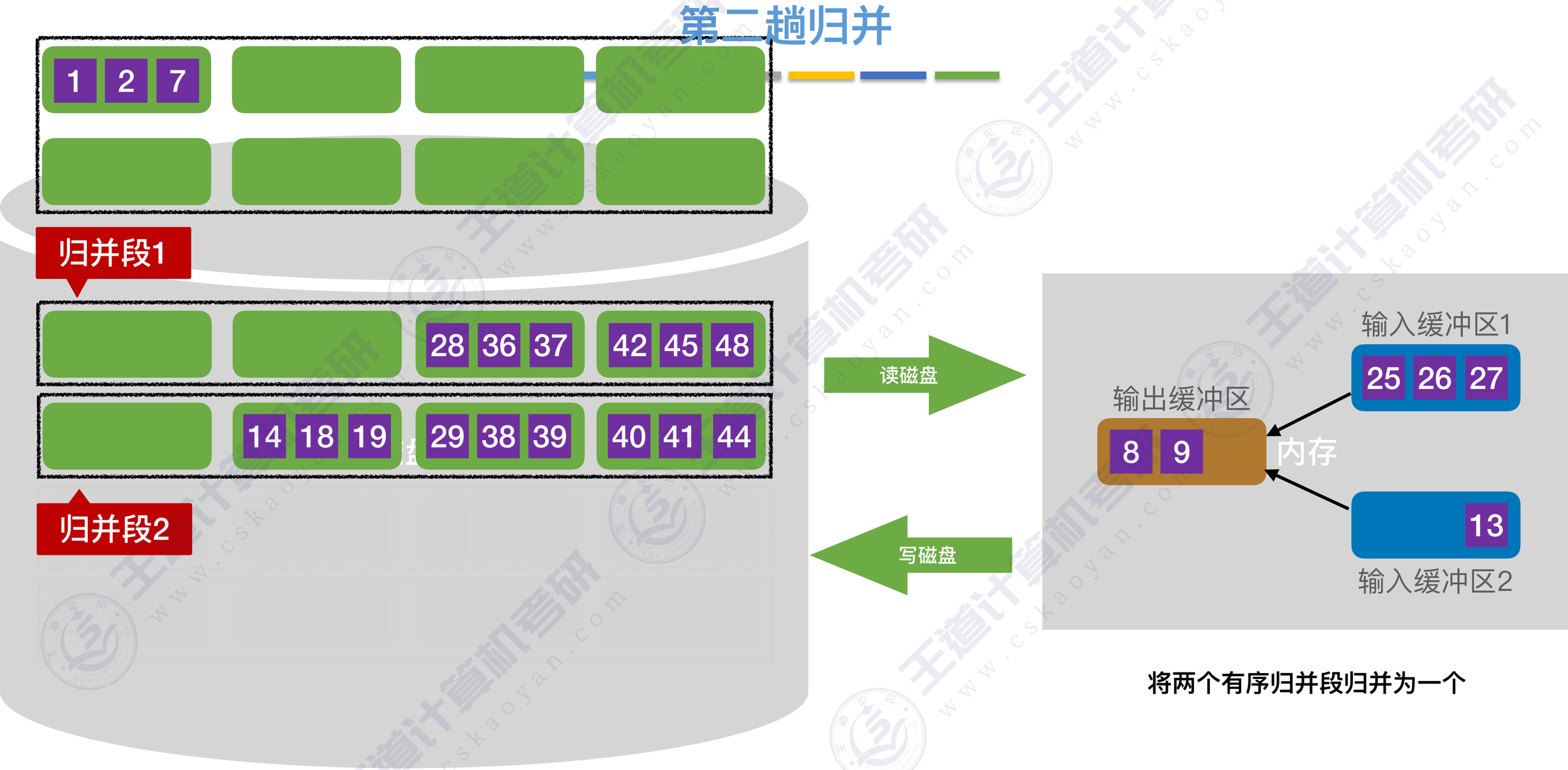
第二趟归并



将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

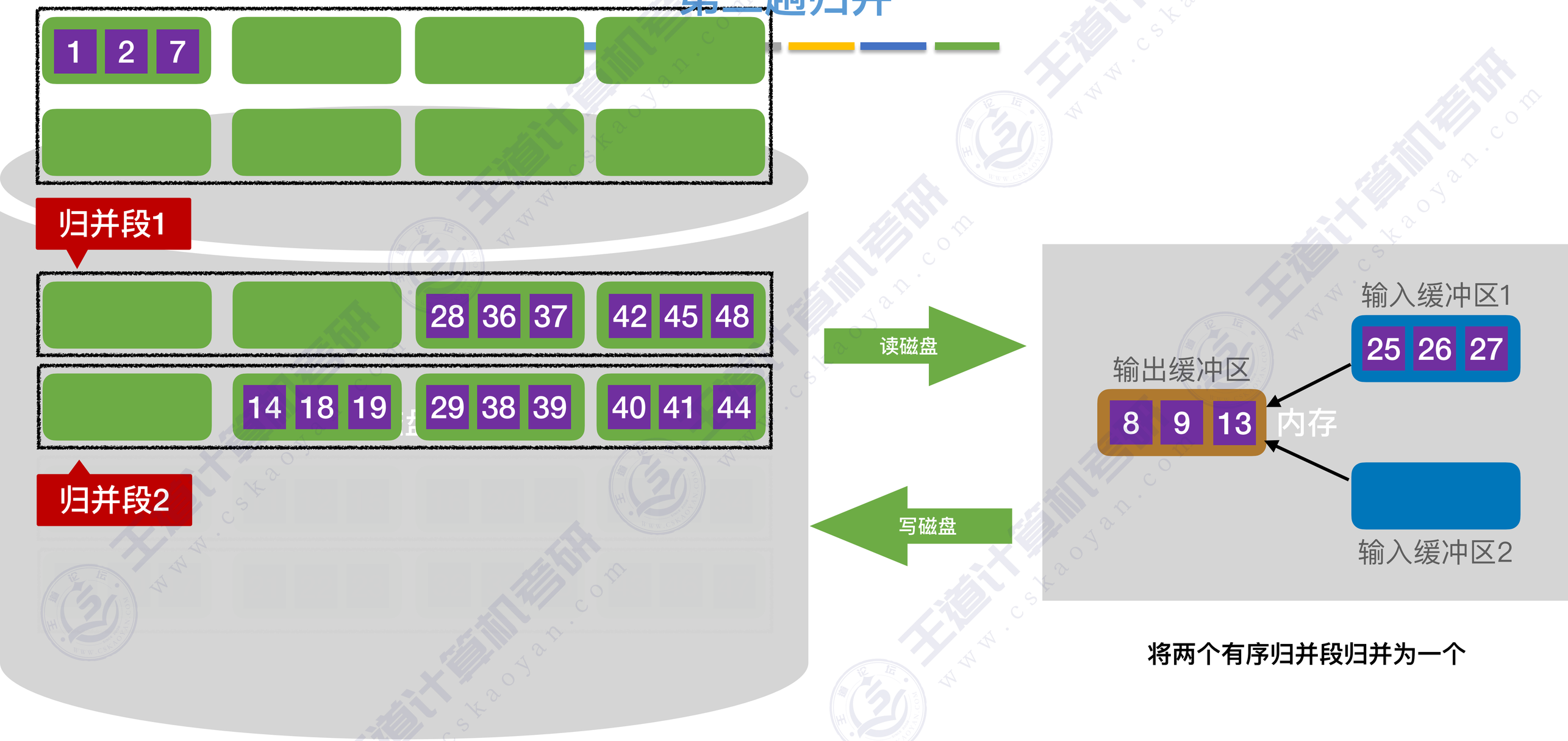
第二趟归并



将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

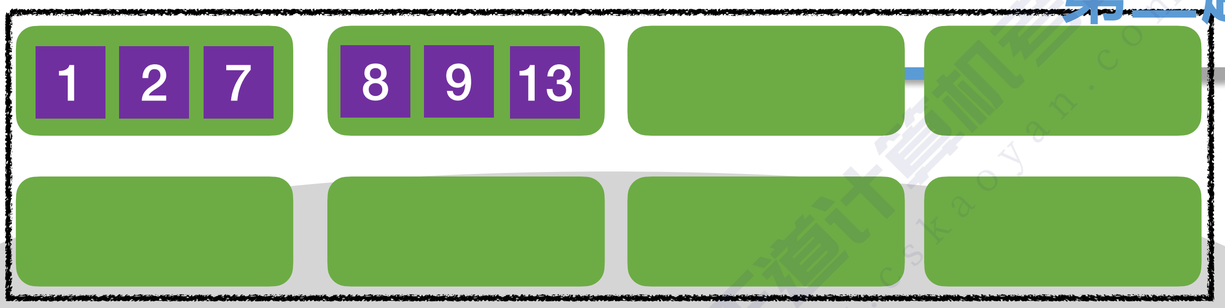
第二趟归并



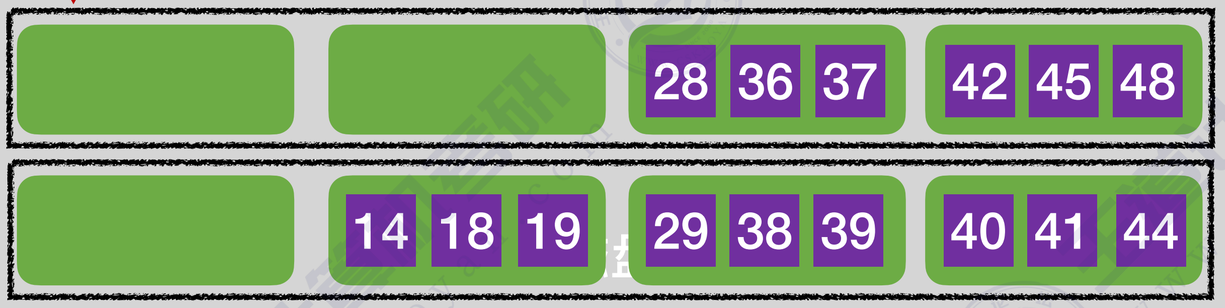
将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并



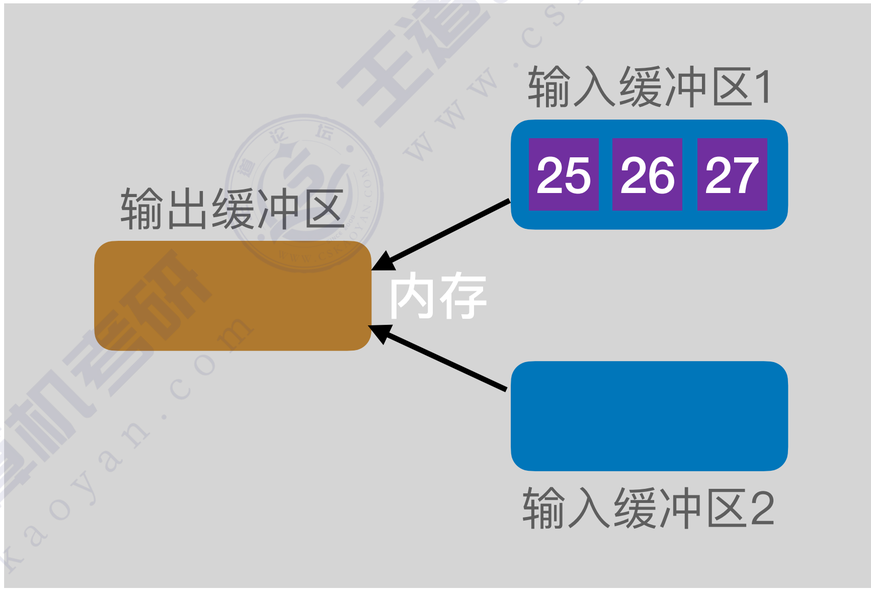
归并段1



归并段2

读磁盘

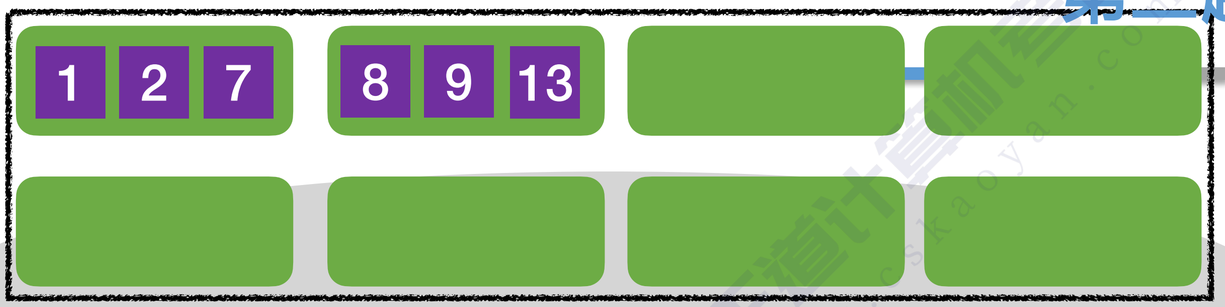
写磁盘



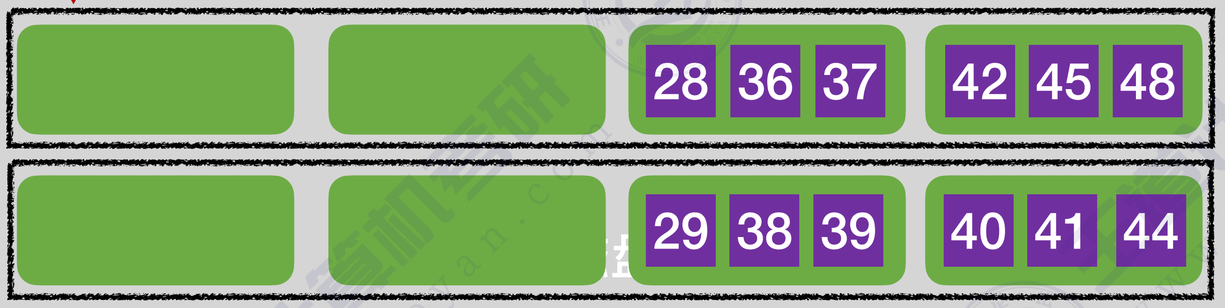
将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并



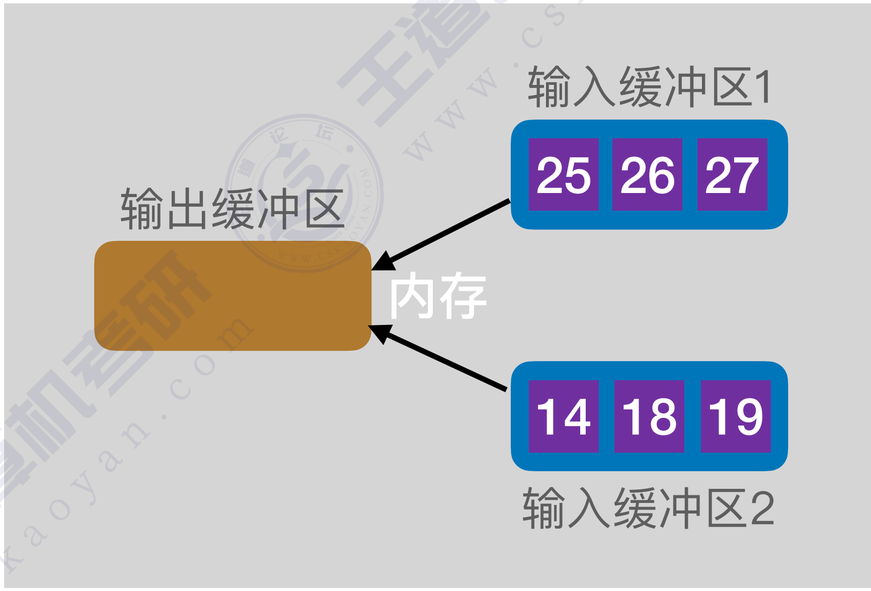
归并段1



归并段2

读磁盘

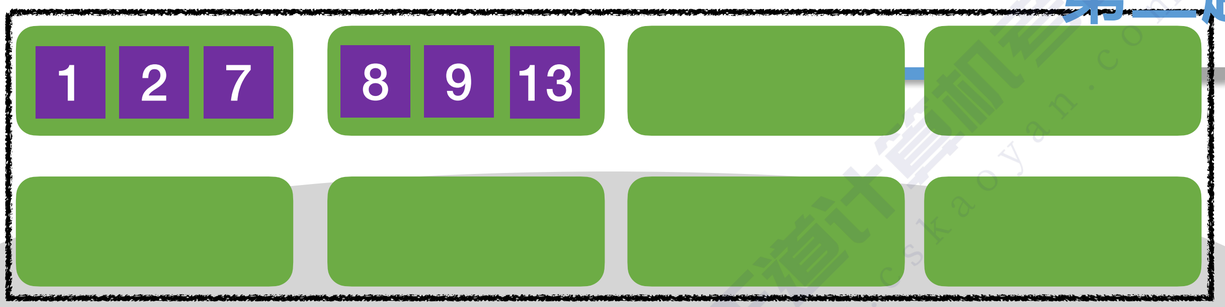
写磁盘



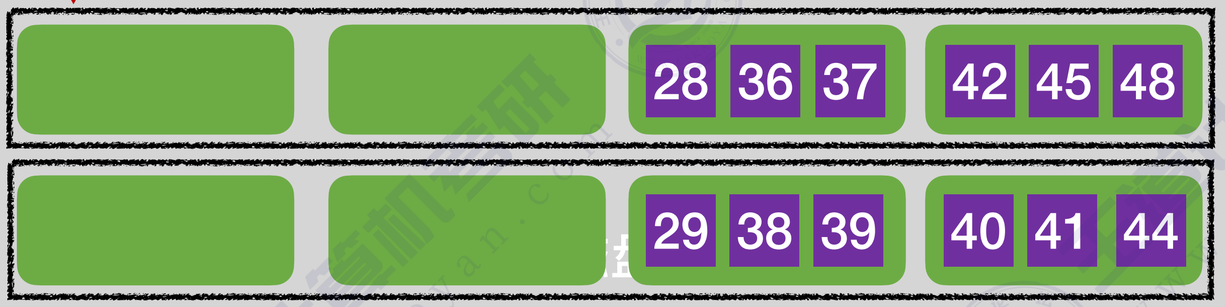
将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并



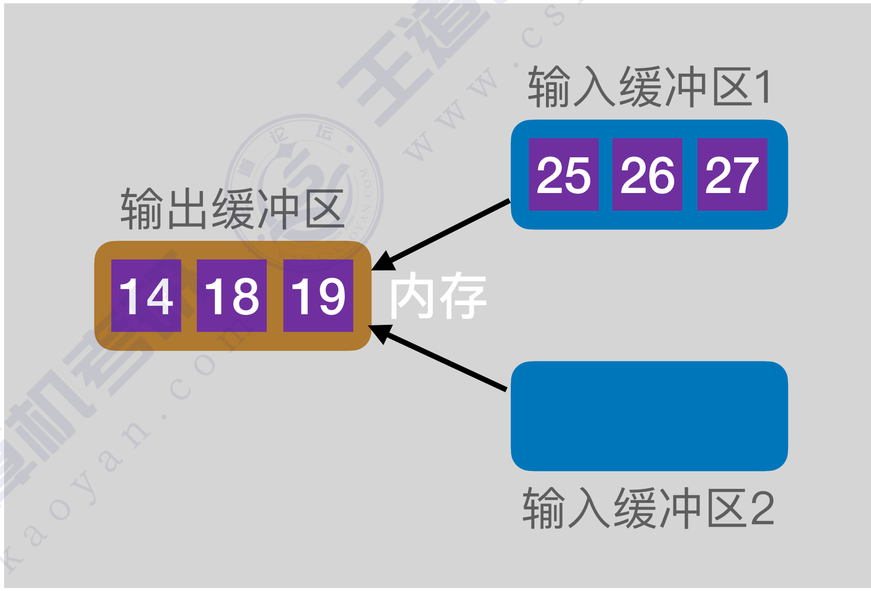
归并段1



归并段2

读磁盘

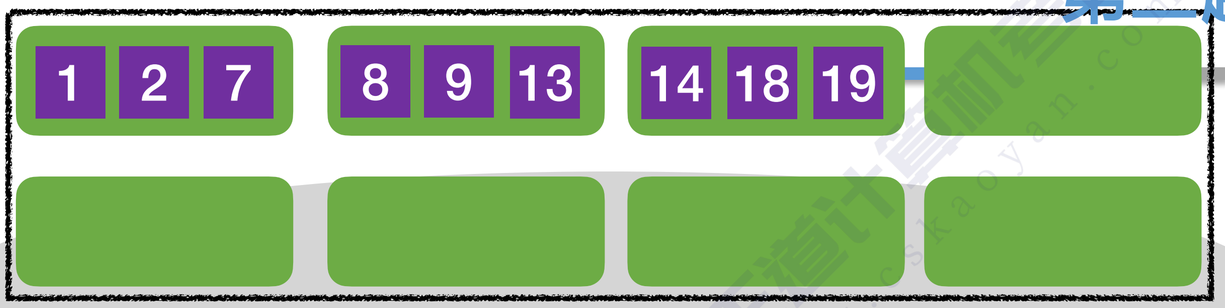
写磁盘



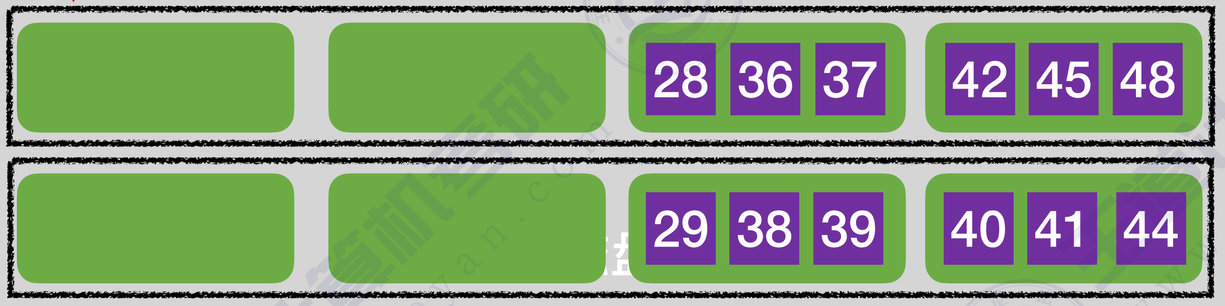
将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并



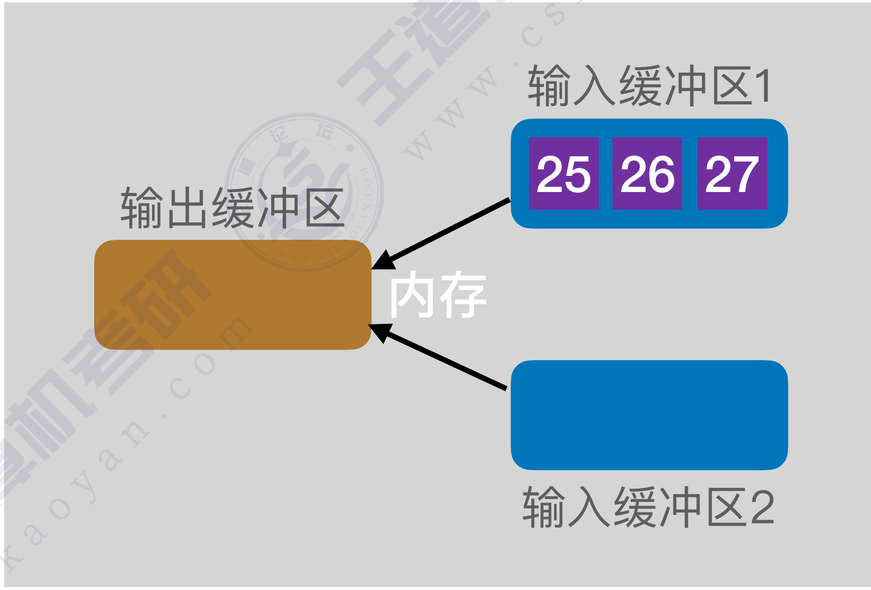
归并段1



归并段2

读磁盘

写磁盘



将两个有序归并段归并为一个

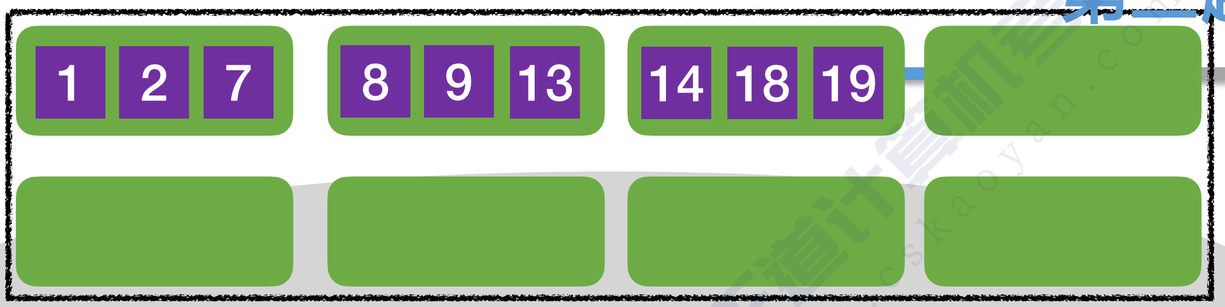
王道考研/CSKAOYAN.COM

第二趟归并

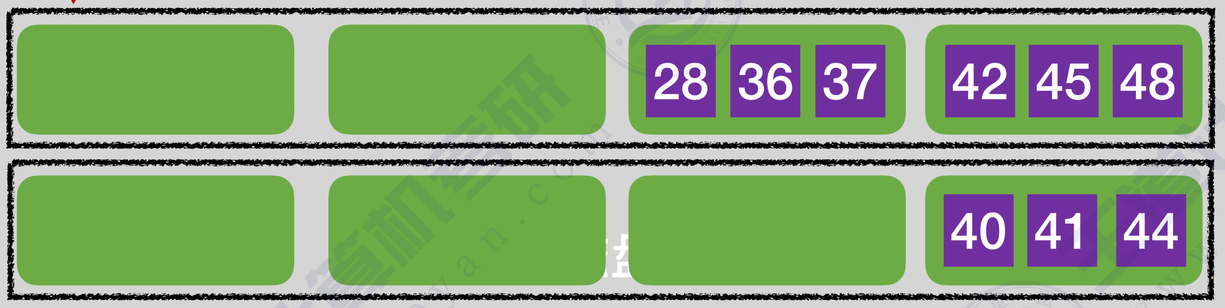
将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并



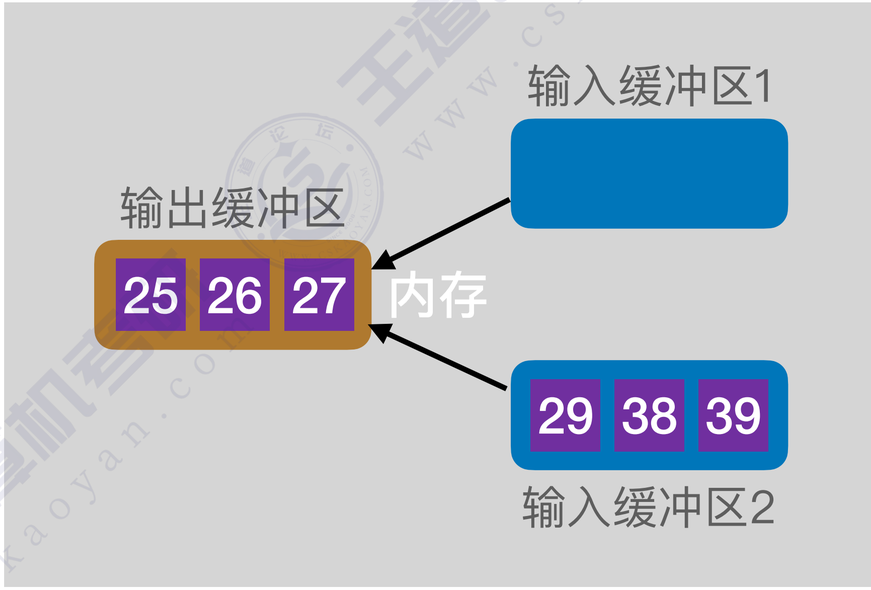
归并段1



归并段2

读磁盘

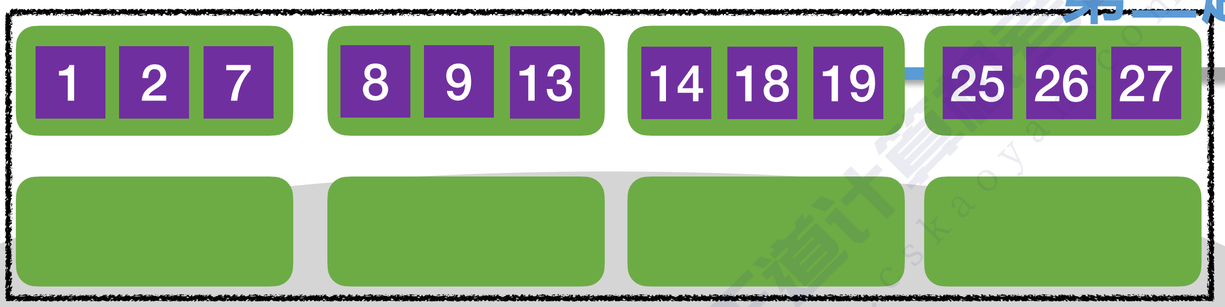
写磁盘



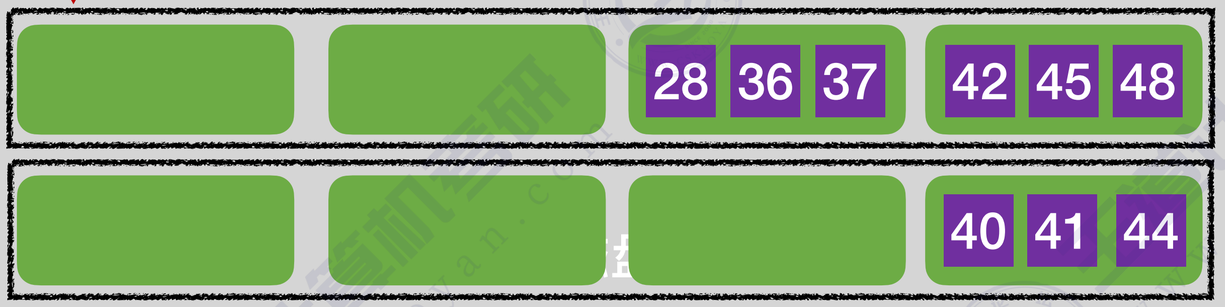
将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并



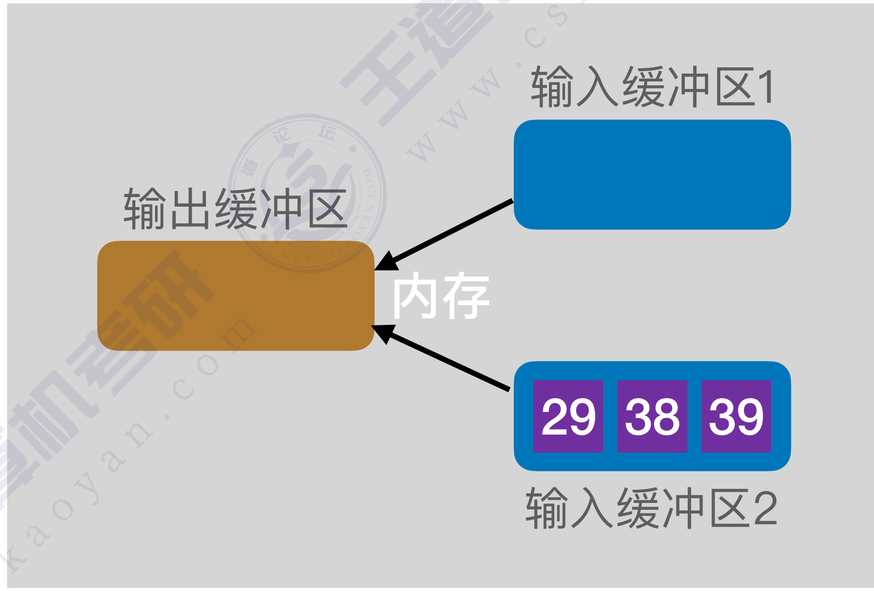
归并段1



归并段2

读磁盘

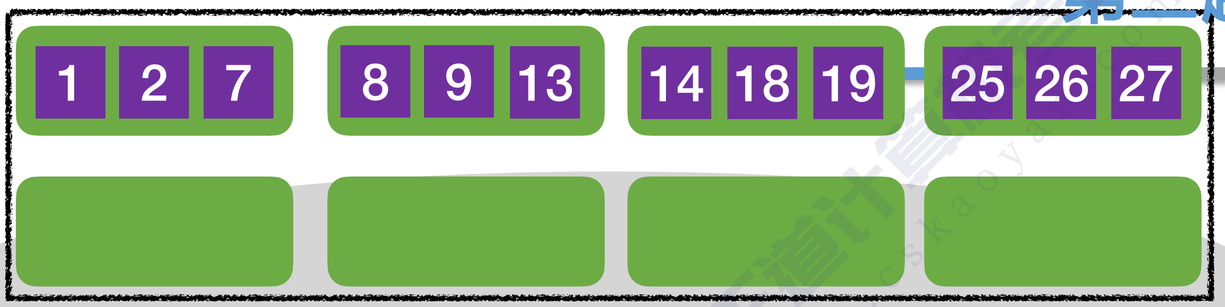
写磁盘



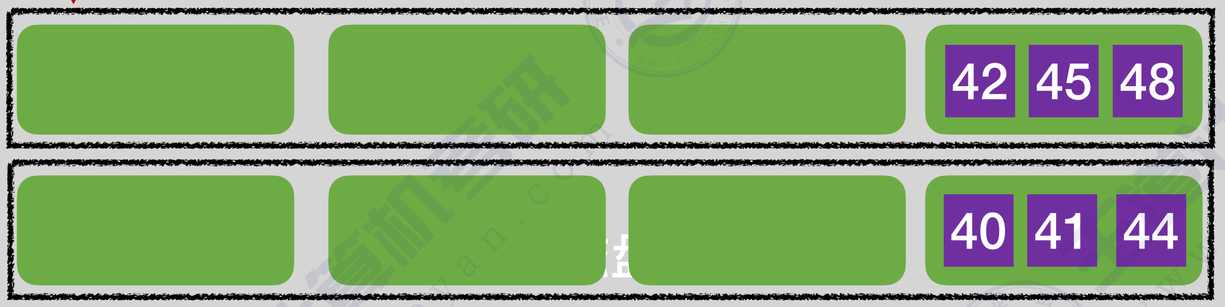
将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并



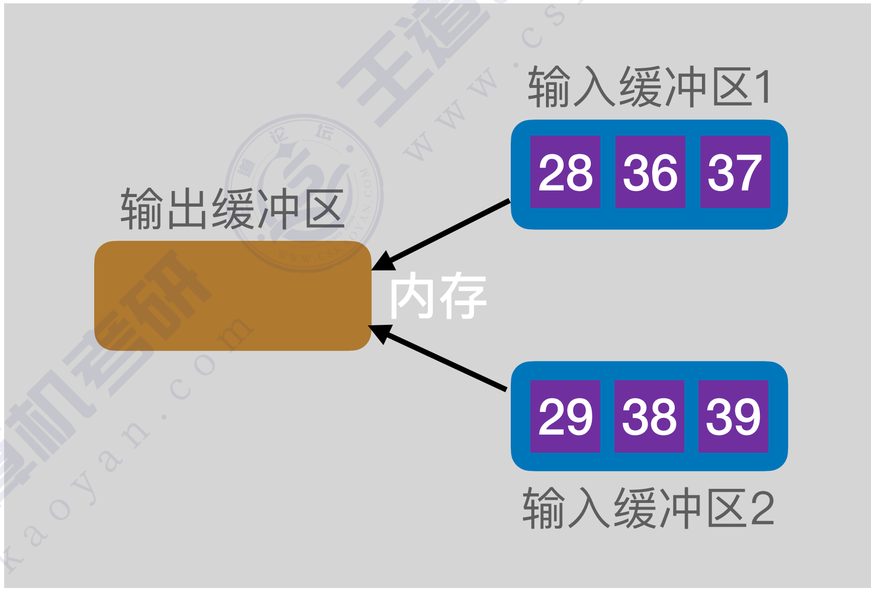
归并段1



归并段2

读磁盘

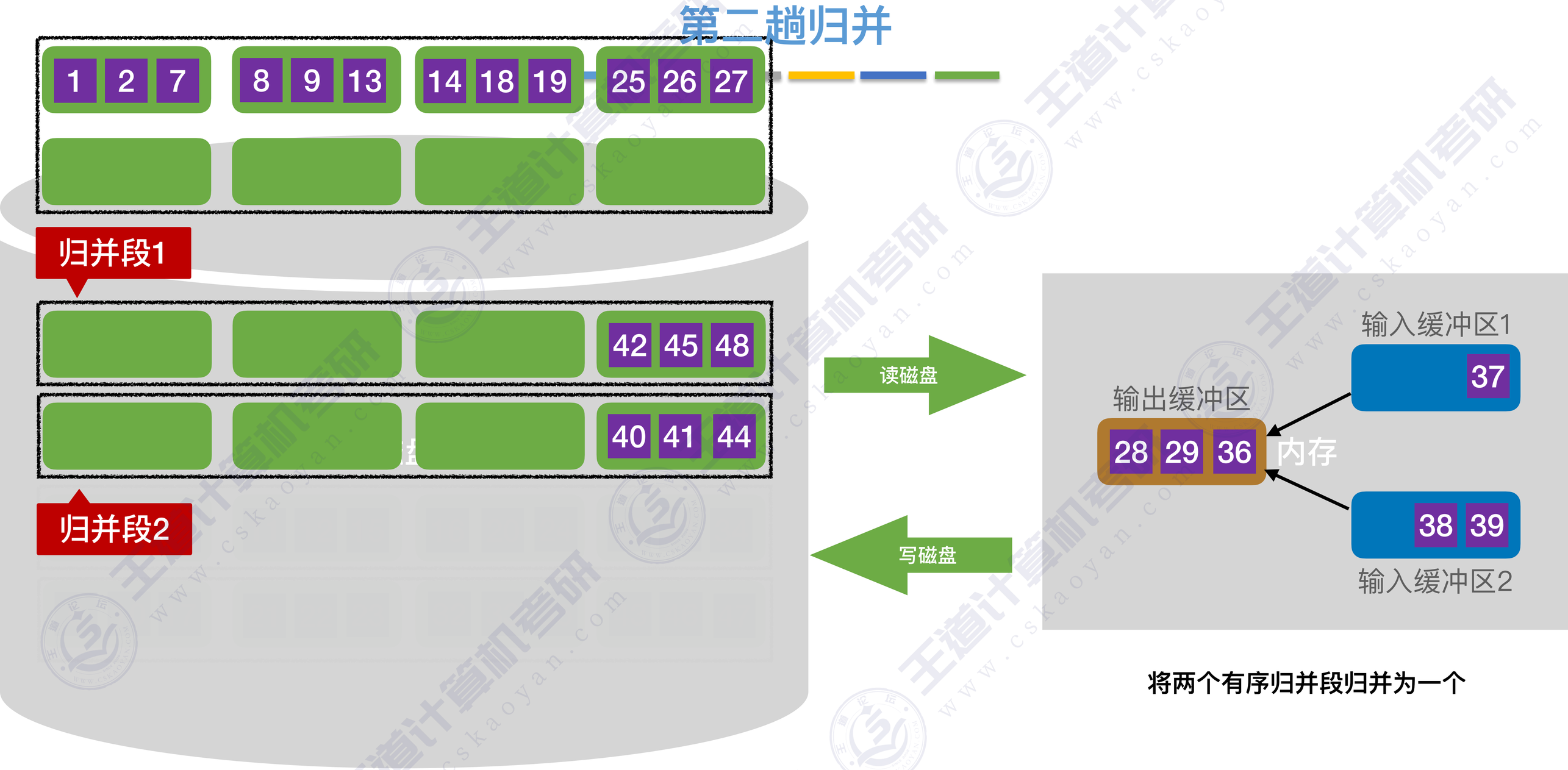
写磁盘



将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

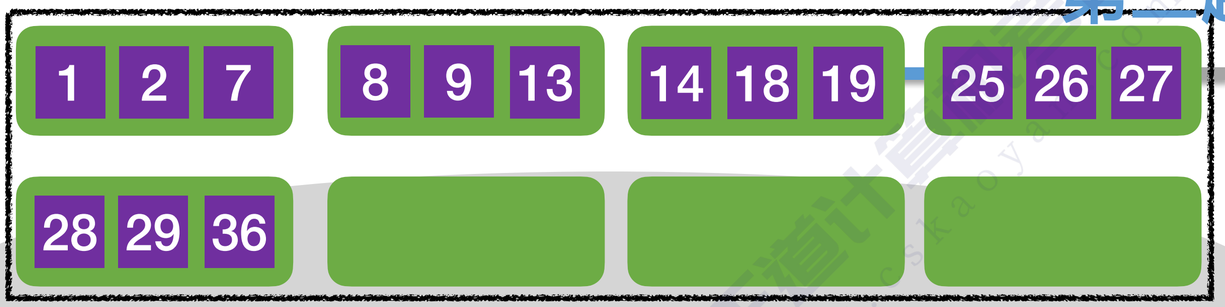
第二趟归并



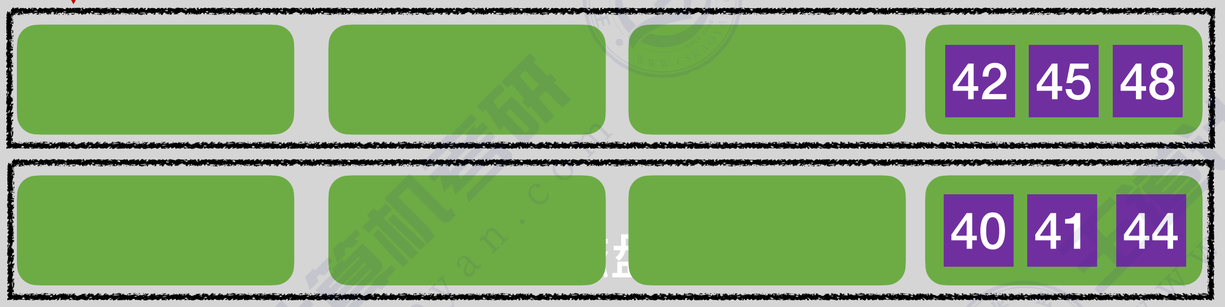
将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并



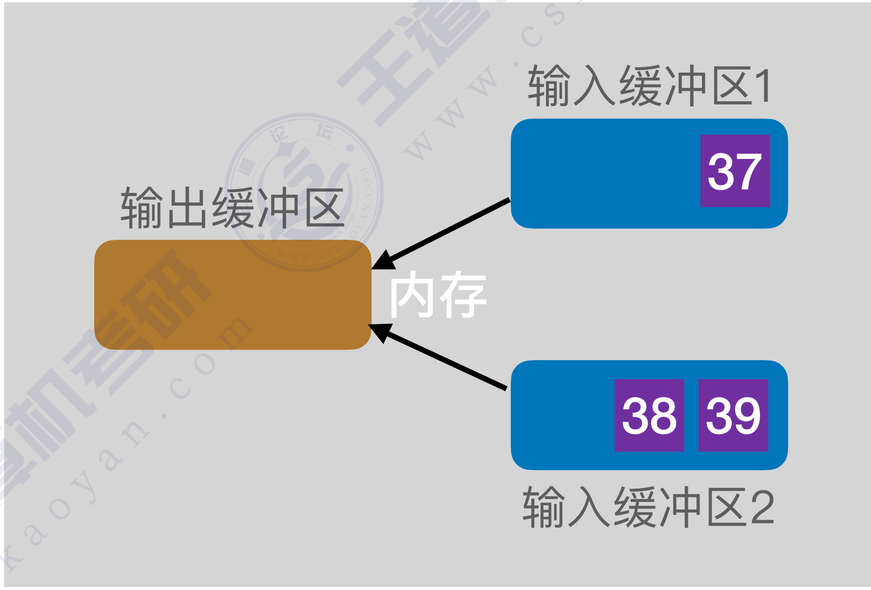
归并段1



归并段2

读磁盘

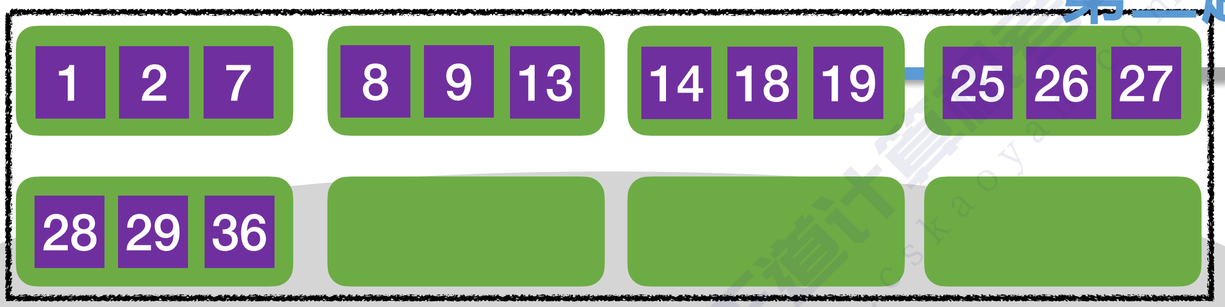
写磁盘



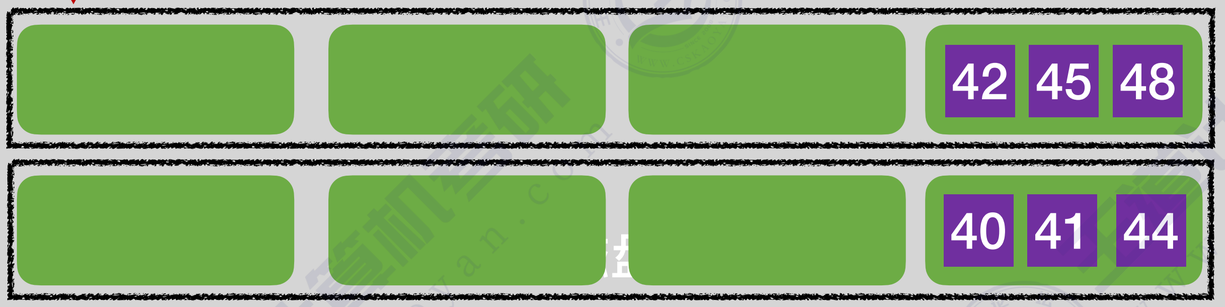
将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并



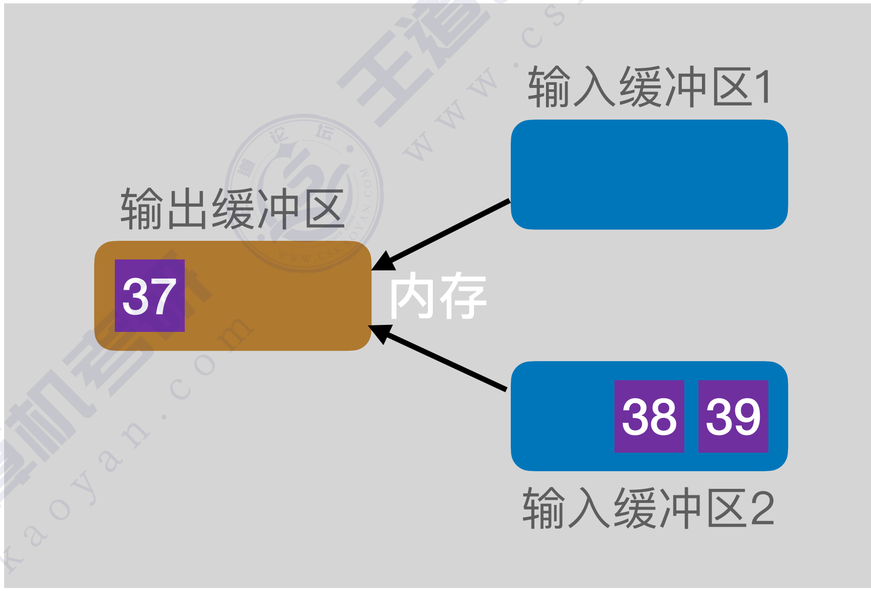
归并段1



归并段2

读磁盘

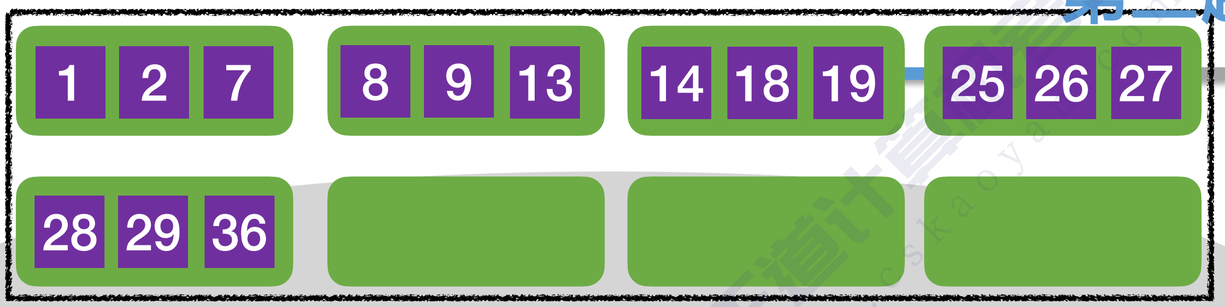
写磁盘



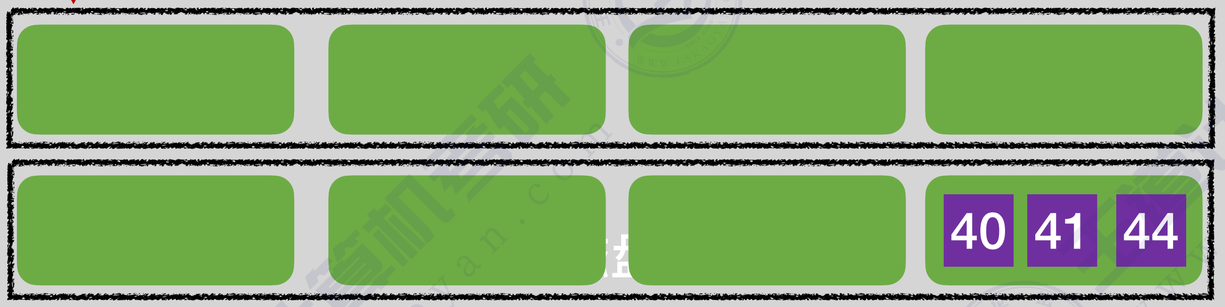
将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并



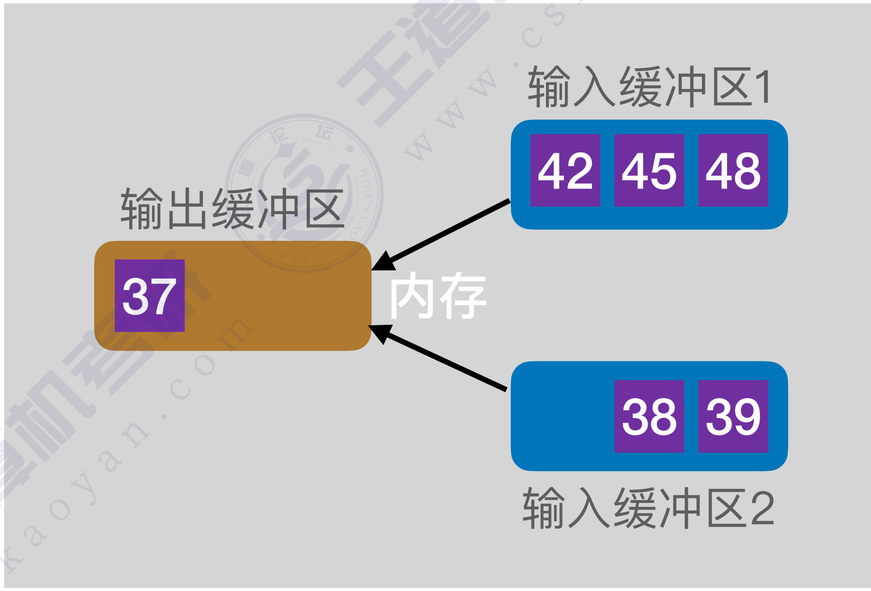
归并段1



归并段2

读磁盘

写磁盘



将两个有序归并段归并为一个

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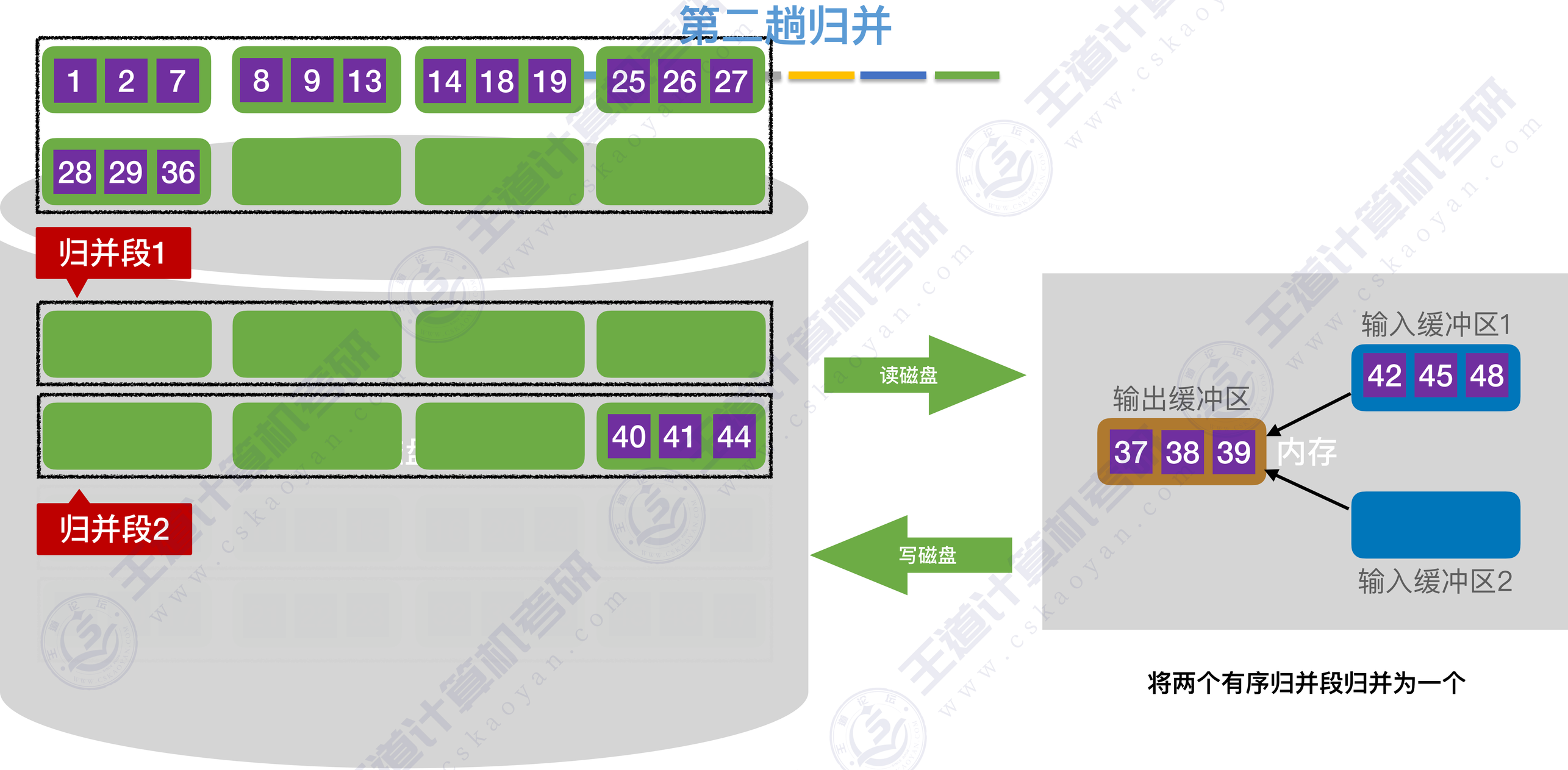
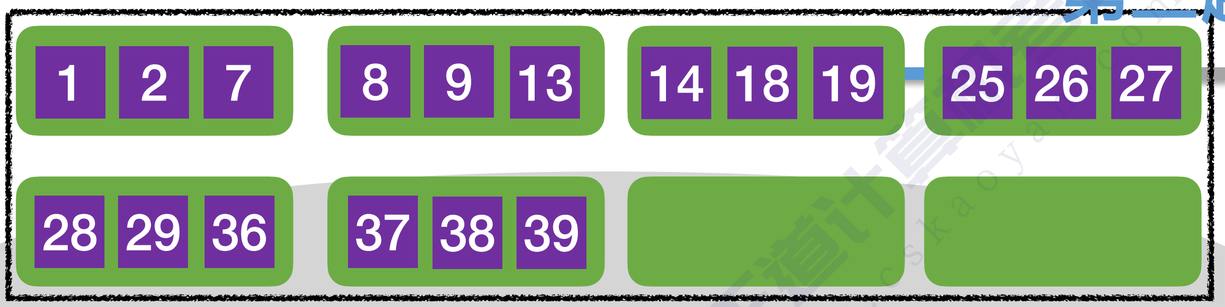
第二趟归并



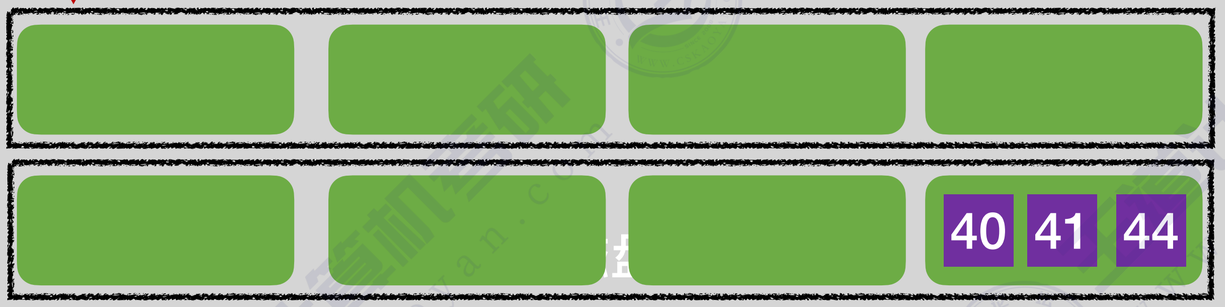
将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并



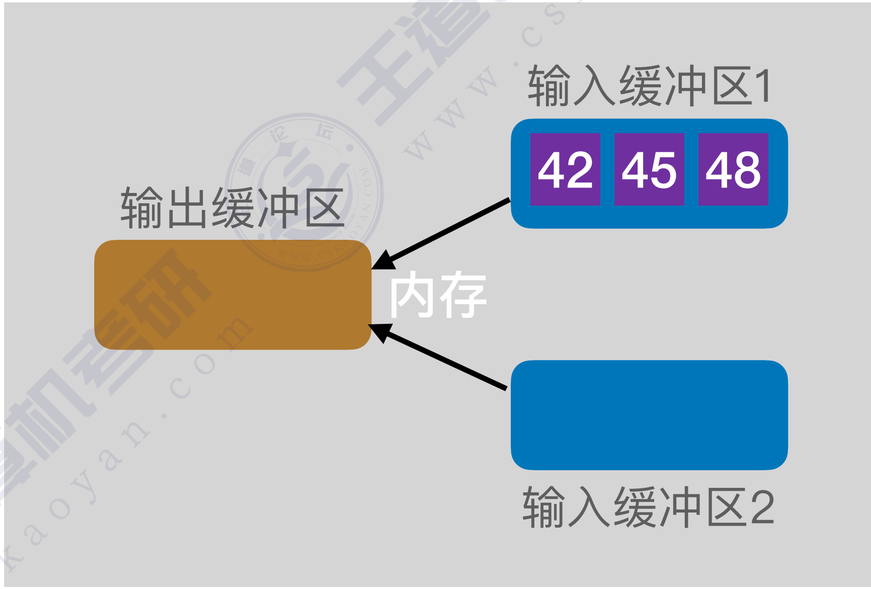
归并段1



归并段2

读磁盘

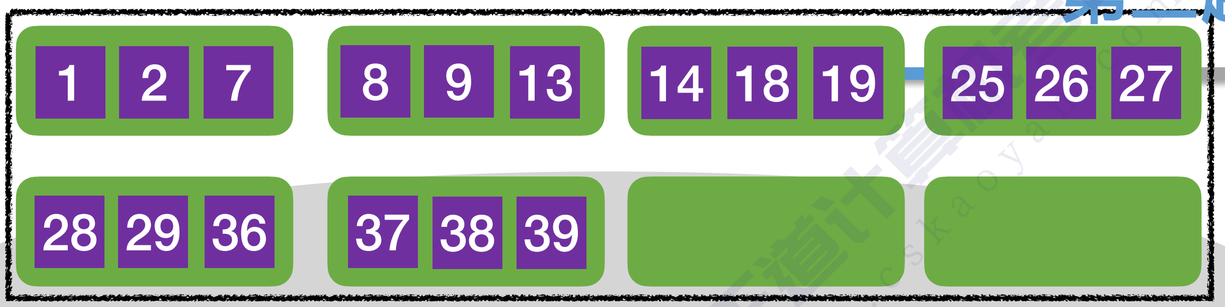
写磁盘



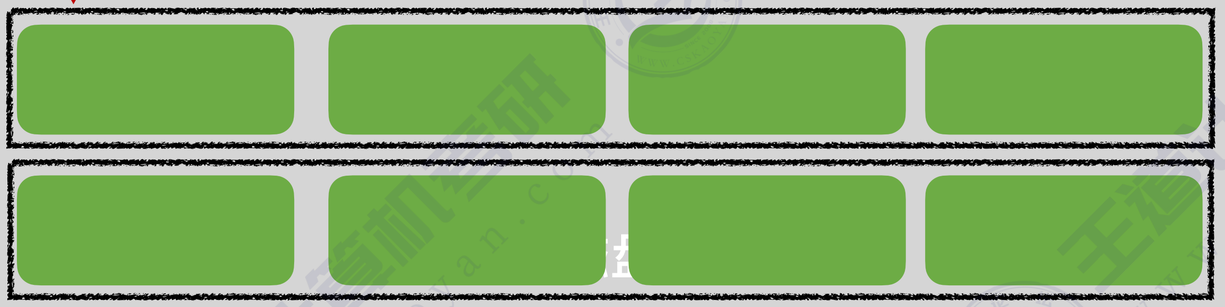
将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并



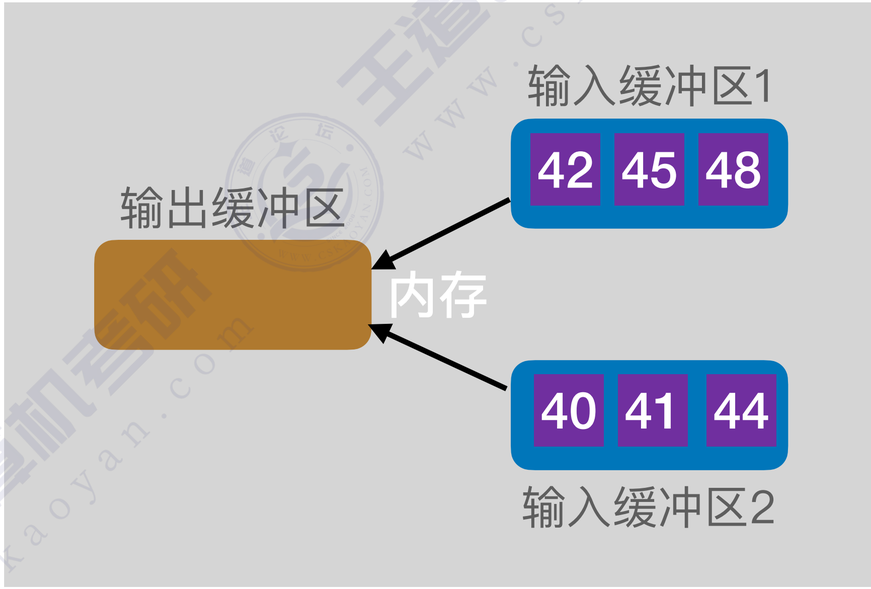
归并段1



归并段2

读磁盘

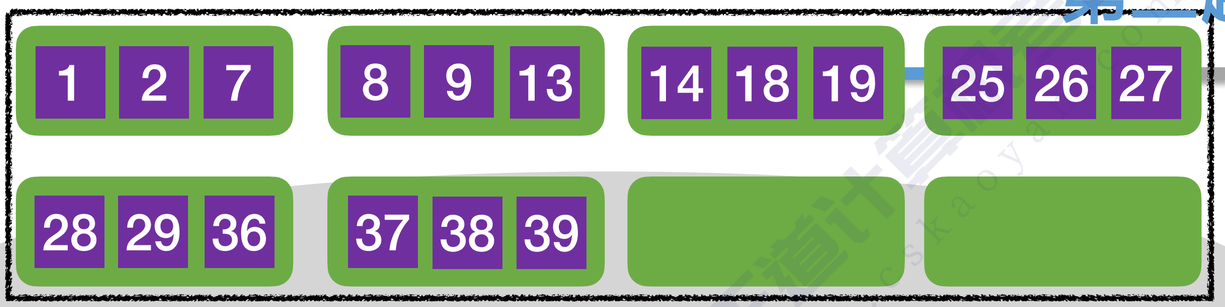
写磁盘



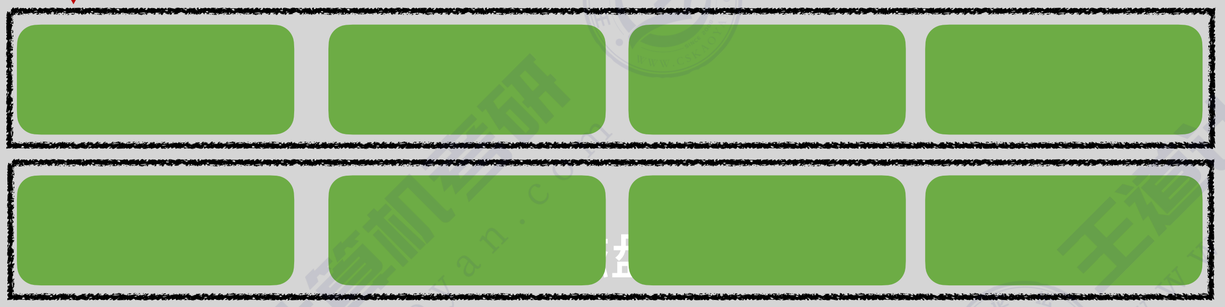
将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并



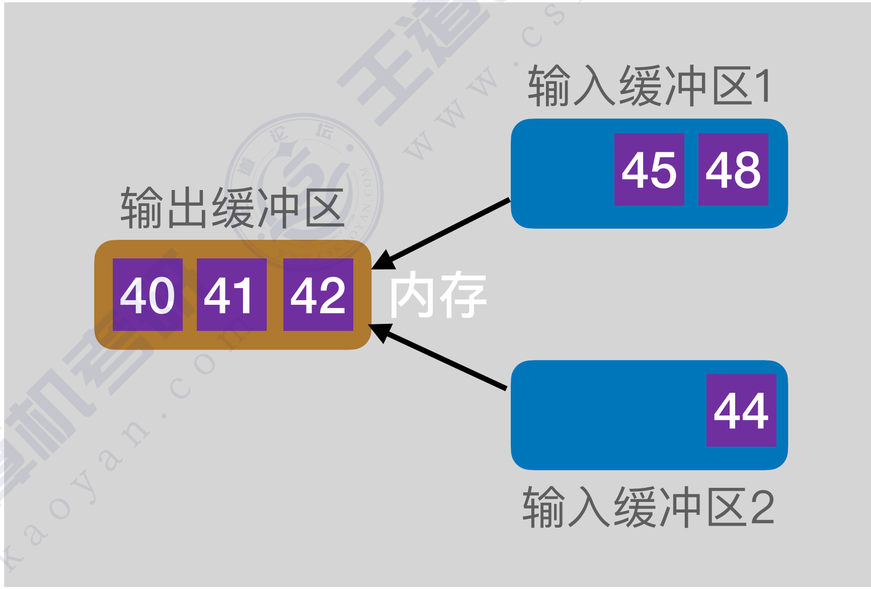
归并段1



归并段2

读磁盘

写磁盘



将两个有序归并段归并为一个

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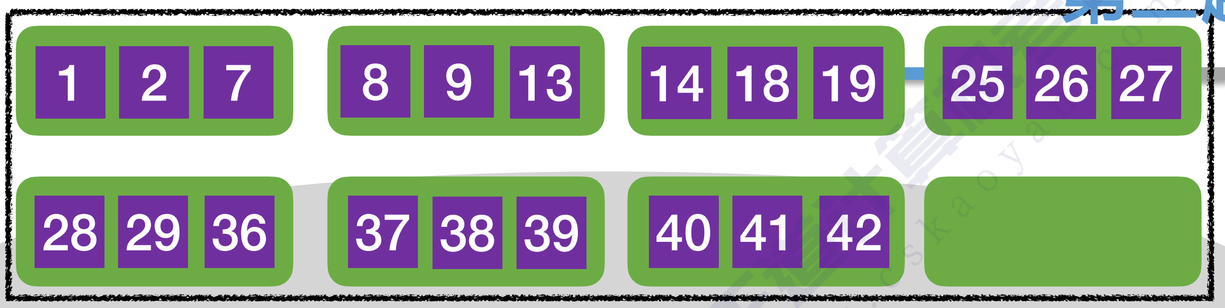
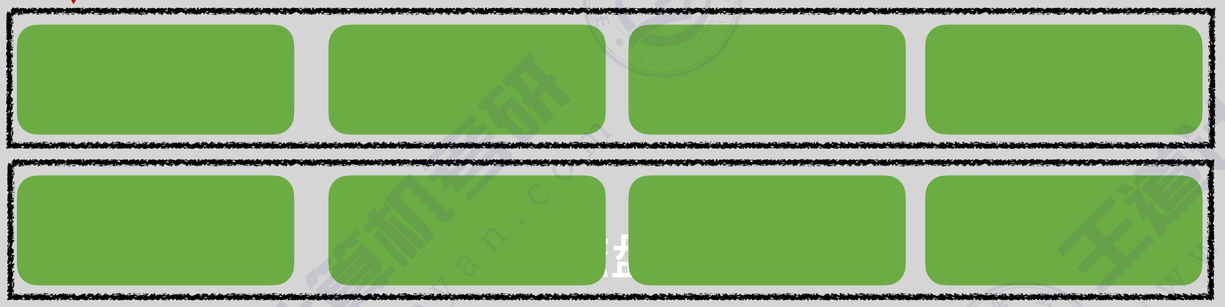
第二趟归并



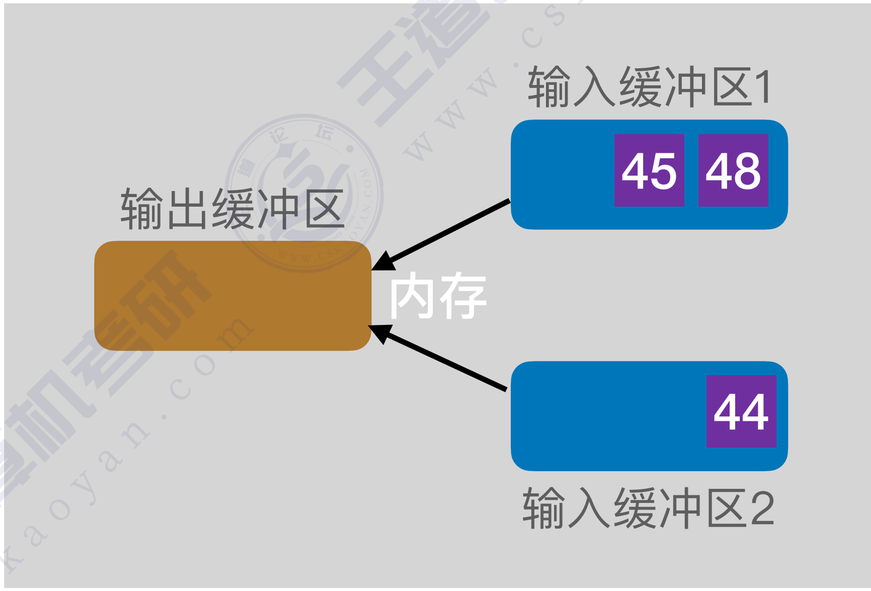
归并段1



归并段2

读磁盘

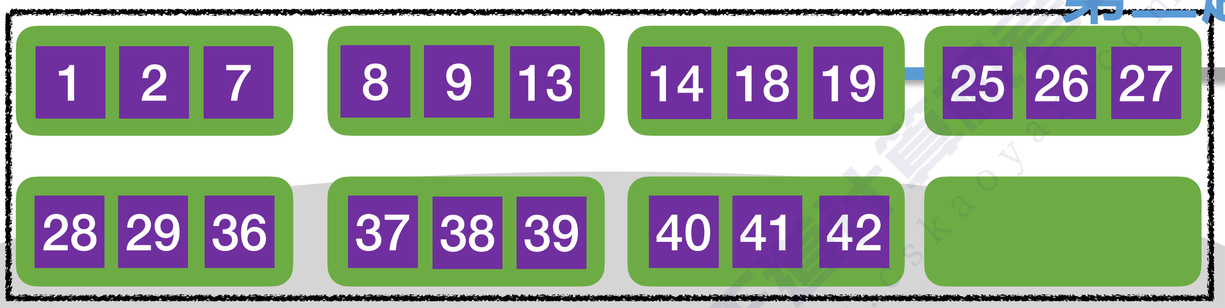
写磁盘



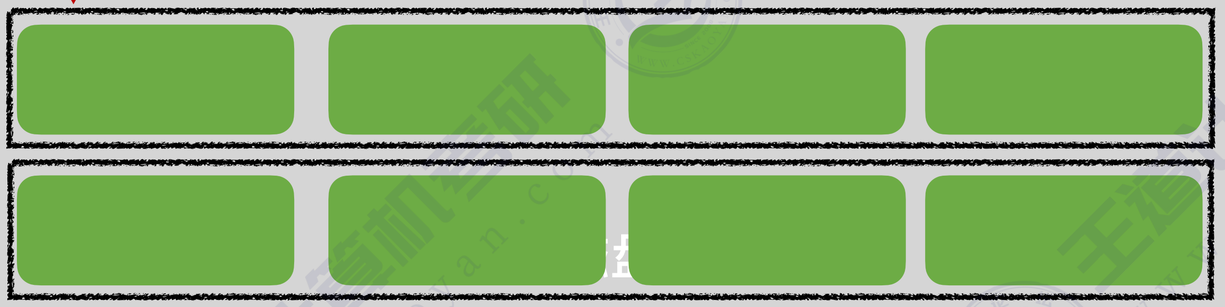
将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并



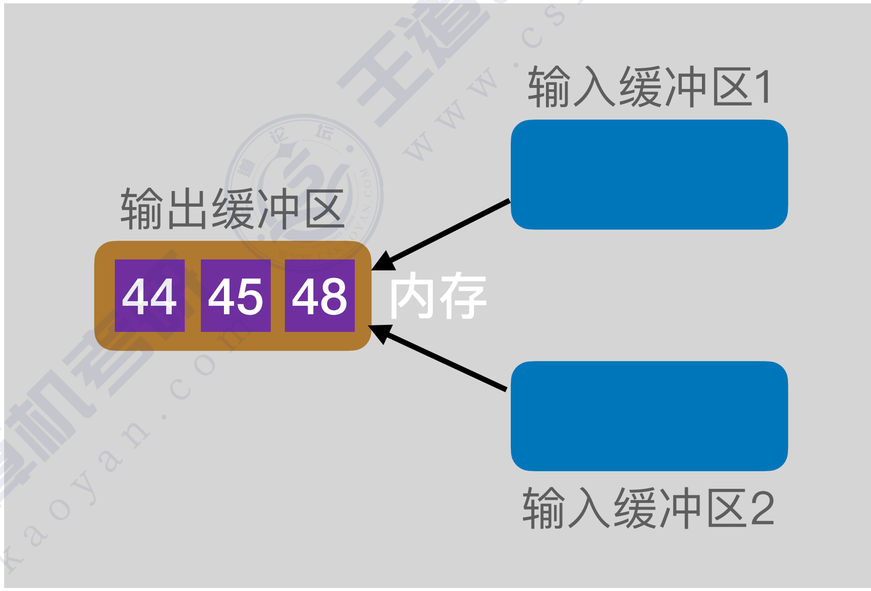
归并段1



归并段2

读磁盘

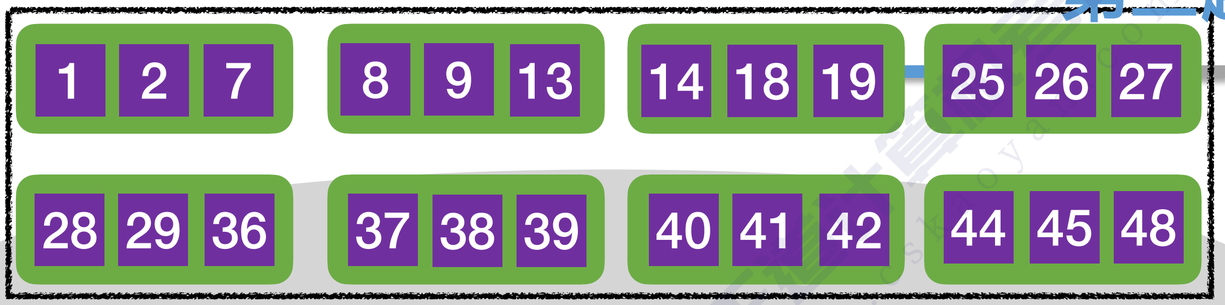
写磁盘



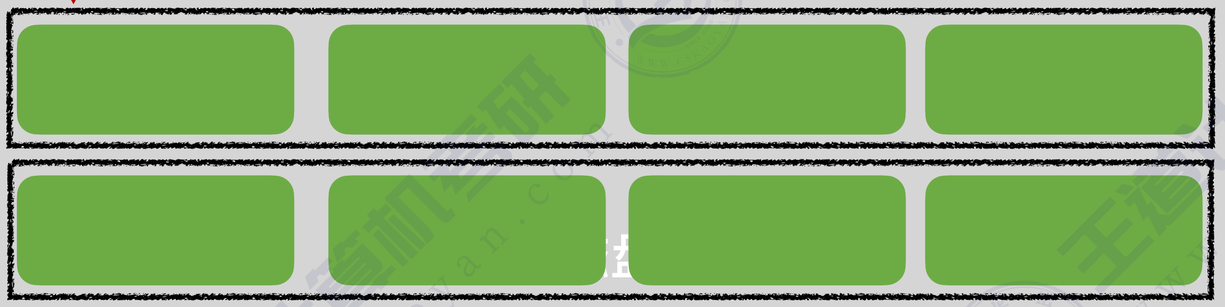
将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并



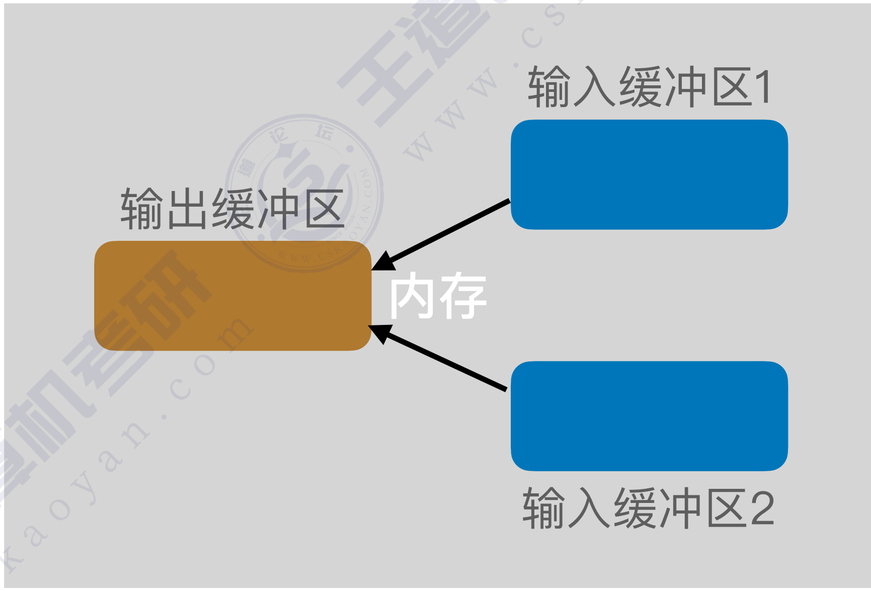
归并段1



归并段2

读磁盘

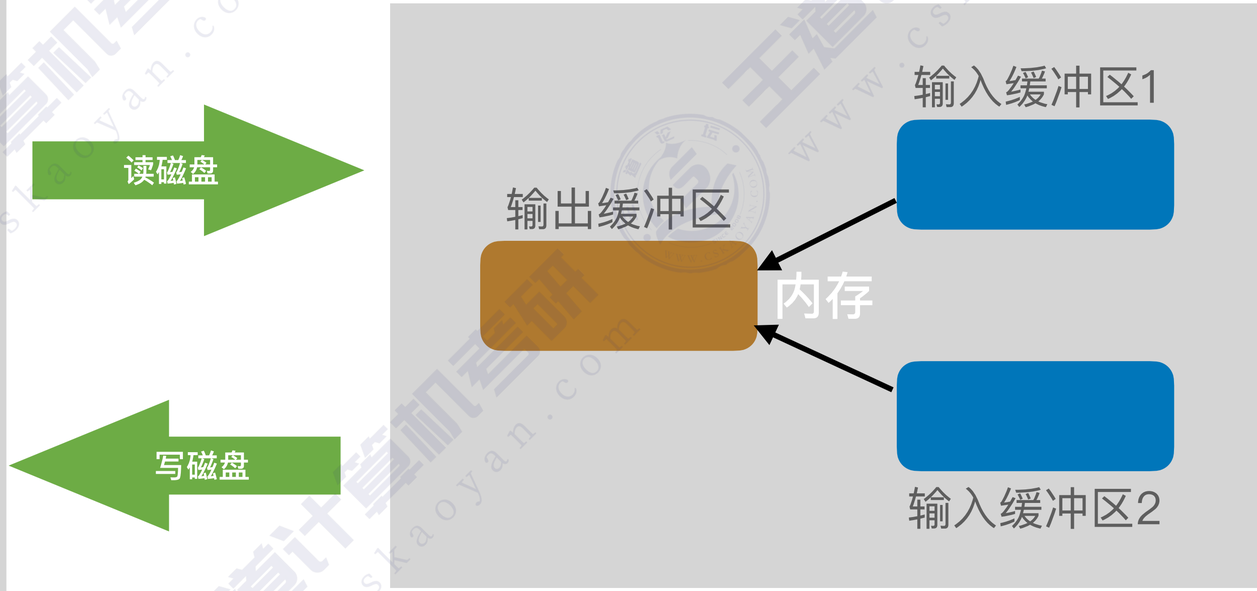
写磁盘



将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

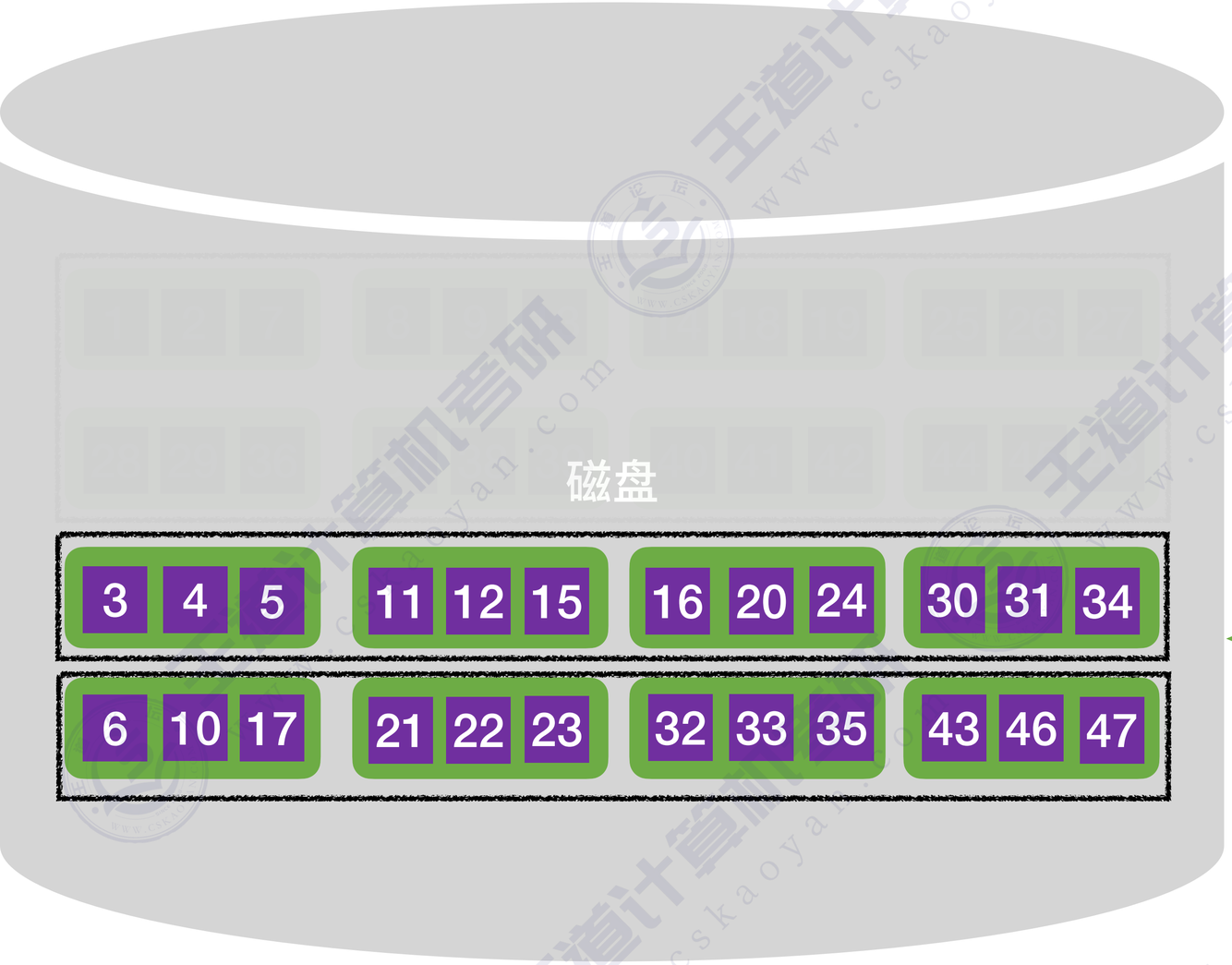
第二趟归并

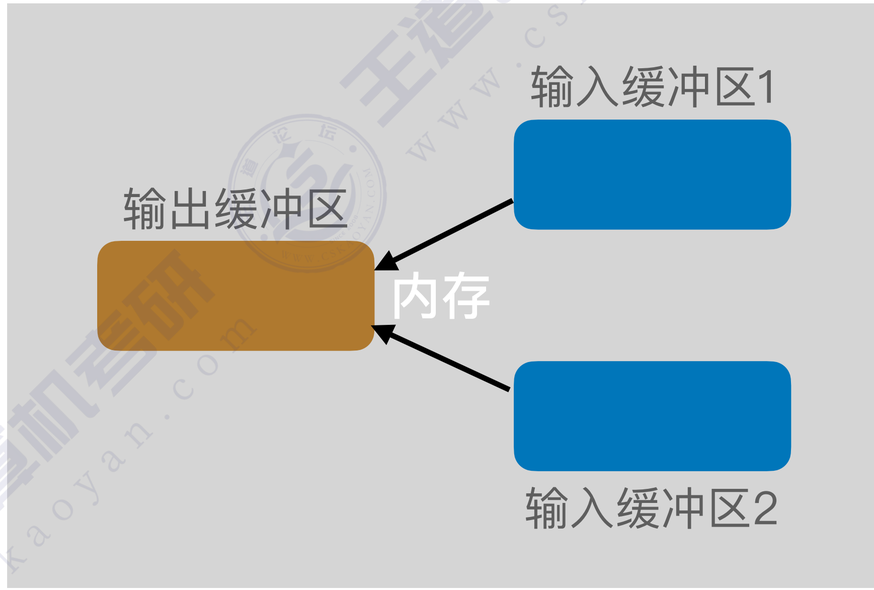


将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并

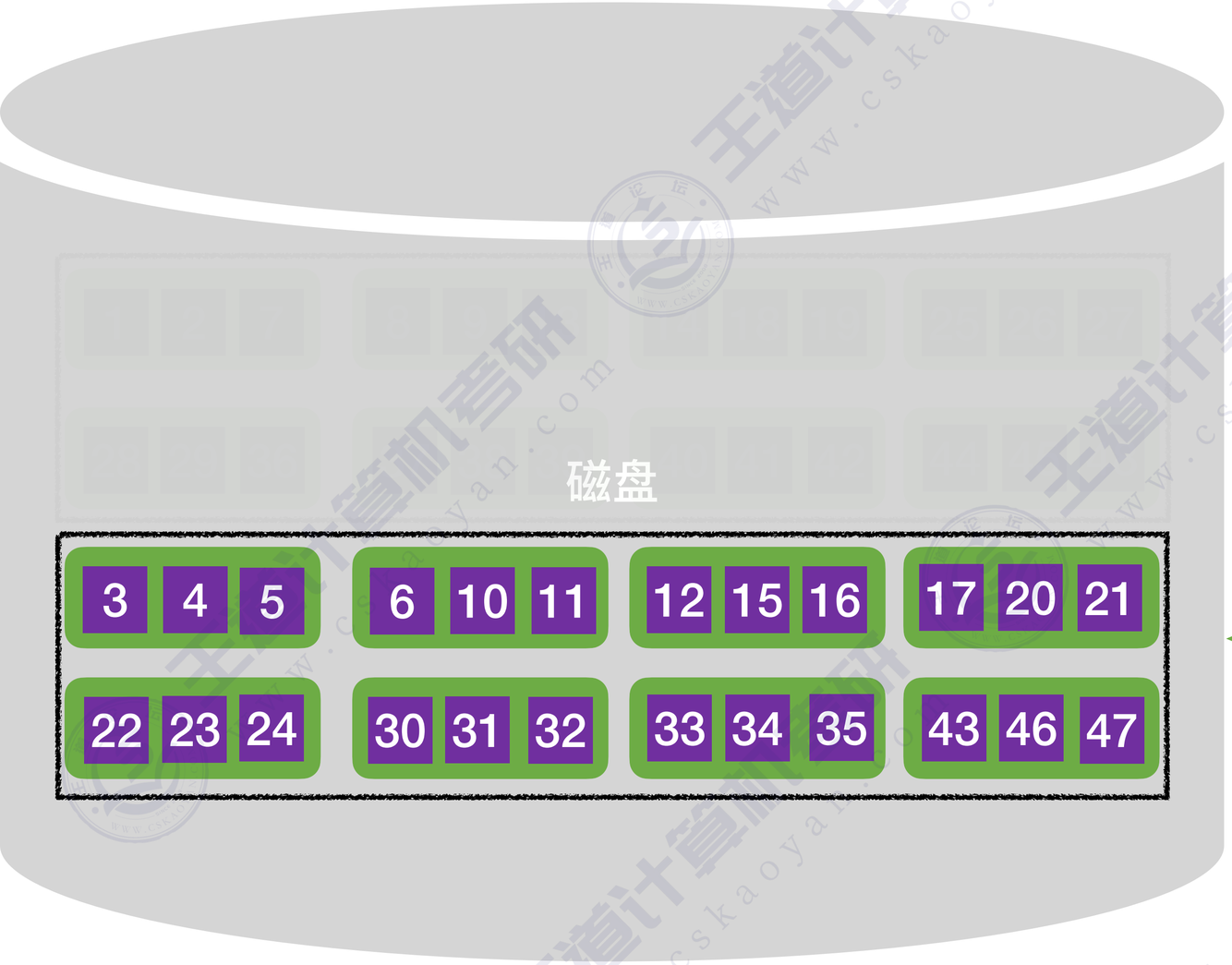




将两个有序归并段归并为一个

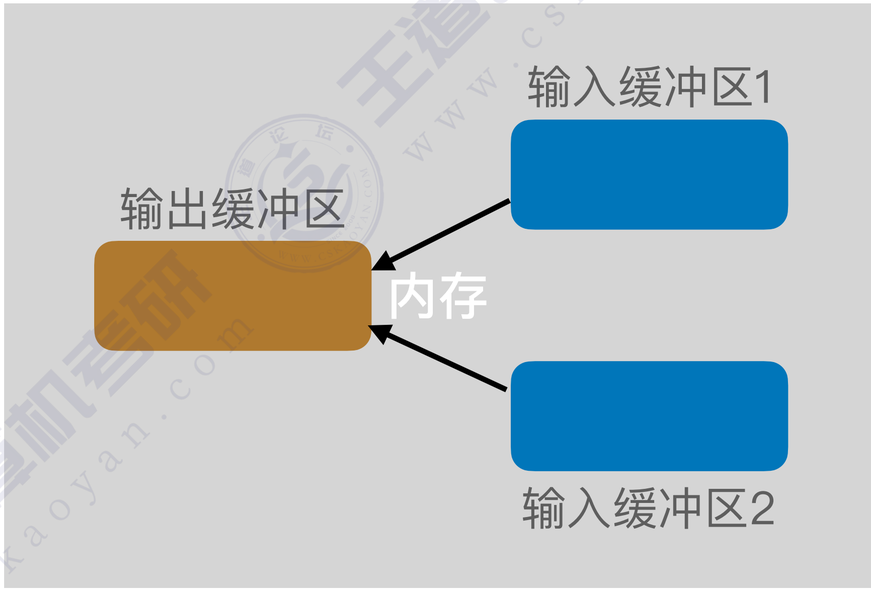
王道考研/CSKAOYAN.COM

第二趟归并



读磁盘

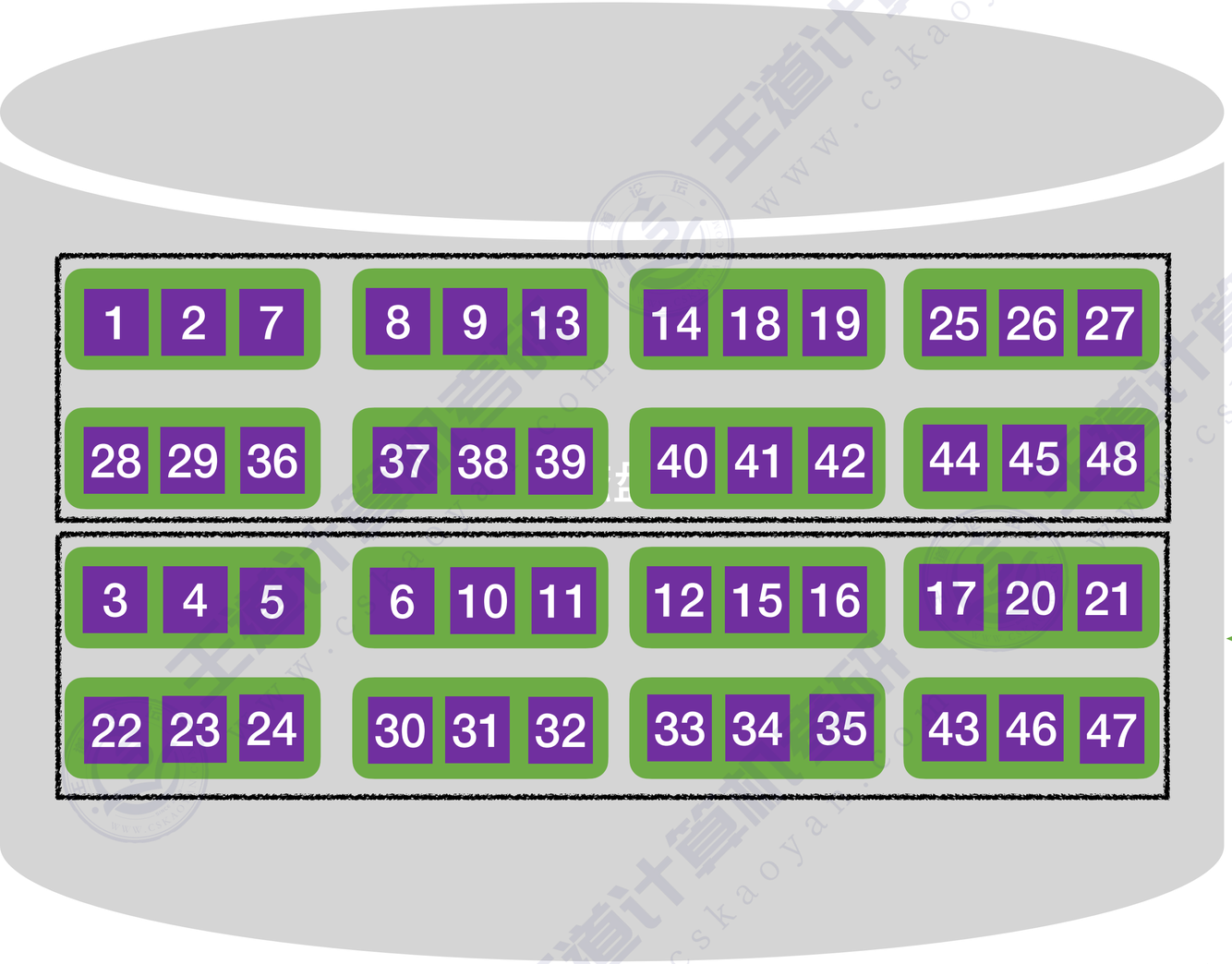
写磁盘

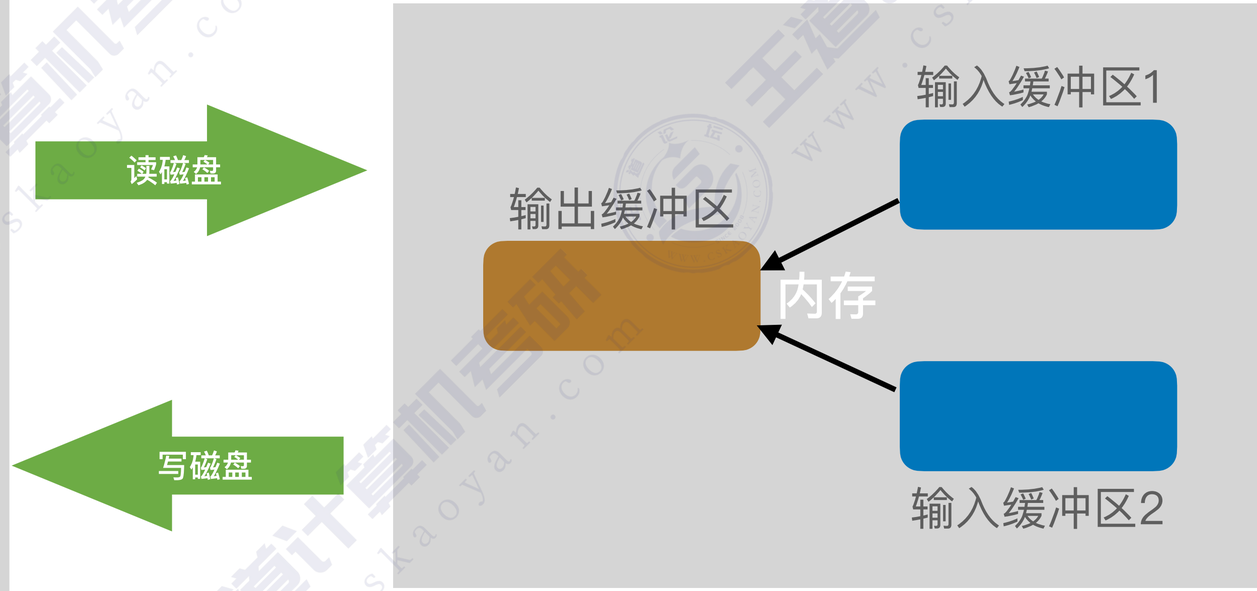


将两个有序归并段归并为一个

王道考研/CSKAOYAN.COM

第二趟归并





将两个有序归并段归并为一个

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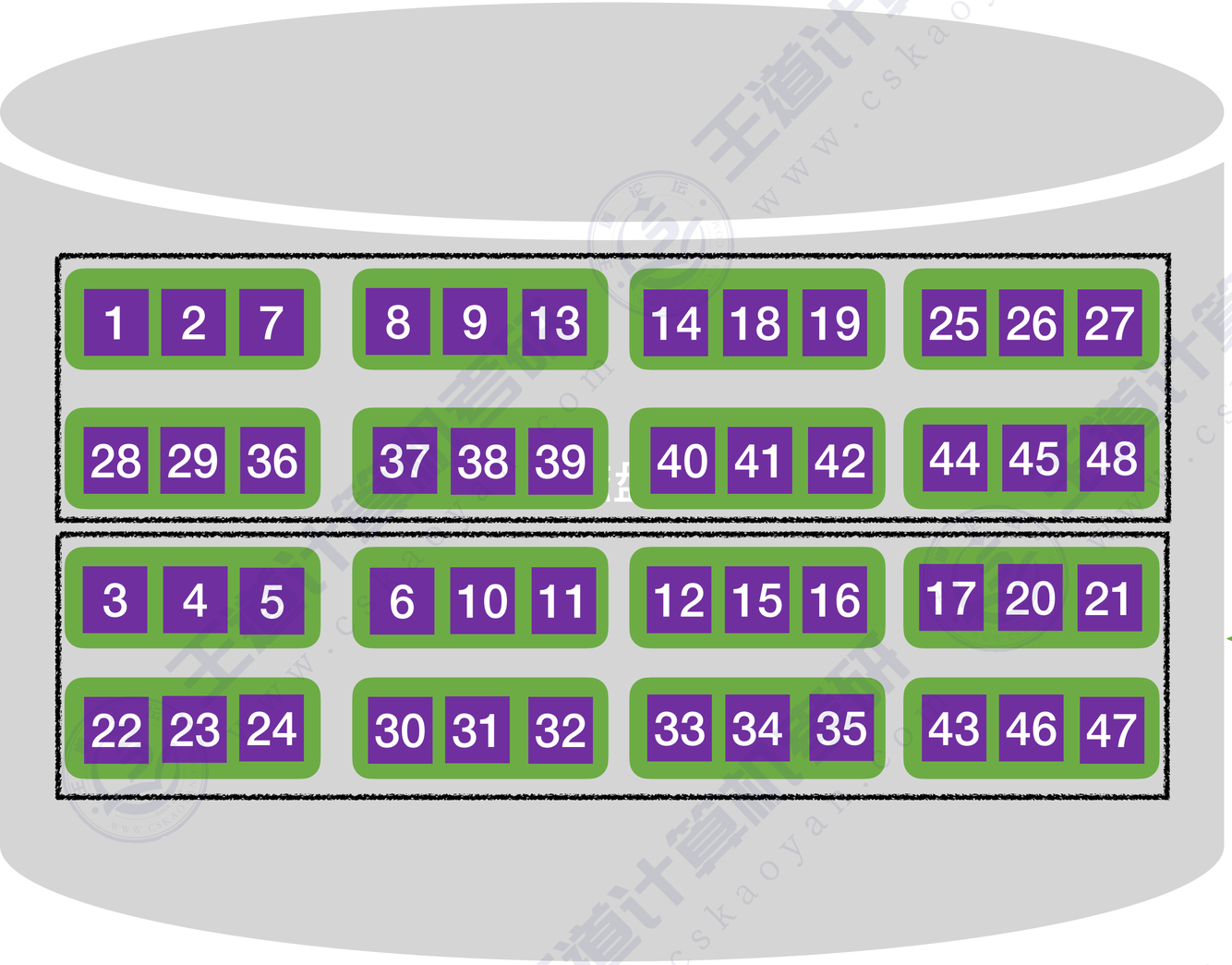
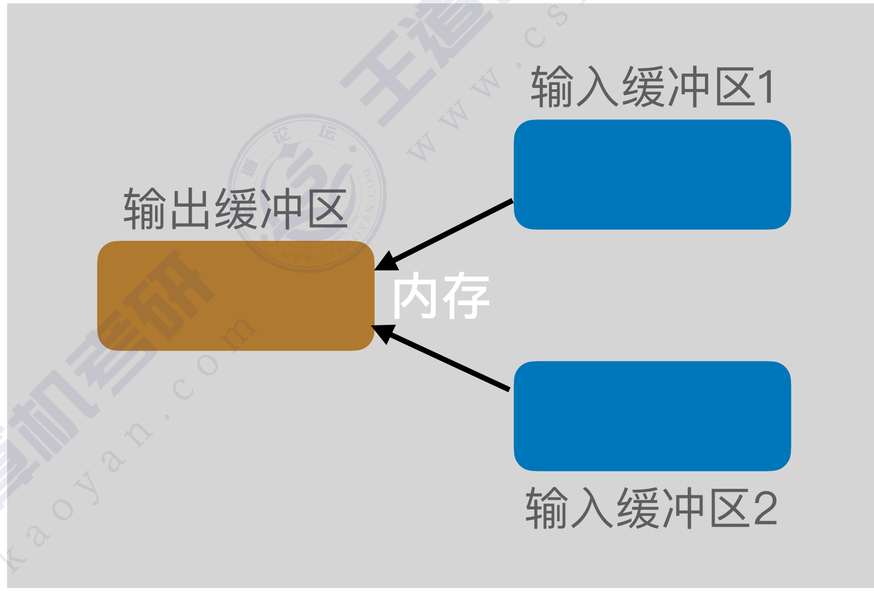
第三趟归并

把2个有序子序列（归并段）归并





将两个有序归并段归并为一个

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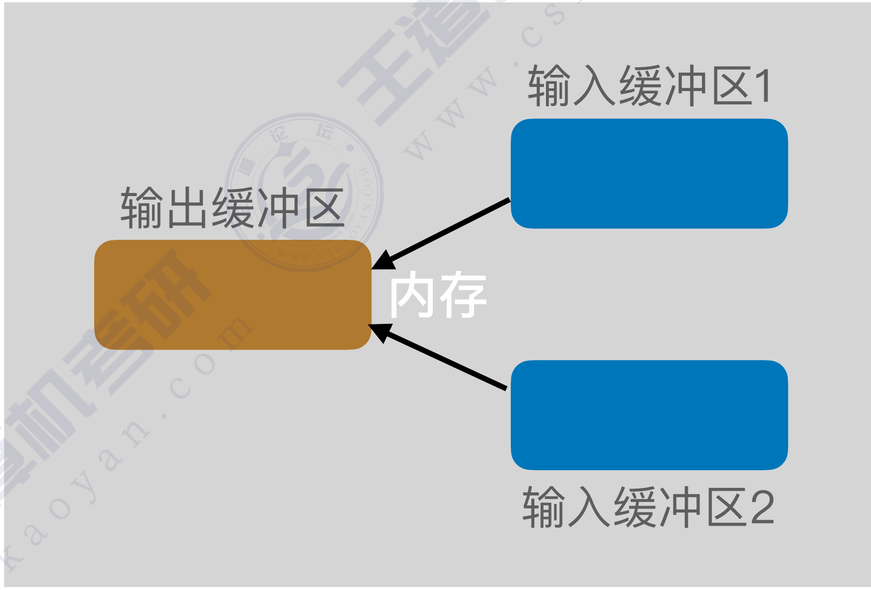
第三趟归并

经过3趟归并，整体有序

1 2 3	4 5 6	7 8 9	10 11 12
13 14 15	16 17 18	19 20 21	22 23 24
25 26 27	28 29 30	31 32 33	34 35 36
37 38 39	40 41 42	43 44 45	46 47 48

读磁盘

写磁盘



将两个有序归并段归并为一个

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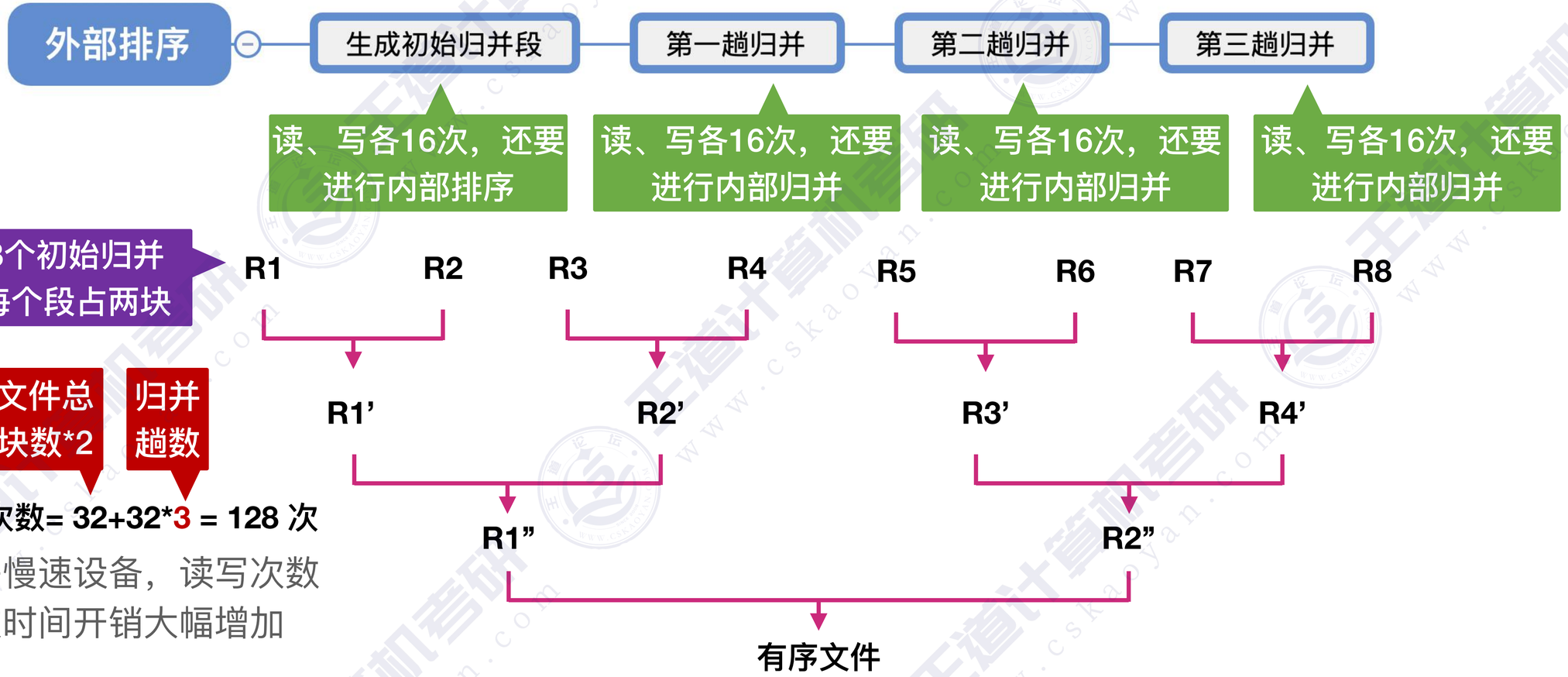
时间开销分析



文件总块数*2

归并趟数

读、写磁盘次数= 32+32*3 = 128 次

注：磁盘是慢速设备，读写次数太多导致时间开销大幅增加

外部排序时间开销=读写外存的时间+内部排序所需时间+内部归并所需时间

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如何优化?

外部排序时间开销=读写外存的时间+内部排序所需时间+内部归并所需时间

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优化：多路归并

将4个有序归并段归并为一个

王道考研/CSKAOYAN.COM

优化：多路归并

读磁盘

写磁盘

将4个有序归并段归并为一个

王道考研/CSKAOYAN.COM

优化：多路归并

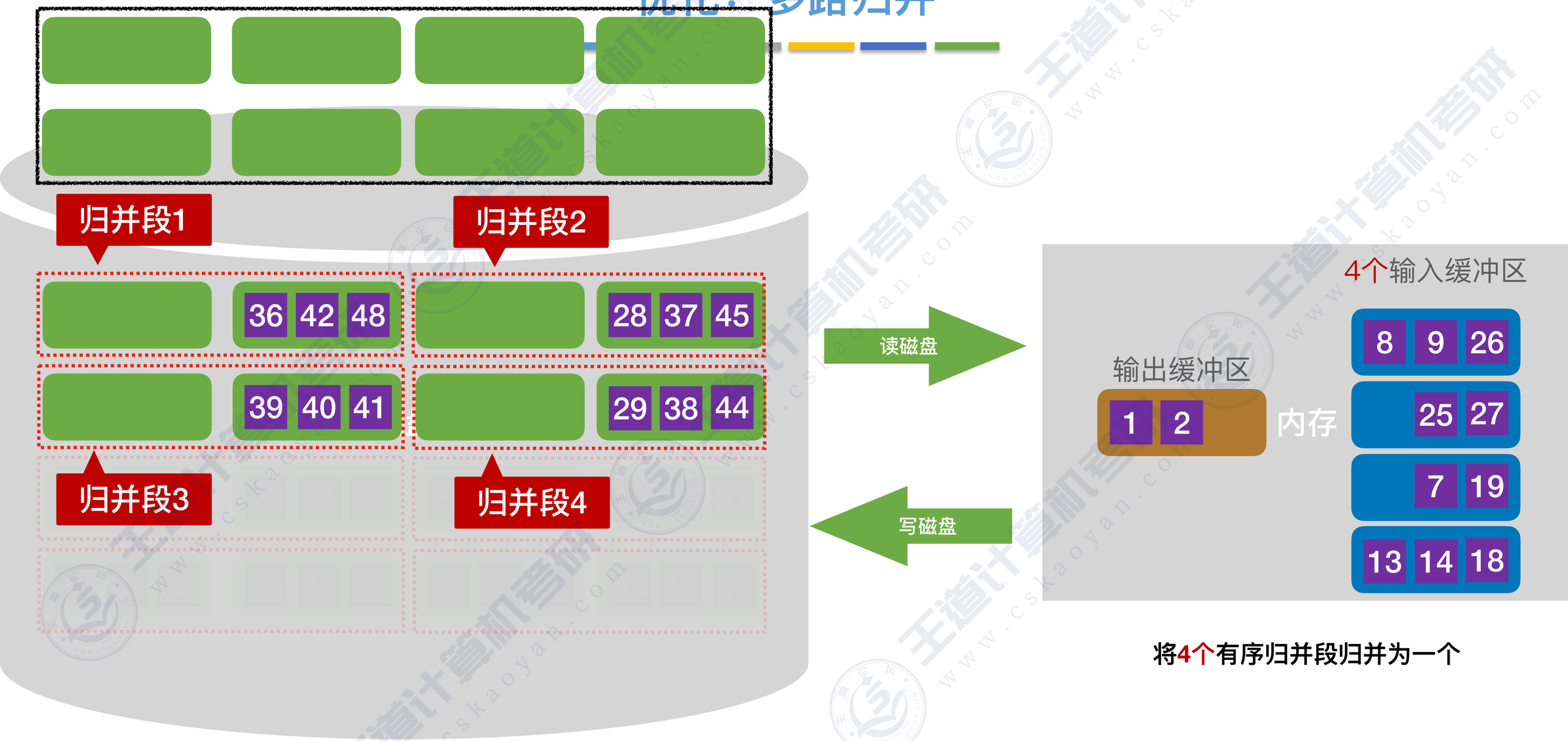
读磁盘

写磁盘

将4个有序归并段归并为一个

王道考研/CSKAOYAN.COM

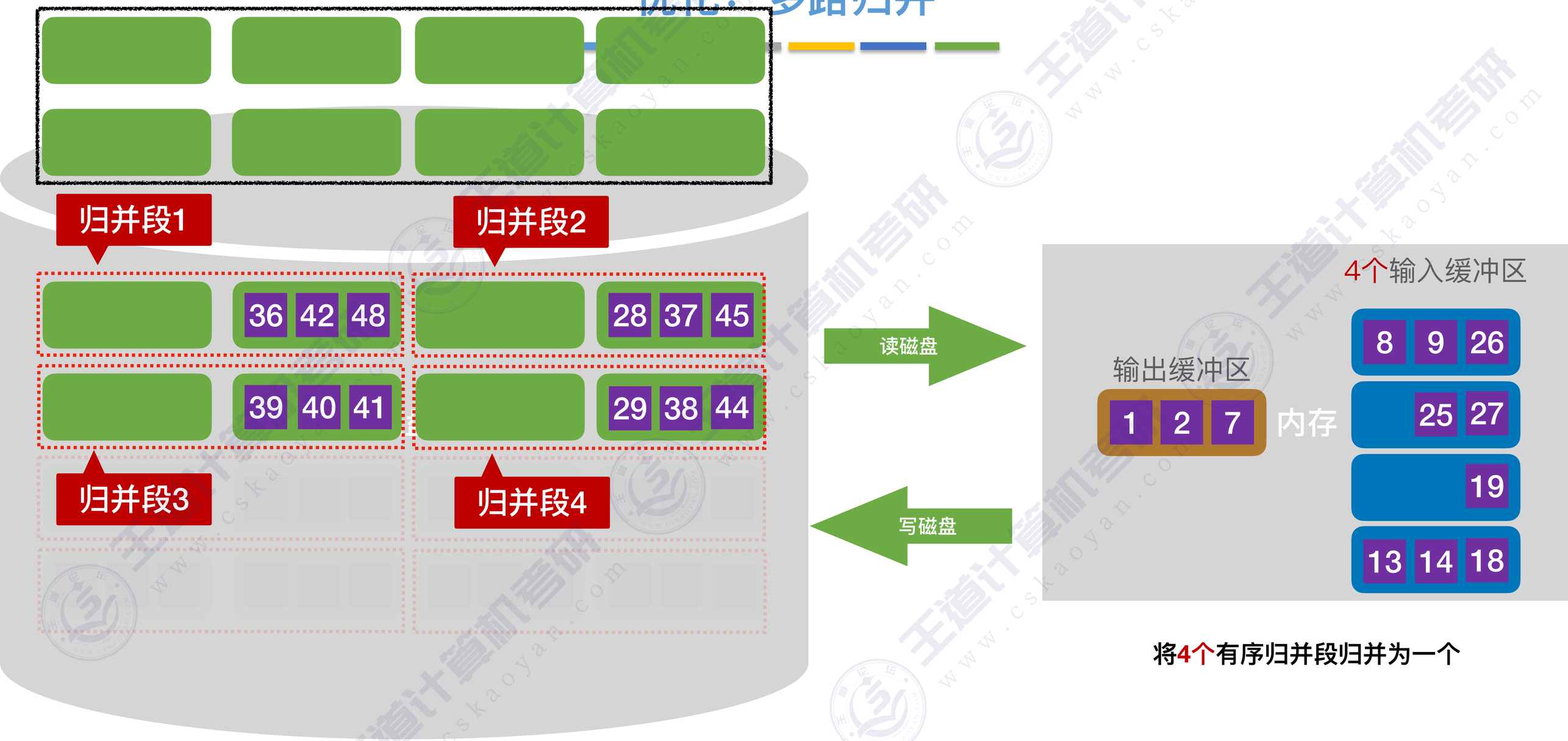
优化：多路归并



将4个有序归并段归并为一个

王道考研/CSKAOYAN.COM

优化：多路归并



将4个有序归并段归并为一个

王道考研/CSKAOYAN.COM

优化：多路归并

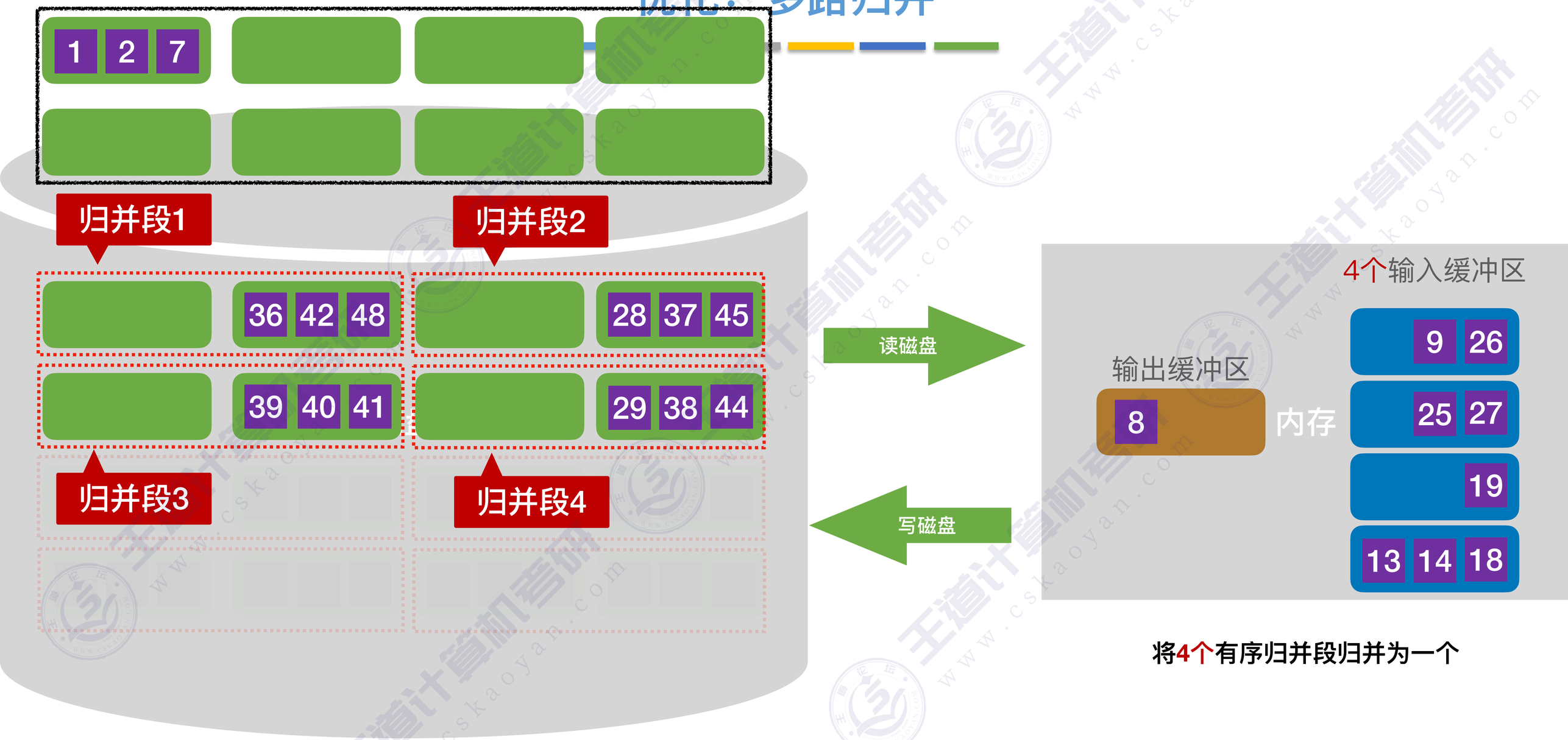
读磁盘

写磁盘

将4个有序归并段归并为一个

王道考研/CSKAOYAN.COM

优化：多路归并



将4个有序归并段归并为一个

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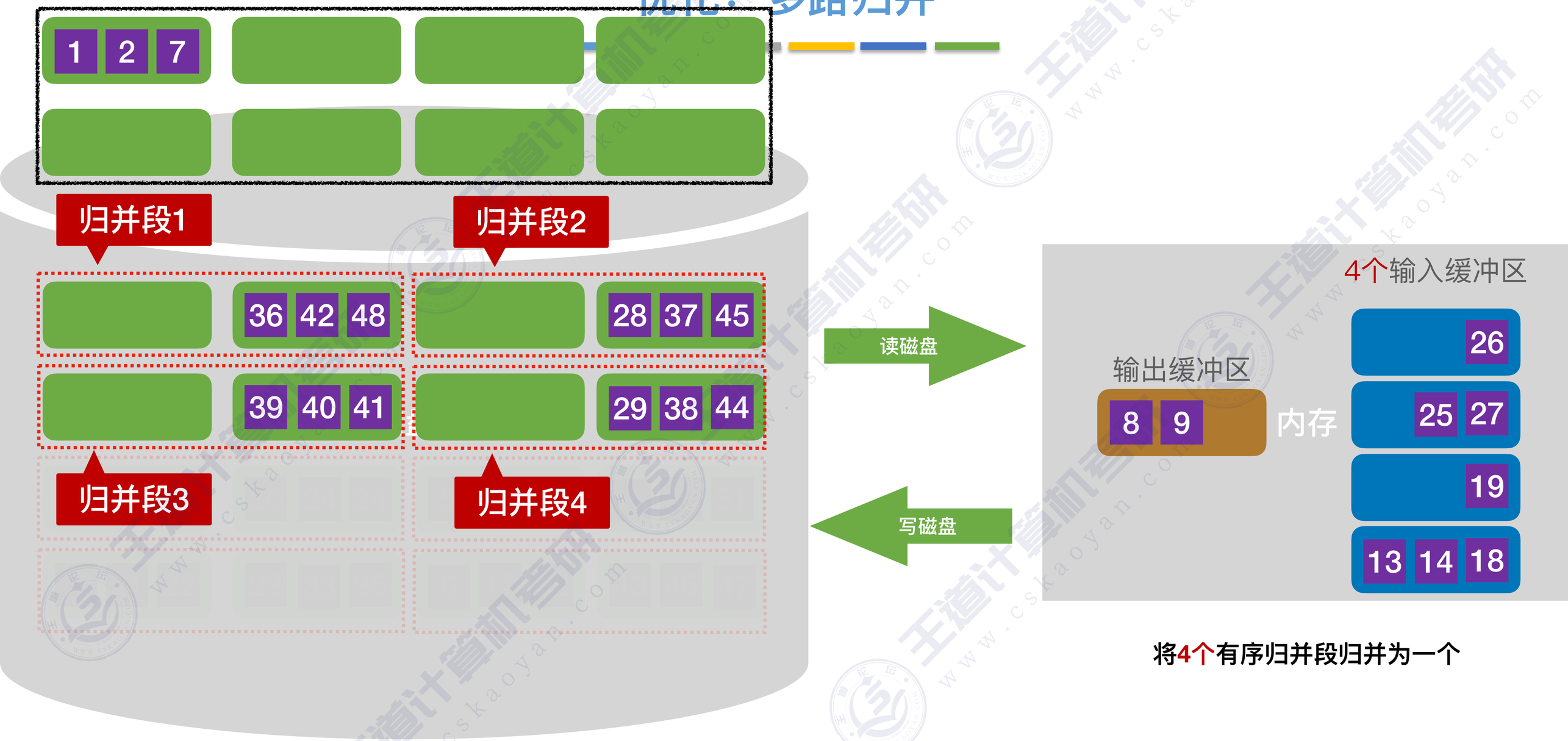
优化：多路归并



将4个有序归并段归并为一个

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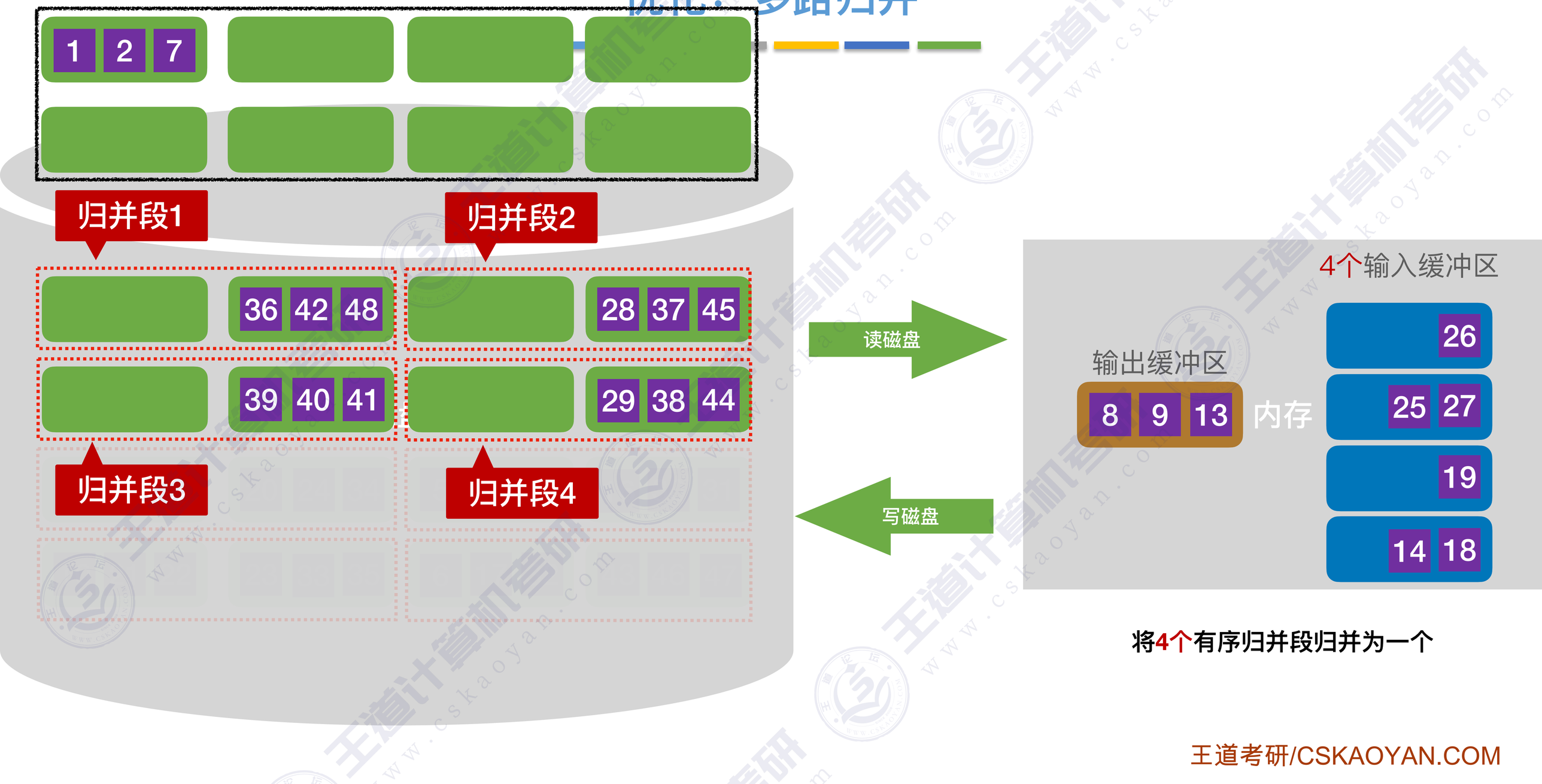
优化：多路归并



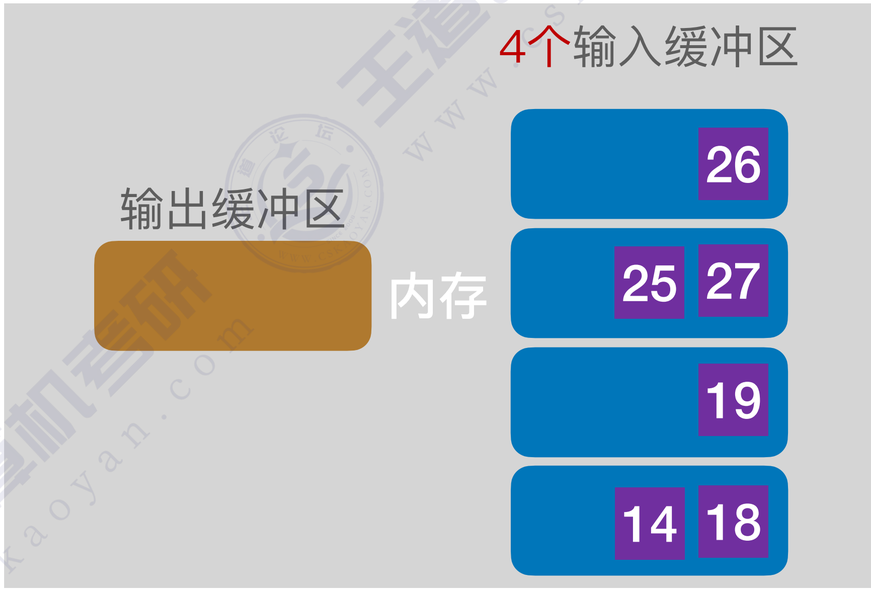
将4个有序归并段归并为一个

王道考研/CSKAOYAN.COM

优化：多路归并

读磁盘

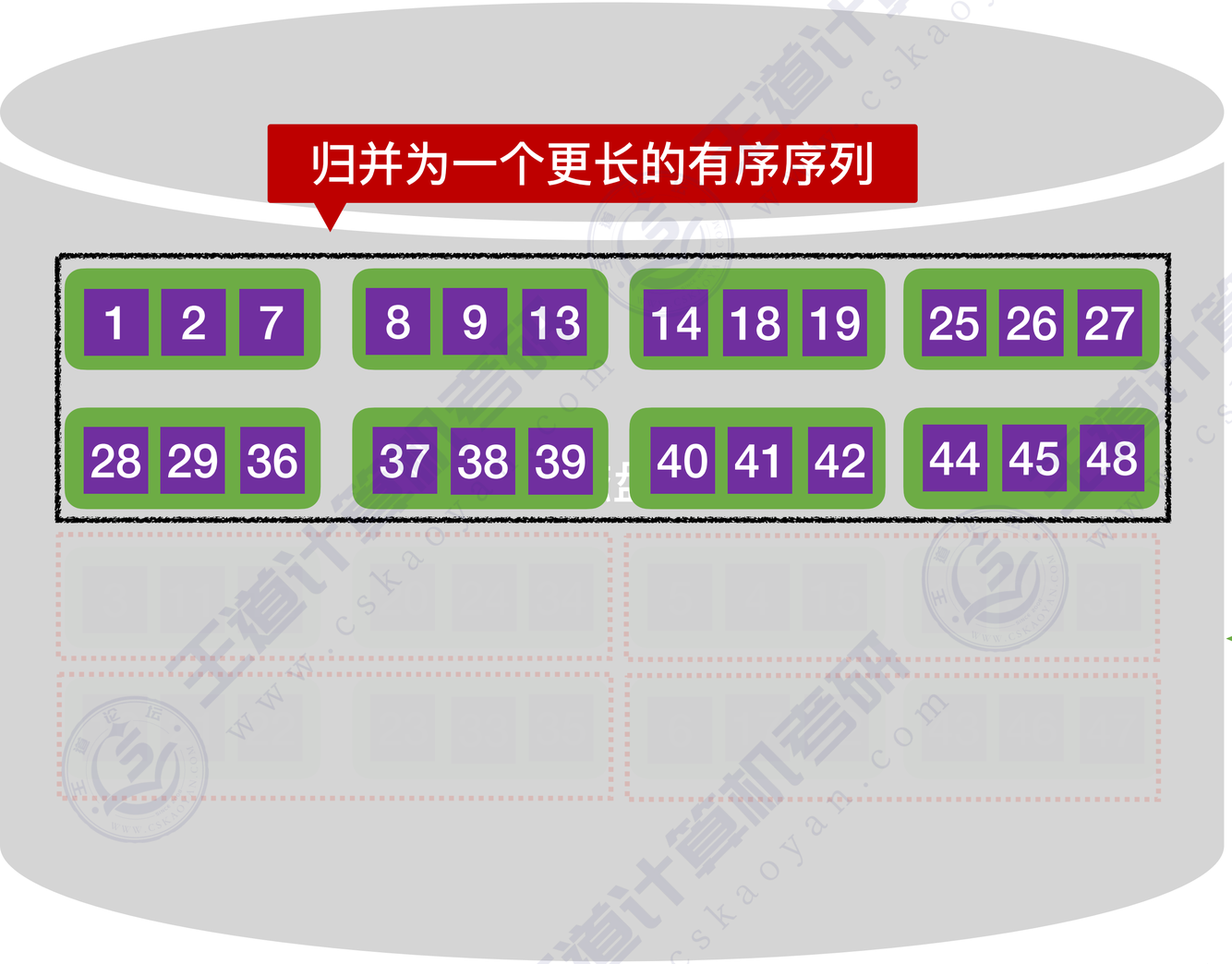
写磁盘



将4个有序归并段归并为一个

王道考研/CSKAOYAN.COM

优化：多路归并



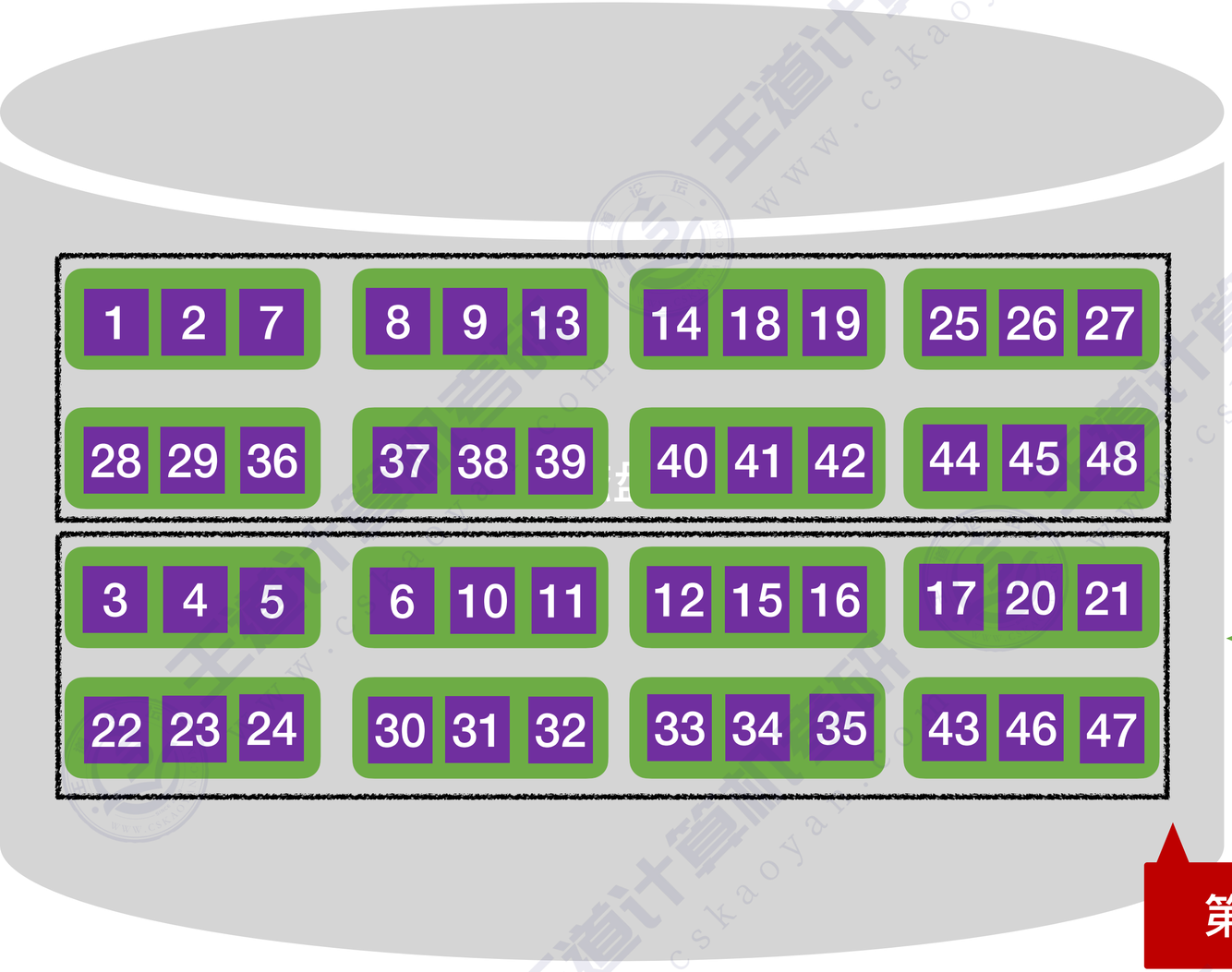
读磁盘

写磁盘

将4个有序归并段归并为一个

王道考研/CSKAOYAN.COM

优化：多路归并



读磁盘

写磁盘

第一趟归并之后

将4个有序归并段归并为一个

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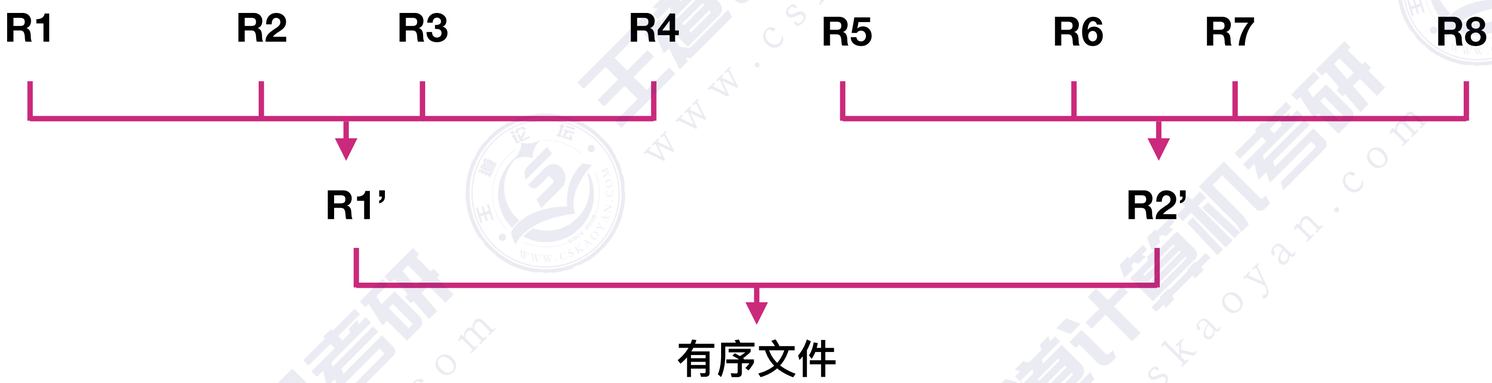
优化：多路归并

外部排序时间开销=读写外存的时间+内部排序所需时间+内部归并所需时间



采用4路归并，只需进行两趟归并即可

读、写磁盘次数= 32+32*2 = 96 次

重要结论：采用多路归并可以减少归并趟数，从而减少磁盘I/O(读写)次数

对 r 个初始归并段，做k路归并，则归并树可用 k 叉树表示

若树高为h，则归并趟数 = h-1 = $\lceil log_k r \rceil$

k越大，r越小，归并趟数越少，读写磁盘次数越少

推导：k叉树第h层最多有k^{h-1} 个结点

则 $r \le k^{h-1}$，$(h-1)_{最小} = \lceil log_k r \rceil$

多路归并带来的负面影响：

①k路归并时，需要开辟k个输入缓冲区，内存开销增加。

②每挑选一个关键字需要对比关键字（k-1）次，内部归并所需时间增加

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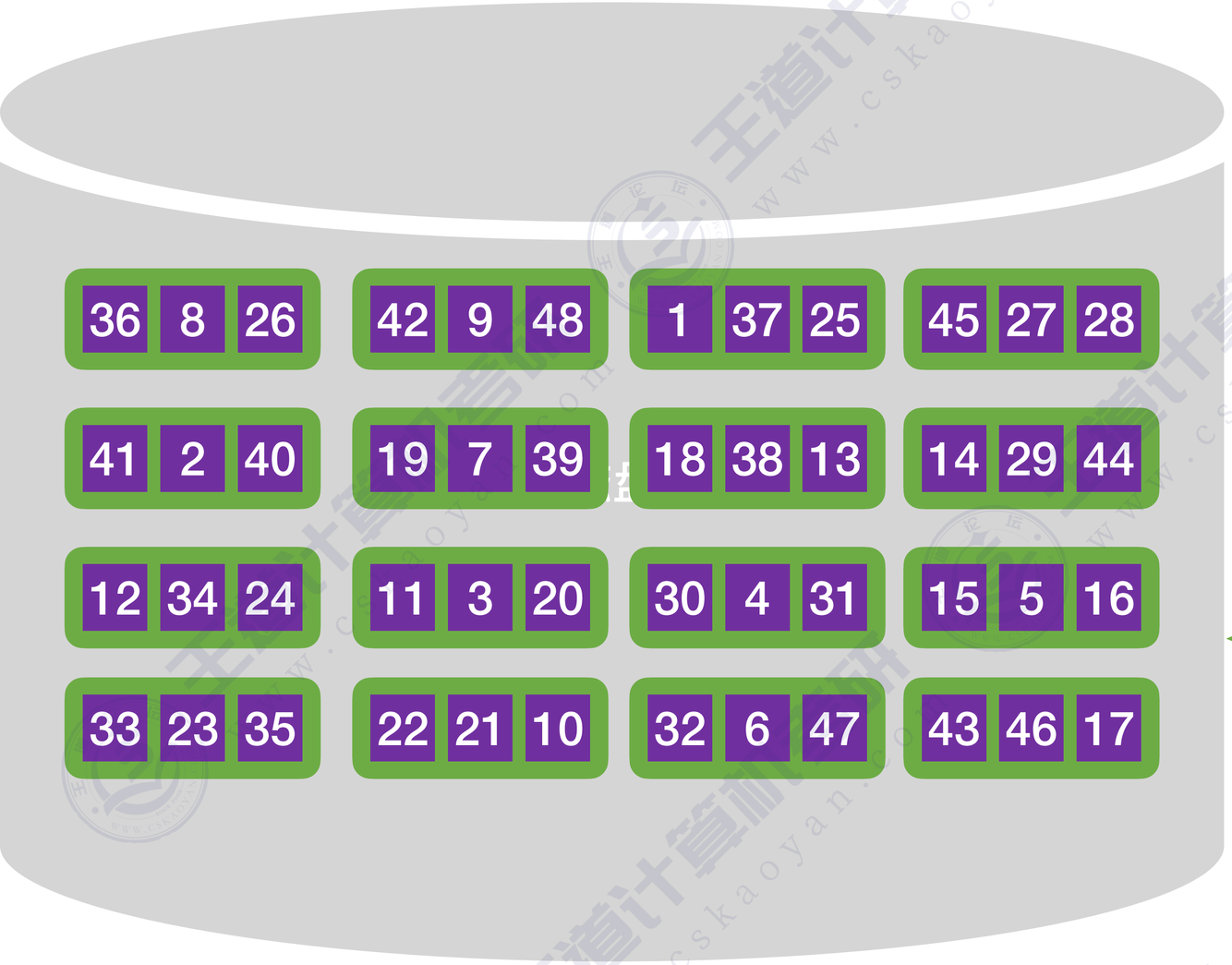
优化：减少初始归并段数量



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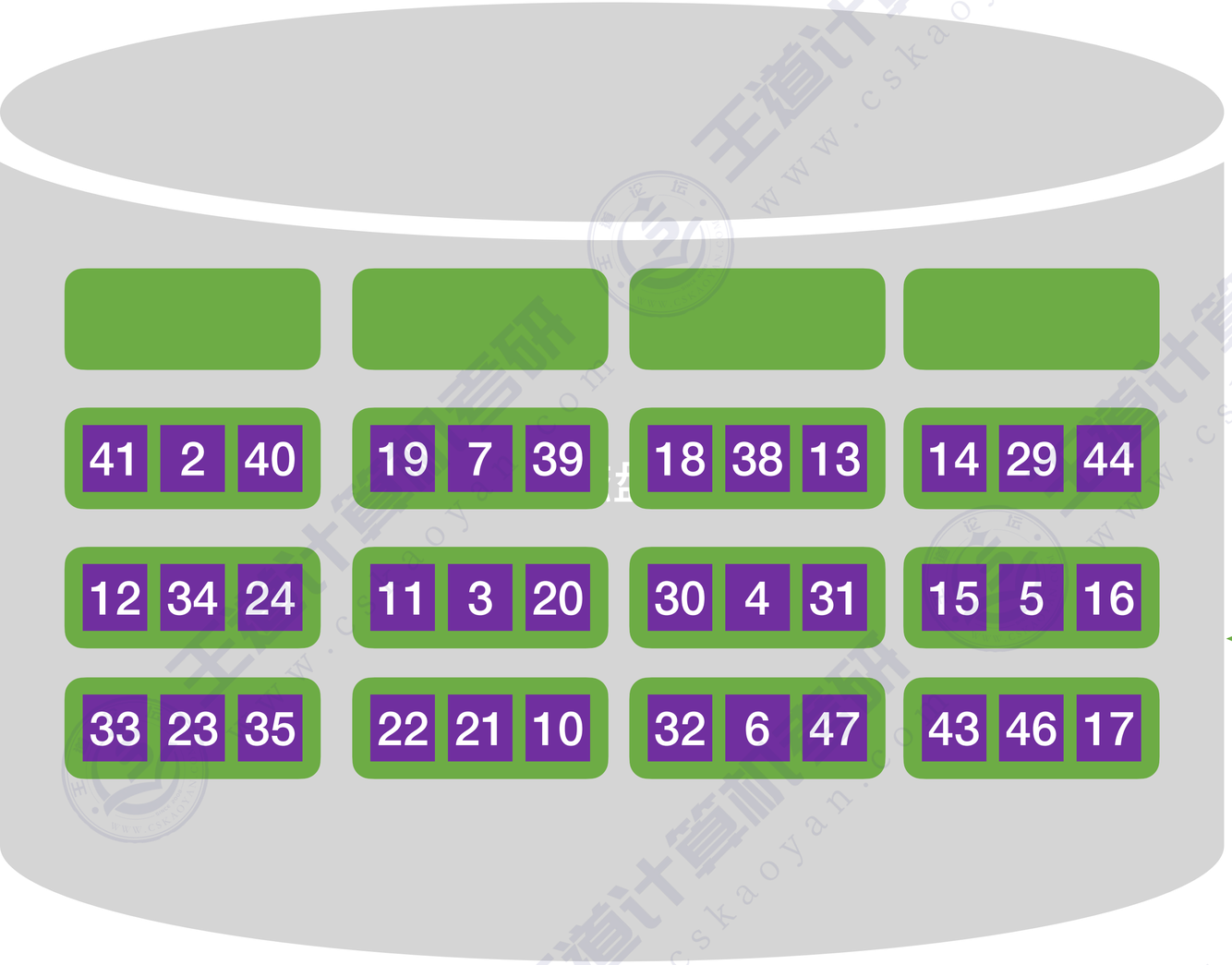
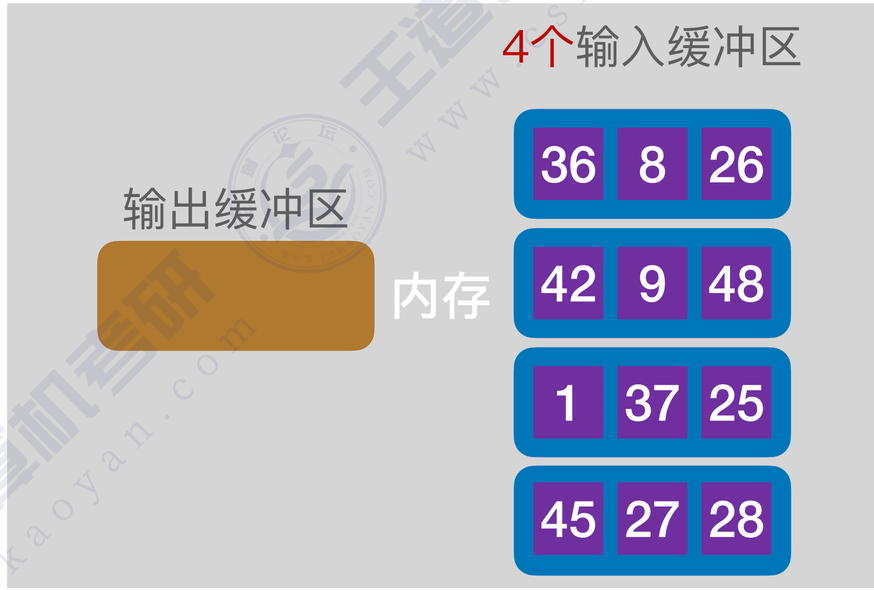
优化：减少初始归并段数量





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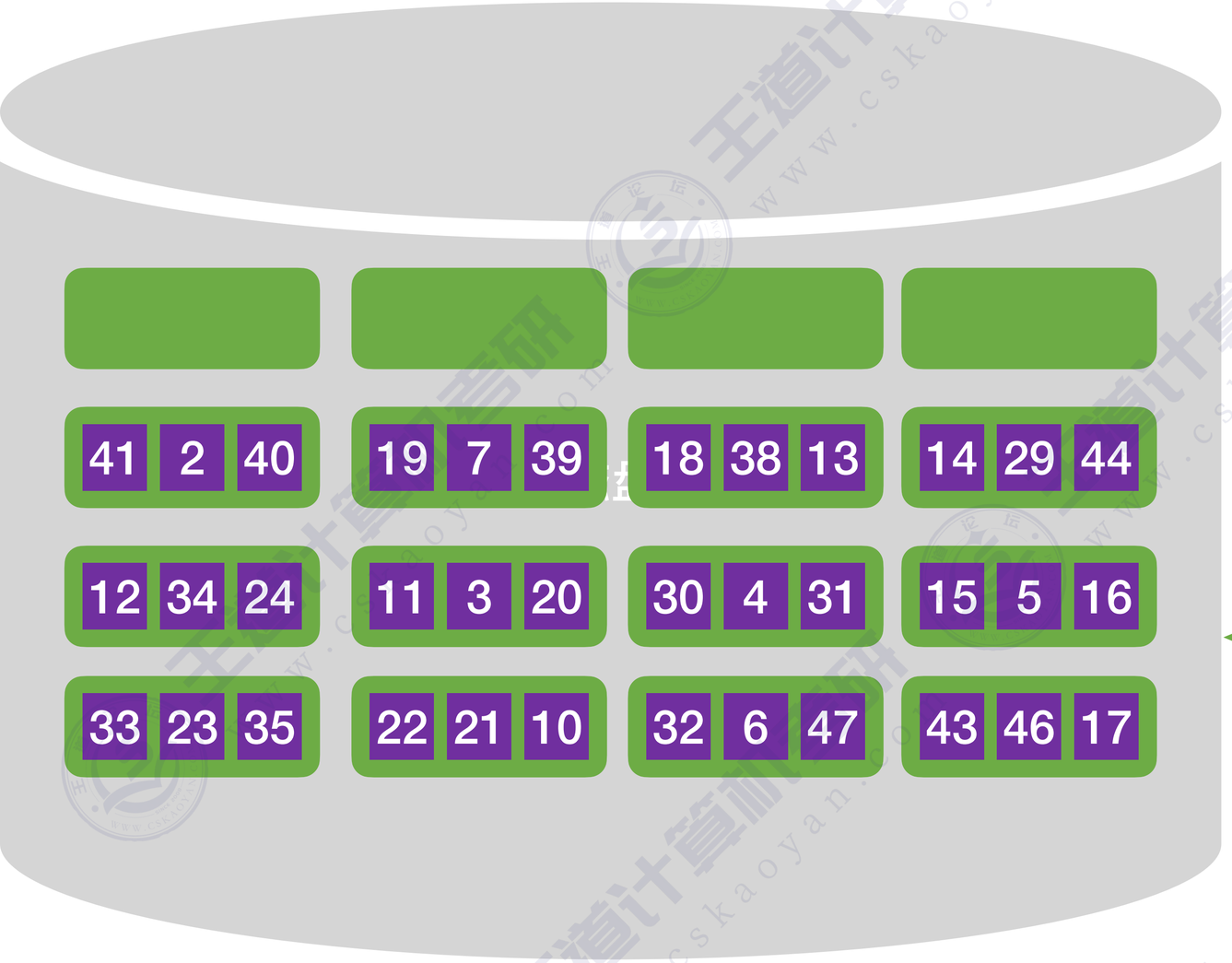
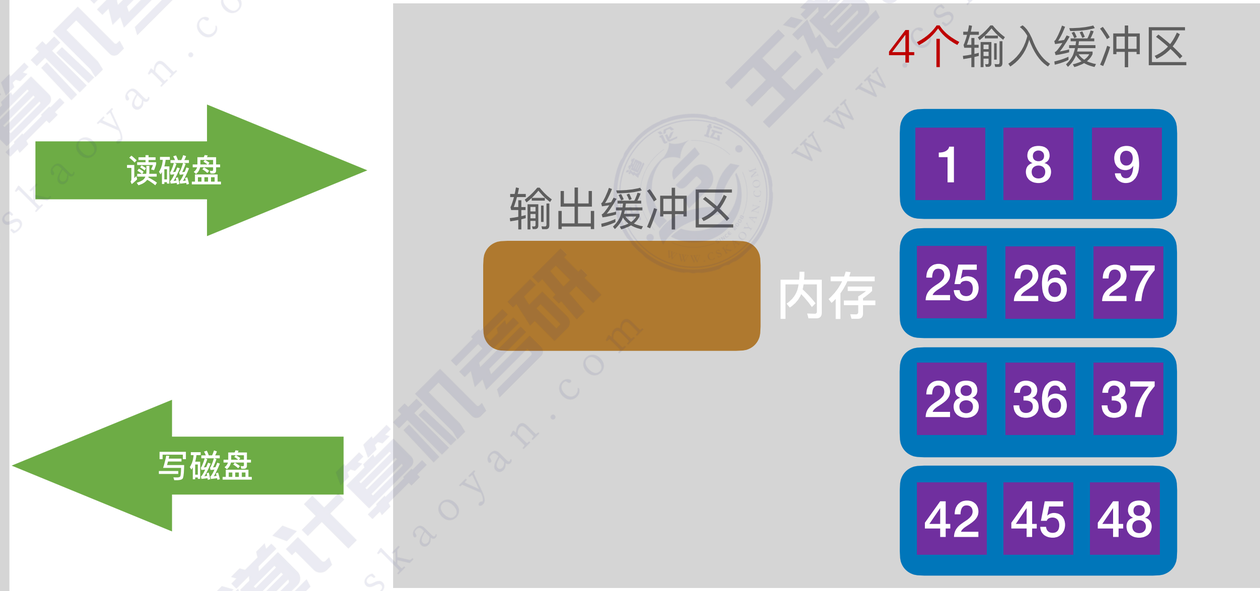
优化：减少初始归并段数量





生成初始归并段的“内存工作区”越大，初始归并段越长

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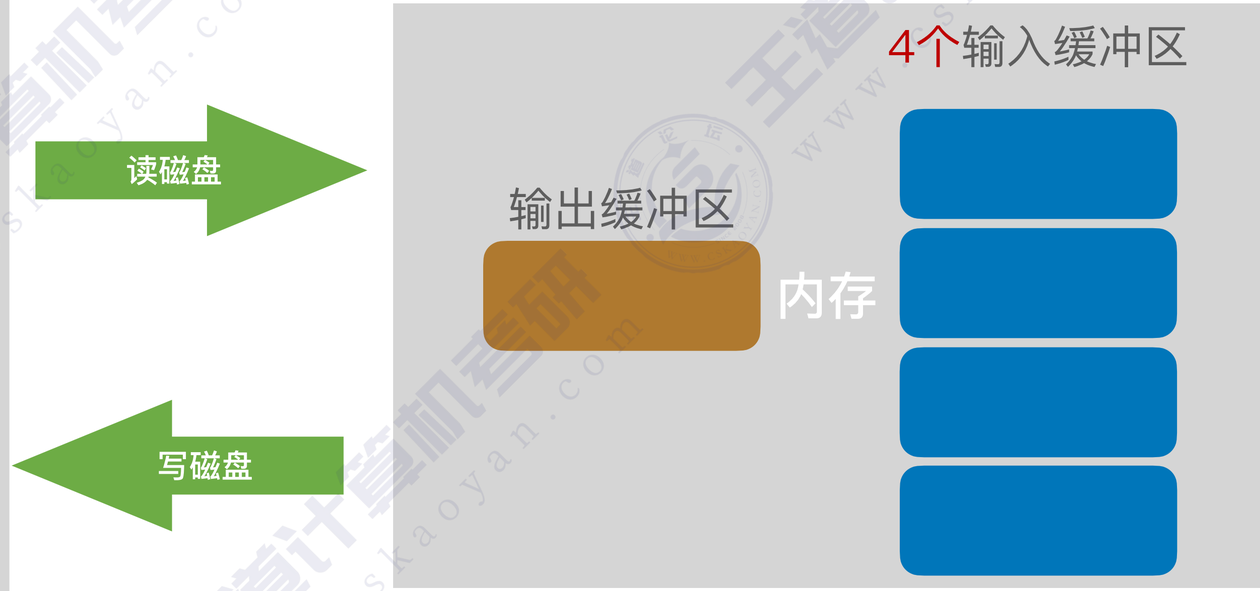
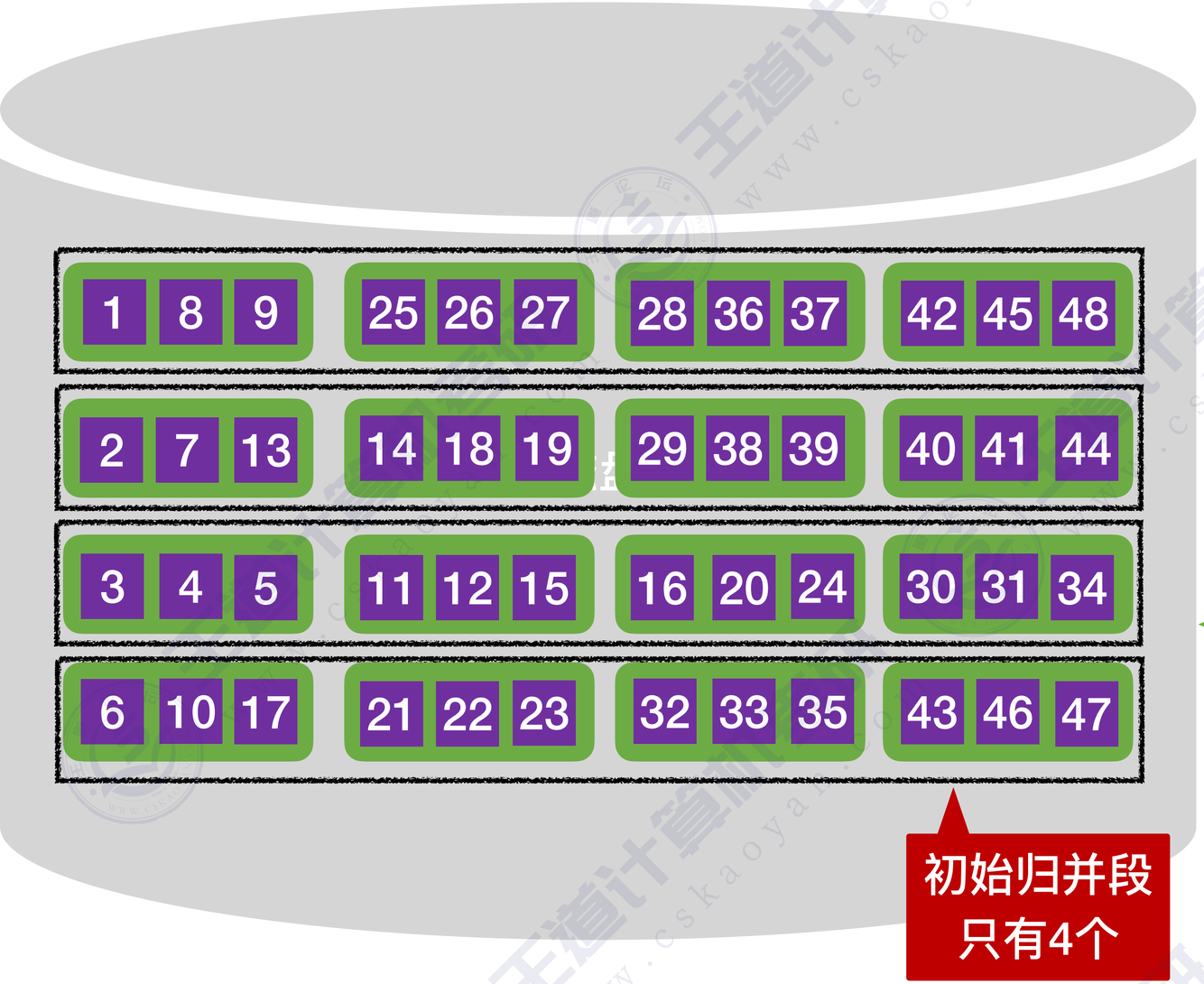
优化：减少初始归并段数量

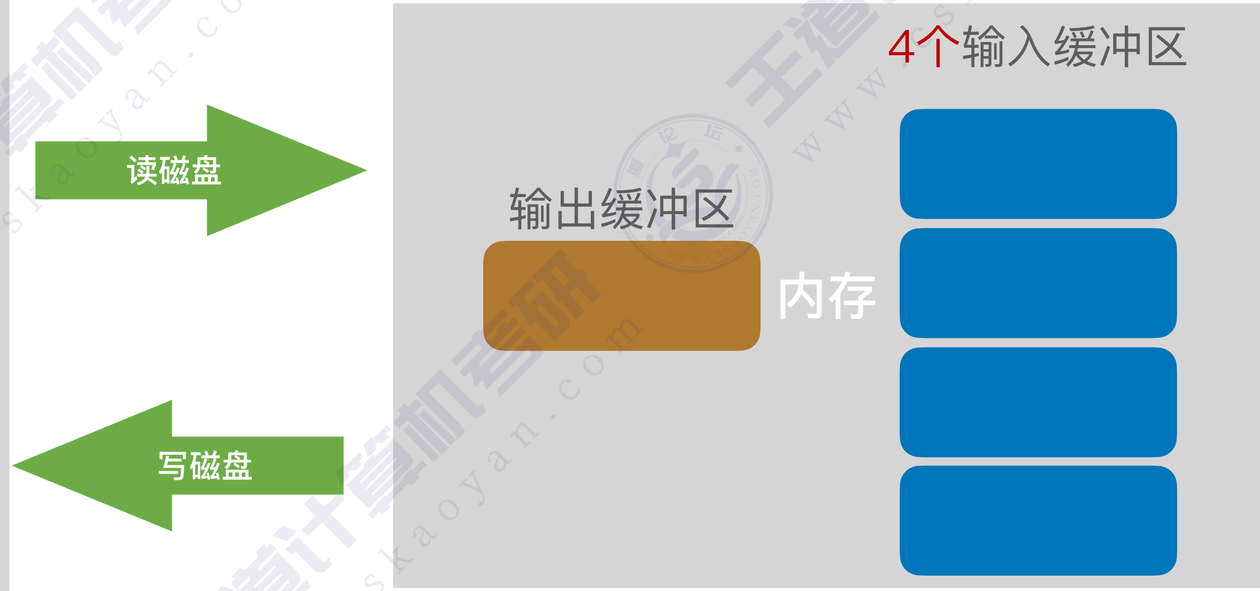


生成初始归并段的“内存工作区”越大，初始归并段越长

王道考研/CSKAOYAN.COM

优化：减少初始归并段数量





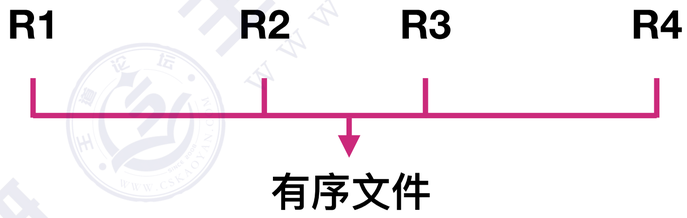
生成初始归并段的“内存工作区”越大，初始归并段越长

优化：减少初始归并段数量

对 r 个初始归并段，做k路归并，则归并树可用 k 叉树表示

若树高为h，则归并趟数 = h-1 = $\lceil log_k r \rceil$

k越大，r越小，归并趟数越少，读写磁盘次数越少



结论：若能增加初始归并段的长度，则可减少初始归并段数量 r

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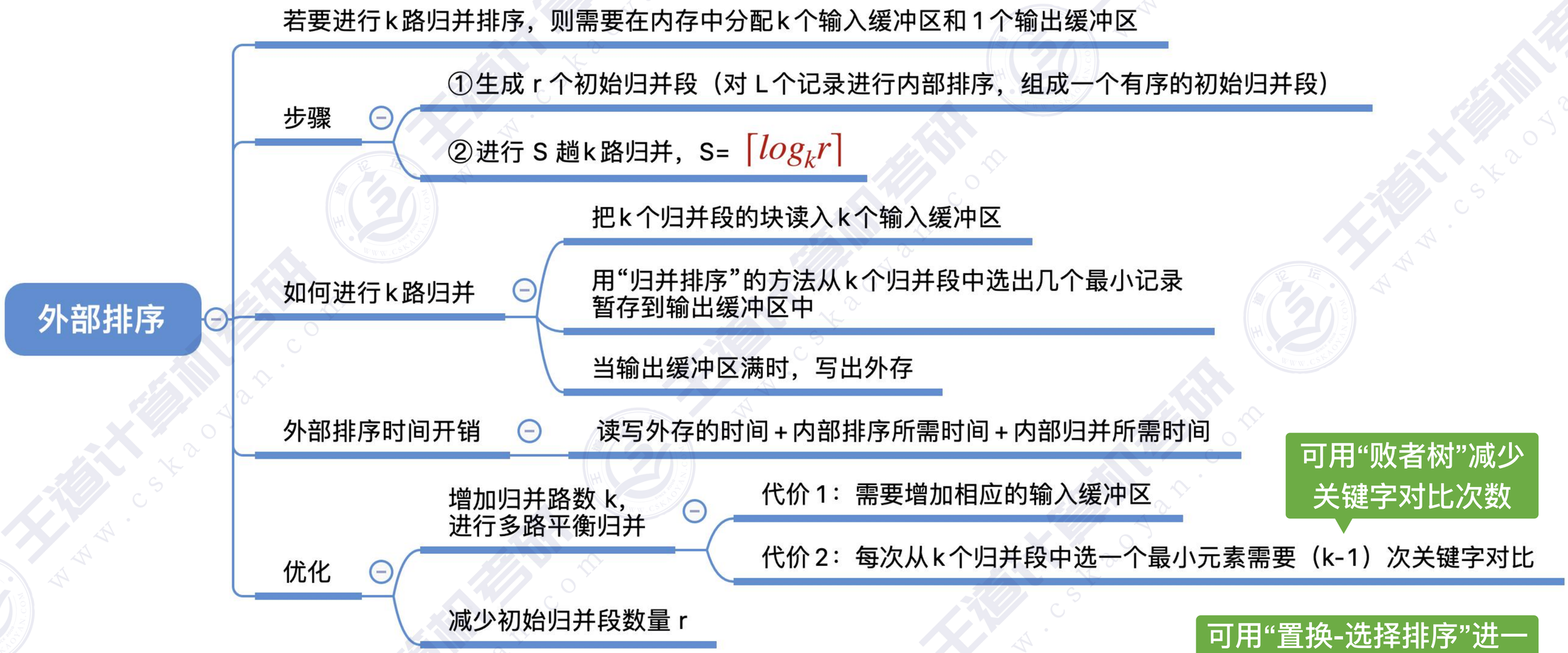
知识回顾与重要考点



注：按照本节介绍的方法生成的初始归并段，若共 N 个记录，内存工作区可以容纳 L 个记录，则初始归并段数量 r = $\lceil N/L \rceil$

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纠正：什么是多路平衡归并?

误：对r个初始归并段，做k路平衡归并，归并树可用严格k叉树（即只有度为k与度为0的结点的k叉树）来表示。

知道错哪了不?

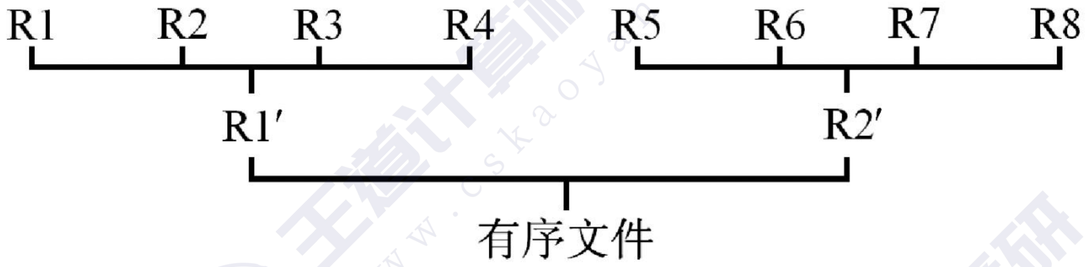

图 7.8　4 路平衡归并的排序过程

k路平衡归并：①最多只能有k个段归并为一个；

②每一趟归并中，若有 m 个归并段参与归并，则经过这一趟处理得到$\lceil m/k \rceil$个新的归并段

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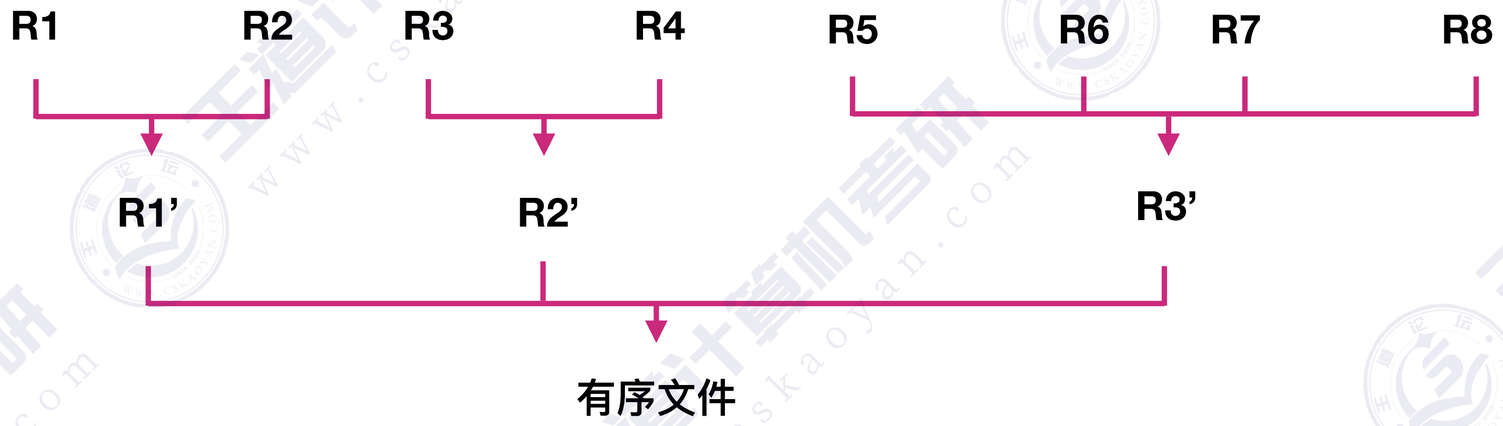
纠正：什么是多路平衡归并?

8个归并段经过一趟处理后得到3个新的归并段



这个例子是不是4路归并排序？—— 是！

这个例子是不是4路平衡归并排序？—— 不是！

$\lceil 8/4 \rceil = 2 \neq 3$

k路平衡归并：①最多只能有k个段归并为一个；

②每一趟归并中，若有 m 个归并段参与归并，则经过这一趟处理得到$\lceil m/k \rceil$个新的归并段

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